UNIT- V

5 Server Side Scripting Languages

Syllabus

PHP: *Introduction to PHP, uses of PHP, general syntactic characteristics, Primitives, operations and expressions, output, control statements, arrays, functions, pattern matching, form handling, files, cookies, session tracking, using MySQL with PHP, WAP and WML.* ***Introduction to ASP.NET :*** *Overview of the .NET Framework, Overview of C#, Introduction to ASP.NET, ASP.NET Controls, Web Services. Overview of Node JS.*

Contents

Part - I : PHP

5.1 Introduction to PHP

5.1.1 Origins and Uses of PHP

- PHP was developed in 1994 by Apache group.
- PHP stands for PHP : **H**ypertext **P**reprocessor.
- PHP is a server-side scripting language. It is mainly used for form handling and database access.
- It is free to download and use.

5.1.2 Overview

- PHP is a server side scripting language embedded in XHTML. It is an alternative to CGI, ASP, ASP.NET and JSP.
- The extensions to the PHP files are .php, .php3 or .phtml.
- The php processor works in **two modes**. If the PHP processor finds XHTML tags in the PHP script then the code is simply copied to the output file. But when the PHP processor finds the PHP code in the script then that code is simply interpreted and the output is copied to the output file.
- If you click for view source on the web browser you can never see the PHP script because the output of PHP script is send directly to the browser but you can see the XHTML tags.
- PHP makes use of dynamic typing that means there is no need to declare variables in PHP. The type of variable gets set only when it is assigned with some value.
- PHP has large number of library functions which makes it flexible to develop the code in PHP.

5.1.3 Installation of PHP

- There are various methods of getting PHP installed on your machine.
- The PHP requires the Apache Web server to execute its code. The Apache web server is open source software and can be easily downloaded from the Internet. The site for getting the Apache web server is **http://httpd.apache.org/download.cgi**.
- Then PHP can be installed on your machine from the web site **http://www.php.net**.
- Another approach which is the most efficient way to install Apache, PHP, MYSQL on your machine is to use the packages like XAMPP (The X stands for any OS) or

WAMPP (the W stands for Windows OS). These packages support the Apache, PHP, MYSQL, PERL.

- I have installed the XAMPP package. The root directory can be accessed using the **http://localhost/**

5.2 General Syntactic Characteristics

- PHP code can be embedded in the XHTML document. The code must be enclosed within **<?php** and **>**
- If the PHP script is stored in some another file and if it needs to be referred then **include** construct is used. For instance :

```
Include("myfile.inc")
```

- The variable names in PHP begin with the $ sign.
- Following are some reserved keywords that are used in PHP.

and	default	false	if	or	this
break	do	For	include	require	true
case	else	foreach	list	return	var
class	elseif	function	new	static	virtual
continue	extends	gloabal	not	switch	while
					xor

- The **comments** in PHP can be #,//, /* …*/
- The PHP statements are terminated by semicolon.

Open some suitable text editor like Notepad and type the following code. Save your code by the extension .php.

It is expected that the PHP code must be stored in **htdocs** folder of Apache.

As I have installed **xampp,** I have got the directory c:\xampp\htdocs. I have created a folder named php-examples inside the htdocs and stored all my PHP documents in that folder.

Hence when I want to get the output of the PHP code I always give the URL

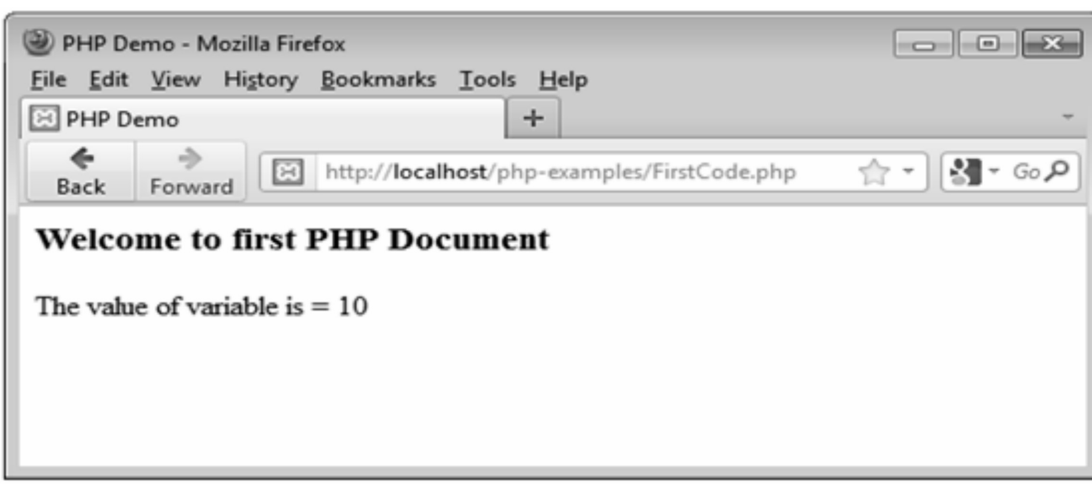
```
http://localhost/php-examples/programmName.php
The http://localhost refers to the path c:\xampp\htdocs
```

Following is the first example of PHP script

PHP Document[FirstCode.php]

```
<html>
 <head>
  <title> PHP Demo </title>
 </head>
 <body>
  <?php
   $i=10;
   echo "<h3>Welcome to first PHP Document</h3>";
   echo "The value of variable is = $i";
  ?>
 </body>
</html>
```

Output



5.2.1 Variables

- Variables are the entities that are used for storing the values.
- PHP is a dynamically typed language. That is PHP has no type declaration.
- The value can be assigned to the variable in following manner -

```
$variable_name=value
```

- If the value is not assigned to the variable then by default the value is NULL. The unsigned variables are called **unbound variable.**
- If the unbound variable is used in the expression then its NULL value is converted to the value 0.

- Following are some rules that must be followed while using the variables -
  1. The variable must start with letter or underscore _ , but it should not begin with a number.
  2. It consists of alphanumeric characters or underscore.
  3. There should not be space in the name of the variable.
  4. While assigning the values to the variable the variable must start with the $. For example

```
$marks=100;
```

- Using the function **IsSet** the value of the variable can be tested. That means if IsSet($marks) function returns **TRUE** then that means some value is assigned to the variable **marks.**
- If the unbound variable gets referenced then the error reporting can be done with the help of a function **error_reporting(7)**. The default error reporting level is 7.

5.3 Primitives

SPPU : May-18,19, Marks 4

5.3.1 Integer Type

- For displaying the integer value the Integer type is used.
- It is similar to the **long** data type in C.
- The size is 32 bit.

5.3.2 Double Type

- For displaying the real values the double data type is used.
- It includes the numbers with decimal point, exponentiation or both. The exponent can be represented by E or e followed by integer literal.
- It is not compulsory to have digits before and after the decimal point. For instance .123 or 123. is allowed in PHP.

5.3.3 String Type

- There is no character data type in PHP. If the character has to be represented then it is represented using the string type itself; but in this case the string is considered to be of length 1.
- The string literals can be defined using either single or double quotes.
- In single quotes the escape sequence or the values of the literals can not be recognized by PHP but in double quotes the escape sequences can be recognized.

For example -

```
'The total marks are= $marks'
```

will be typed as it is but

```
"The total marks are= $marks"
```

will display the value of $marks variable.

5.3.4 Boolean Type

- There are only two types of values that can be defined by the Boolean type and those are TRUE and FALSE.
- If Boolean values are used in context of integer type variable then TRUE will be interpreted as 1 and FALSE will be interpreted as 0.
- If Boolean values are used in context of double type then the FALSE will be interpreted as 0.0.

Review Question

1. *Classify data types of PHP and describe various data types in each type.*

SPPU : May-18,19, Marks 4

5.4 Type Conversion

- PHP supports for both implicit and explicit type conversion.
- The implicit type conversion is called **coercion**. In implicit type conversion the context of expression determines the data type that is expected or required.

Implicit conversion

- The coercion takes place between integer and double types and between Boolean and other scalar type.
- The coercion is also between the numeric and string types. That means whenever a numeric value appears in string context then it is converted to string type, similarly when the string value appears in numeric context it is automatically converted to numeric type.
- When the double value is converted to integer the fractional part is eliminated and the value is not rounded.

Explicit conversion

- There are three ways by which the explicit conversion takes place.
  1. The syntax is

```
(datatype)$variable_name
```

For example -

```
(int)$marks
```

2. The syntax is -

```
Conversion_function($variable_name)
```

For example -

```
Intval($marks)
```

3. The syntax is -

```
settype($variable_name,"datatype")
```

For example -

```
settype($marks,"integer");
```

- The **gettype** function is useful to obtain the data type of the variable.

5.5 Operations and Expressions

5.5.1 Arithmetic Operators

- PHP supports the collection of arithmetic operators such as +, -, /, *, %, ++ and – with their usual meaning.
- While using the arithmetic operators if both the operands are integer then the result will be integer itself.
- If either of the two operands is double then the result will be double.
- PHP has large number of predefined functions. Some of these functions are enlisted in the following table -

Function	Purpose
floor	The largest integer less than or equal to the parameter is returned. For example floor(4.9) will return 4,
ceil	The smallest integer greater than or equal to the parameter is returned. For example ceil(4.9) will return 5,
round	Nearest integer is returned.
abs	Returns the absolute value of the parameter.
min	It returns the smaller element.
max	It returns the larger element.

5.5.2 Relational Operators

- There are eight relational operators used in PHP.
- These are <, >, <=, >=, !=, == has their usual meaning. These are six traditional operators.
- The operator === is used in PHP. It returns true if both operands that are using === have same type and have same value.
- The operator !== is opposite of ===.
- If one of the operand in the six operators is not same then the coercion will occur automatically.

5.5.3 Boolean Operators

- The Boolean operators are

<table>
<tr><th>Operator</th><th>Meaning</th></tr>
<tr><td>and
&&</td><td>The binary AND operation is performed.</td></tr>
<tr><td>or
||</td><td>The binary OR operation is performed.</td></tr>
<tr><td>xor</td><td>The XOR operation will be performed.</td></tr>
</table>

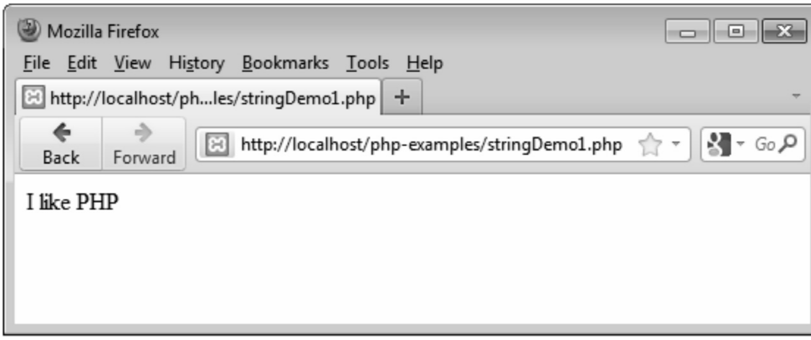
5.5.4 String Operators

- Concatenation is the only one **operator** used in string. It is denoted by dot.
- Strings are treated as the array of characters. The first position of the character is indexed as 0.
- The sample PHP script that stores the string in a variable is as given below -

PHPScript[stringDemo1.php]

```
<?php
$s="I like PHP";
echo $s;
?>
```

Output



Note that the variable **s** is assigned with the string. The string is given in a double quote. Then using the **echo** whatever string is stored in the variable **s** is displayed on the console.

5.6 Output

- The PHP is a server side scripting language and is used to submit the web page to the client browser. Hence it is a standard method of embedding the PHP code within the XHTML document. That means we can use the XHTML tags in PHP while displaying the output.
- The **print** function is used to create simple unformatted output. For example : The string can be displayed as follows

```
print "I am proud of my <b>country</b>"
```

The numeric value can also be displayed using the **print**. For example -

```
print(100);
```

It will display the output as 100.

- PHP also makes use of the **printf** function used in C. For example

```
printf("The student %d has %f marks",$roll_no,$marks);
```

- Following is a simple PHP document which makes use of the statements for displaying the output.

PHP Document[OutputDemo.php]

```
<html>
<head>
    <title> Output Demo</title>
</head>
<body>
 <?php
    print "<h2>Welcome to my Website </h2>";
```

```
    print "<hr/>";
    $roll_no=1;
    $name="AAA";
    printf("<b>The roll number: %d</b>",$roll_no);
   print "<br/><b>";
    printf("The name: %s",$name);
    print "</b>";
?>
</body>
</html>
```

Output

5.7 Statements Control

- The control statements in PHP are similar to the control statements that are used in C.
- The control statements include if statements, while, do while and for loops.

5.7.1 The Selection Statement

- The **if** statement, the **if … else** statement or **if…elseif** statements are used in selection statements. Following is an example of it.

PHPDocument[selection.php]

```
<html>
<head>
    <title>Selection Demo</title>
</head>
```

```
<body>
<?php
    print "<h2>Selection Statement </h2>";
    $a=10;
    $b=20;
    $c=30;
    if($a>$b)
    if($a>$c)
    print "<b><I>a is the largest number </I></b>";
else
        print "<b><I> c is the largest number</I></b>";
    else
    if($b>$c)
        print "<b><I>b is the largest number</I></b>";
else
    print "<b><I>c is the largest number</I></b>";
?>
</body>
</html>
```

Output

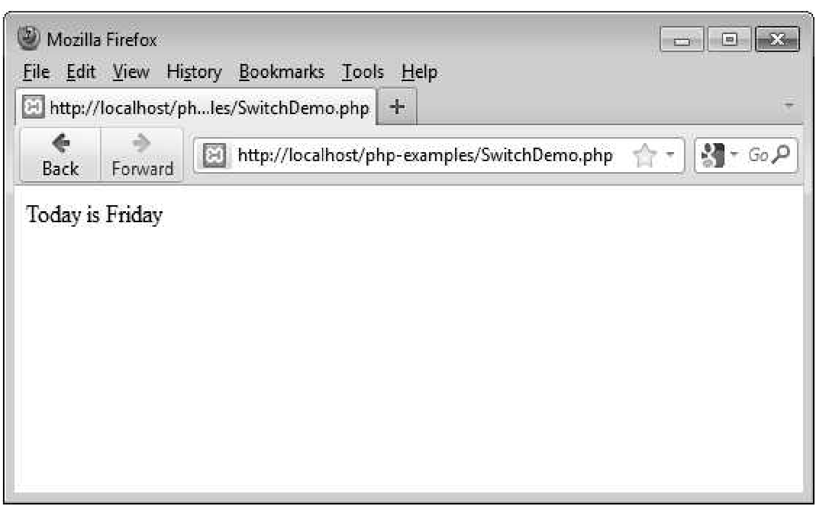
- Similar to **if** statement the **switch** statement can also be used for selection. Following is a simple PHP script for demonstrating switch statement -

PHPDocument[SwitchDemo.php]

```
<?php
$today = getdate();
switch($today['weekday'])
{
```

```
    case "Monday":print "Today is Monday";
        break;
    case "Tuesday":print "Today is Tuesday ";
        break;
    case "Wednesday":print "Today is Wednesday ";
        break;
    case "Thursday":print "Today is Thursday";
        break;
    case "Friday":print "Today is Friday";
        break;
    case "Saturday":print "Today is Saturday";
        break;
    case "Sunday":print "Today is Sunday ";
        break;
default: print "Invalid input";
}
?>
```

Output



5.7.2 Looping Structure

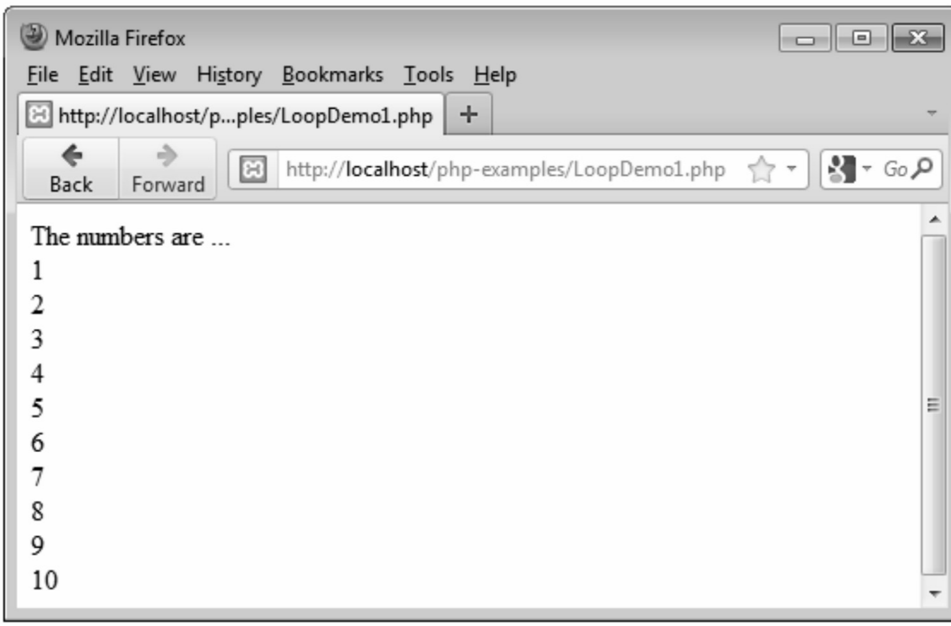
- The **while**, for and **do-while** statements of PHP are similar to JavaScript.
- Following is a simple PHP script which displays the first 10 number.

PHPDocument[LoopDemo1.php]

```
<?php
$i=1;
print "The numbers are ...";
print "<br/>";
```

```
while($i<=10)
{
    print $i;
    print "<br/>";
    $i++;
}
?>
```

Output



Similarly we can modify the above script by using do-while and for loop as follows -

do ...while

```
<?php
$i=1;
print "The numbers are ...";
print "<br/>";
do
{
    print $i;
    print "<br/>";
    $i++;
}while($i<=10);
?>
```

for

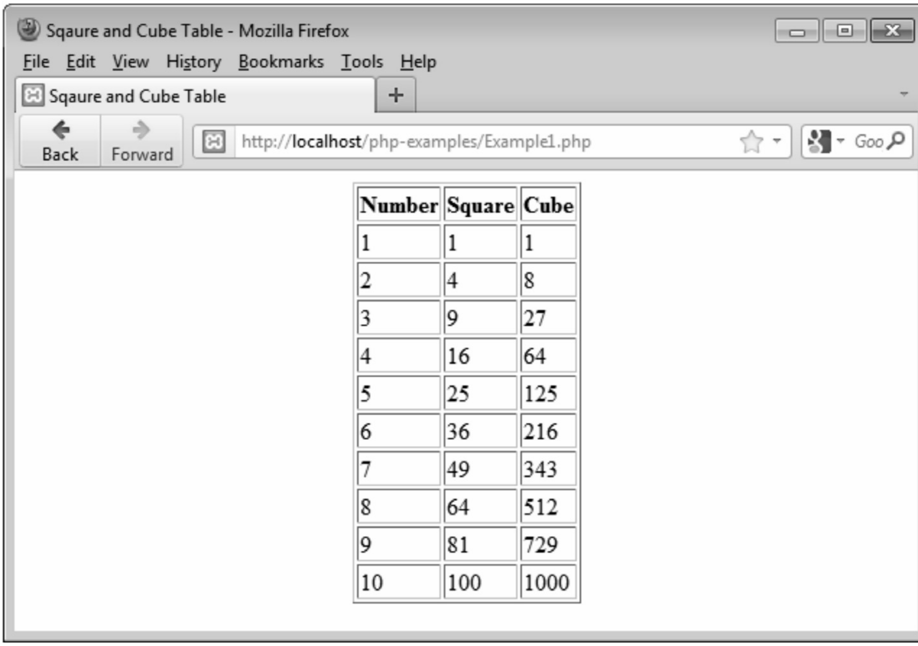
```
<?php
print "The numbers are ...";
print "<br/>";
for($i=1;$i<n)
{
    print $i;
    print "<br/>";
    $i++;
}
?>
```

Example 5.7.1 *Write a PHP script to display the squares and cubes of 1 to 10 numbers.*

Solution :

```
<html>
<head>
<title>Square and Cube Table </title>
</head>
<body>
<center>
<?php
print "<table border =1>";
print "<tr>";
print "<th>Number</th>";
print "<th>Square</th>";
print "<th>Cube</th>";
print "</tr>";
for($i=1;$i<=10;$i++)
{
print "<tr>";
print "<td>$i";
print "</td>";
print "<td>";
print $i*$i;
print "</td>";
print "<td>";
printpow($i,3);
print "</td>";
print "</tr>";
}
print "</table>";
?>
</center>
</body>
</html>
```

Output



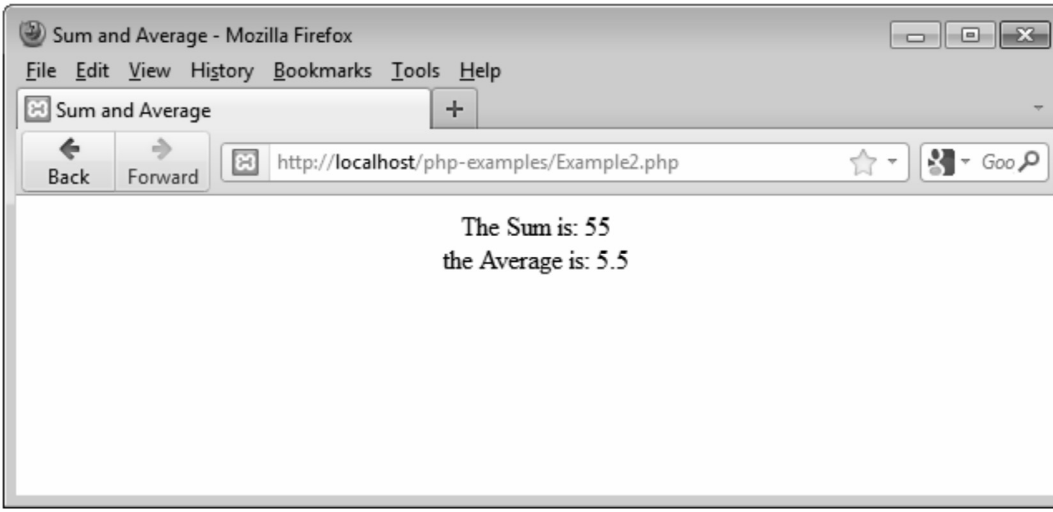
Number	Square	Cube
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
7	49	343
8	64	512
9	81	729
10	100	1000

Example 5.7.2 *Write a PHP script to compute the sum and average of N numbers.*

Solution :

```
<html>
<head>
<title> Sum and Average </title>
</head>
<body>
<center>
<?php
$sum=0;
for($i=1;$i<=10;$i++)
{
    $sum+=$i;
}
$avg=$sum/10;
print "The Sum is:  $sum";
print "<br/>";
print "the Average is: $avg";
?>
</center>
</body>
</html>
```

Output



5.8 Arrays

SPPU : Dec.-18, May-18,19, Marks 6

- An array is a collection of similar type of elements, but in PHP you can have the elements of mixed type together in single array.
- In each PHP, each element has two parts key and value.
- The key represents the index at which the value of the element can be stored.
- The keys are positive integers that are in ascending order.

5.8.1 Creating and Manipulating Arrays

For creating an array in PHP, we use **array()** construct.

Syntax :

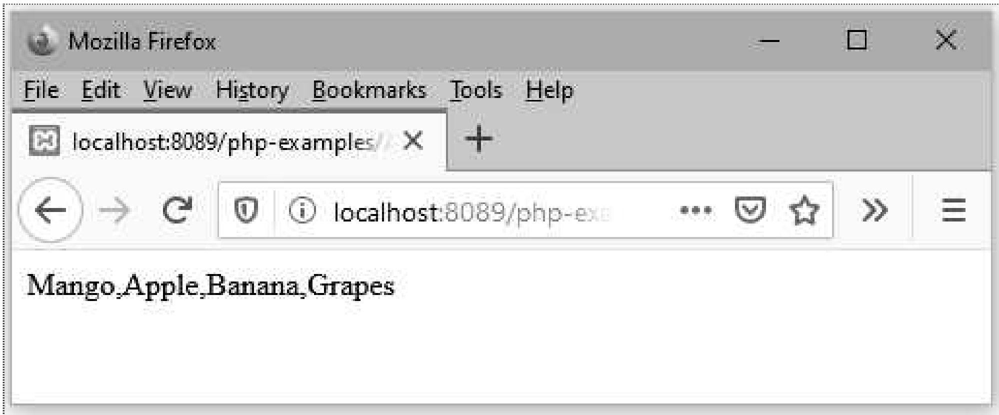
```
$array_name=array(value)
```

Example : Following PHP script shows how to create an array in PHP.

ArrayCreateDemo.php

```
<!DOCTYPE html>
<html>
<body>

<?php
$fruits = array("Mango", "Apple", "Banana","Grapes");
echo $fruits[0] . "," . $fruits[1] . "," . $fruits[2] . ",". $fruits[3];
?>
</body>
</html>
```

Output



Program Explanation : In above program,

1) We have created an array named **fruits** using **array()** construct
2) In the echo statement we are displaying each element of the array with the help of array index such as fruts[0],fruits[1],… and so on.

Adding and Deleting Elements

- In PHP, **arrays are dynamic.** That means they can grow in size or can shrink.
- We can add the element in the array using key/index that hasn't used. For example -

```
$name[5]="CCC";
```

- As there is no current value for index 5, the array will grow. Similarly we can skip the index value and add the element as follows -

```
$name[ ]="CCC";
```

- The advantage to this approach is that we don't have to worry about skipping an index key.
- We can skip some index and can insert the element in the array. For example -

```
$names=array("AAA","BBB","CCC","DDD");
$names[8]="HHH";
```

- If we iterate through the array elements then -

```
Array ([0]=>AAA [1]=>BBB [2]=>CCC [3]=>DDD [8] =>HHH)
```

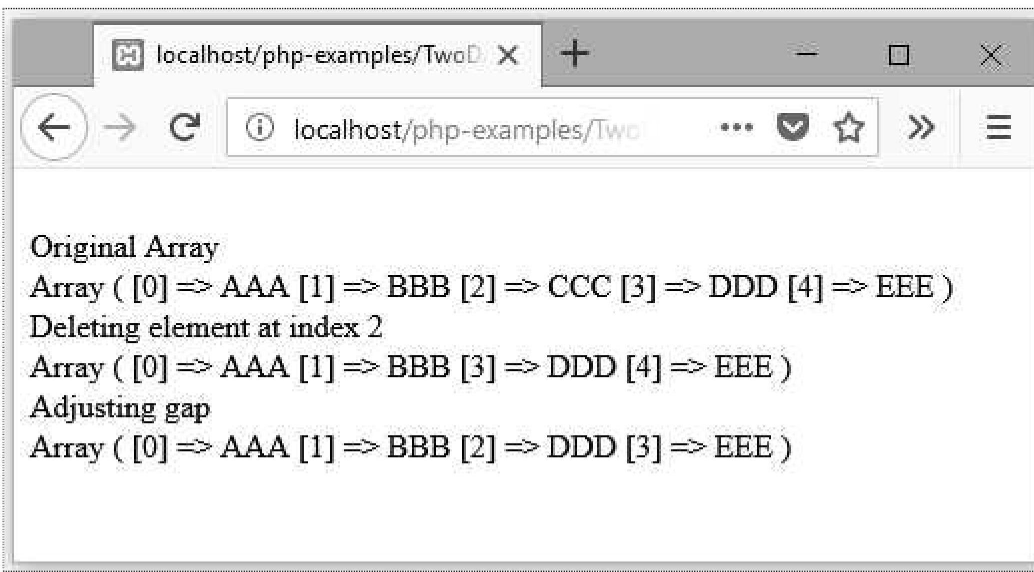
- Thus there is now **gap** in our array. If we try to reference the array[4] then it will return NULL value which represents that there is no value present at that index.
- Using the **unset** function we can create gaps in the PHP array. For example -

PHP Document

```
<!DOCTYPE html>
<html>
<body>
```

```
<?php
$Student =  array("AAA","BBB","CCC","DDD","EEE");
echo "<br/>Original Array<br/>";
print_r($Student);
echo "<br/>Deleting element at index 2<br/>";
unset($Student[2]);
print_r($Student);
echo "<br/>Adjusting gap<br/>";
$Student=array_values($Student);
print_r($Student);
?>
</body>
</html>
```

Output



Script Explanation :

1. In above script we have used **unset** function to create a gap at index 2.
2. Later on we have used **array_values()** to adjust this gap by subsequent element.
3. Using the **print_r** function the array can be displayed on the browser window.

Checking if value exists

- The **isset ()** function is used to check whether **a variable is set or not.**
- If a variable is already unset with **unset()** function, it will no longer be set.
- The isset() function return false if testing variable contains a NULL value.

PHP Document

```
<!DOCTYPE html>
<html>
<body>
<?php
$Student =  array(1=>"AAA",2=>"BBB");
echo "<br/>Original Array<br/>";
print_r($Student);
if(isset($Student[0]))
    echo "<br/>There is no value set at index 0<br/>";
if(isset($Student[1]))
    echo "The value present at index 1 is: ".$Student[1];
?>
</body>
</html>
```

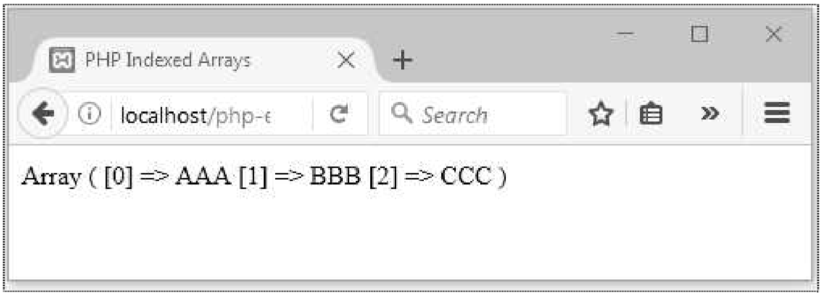
5.8.2 Types of Arrays

There are three types of arrays.

1. Indexed array : Indexed array are the arrays with numeric index. The array values can be stored from index 0. For example -

```
<html>
<head>
    <title>PHP Indexed Arrays</title>
     </head>
    <body>
<?php
$names = array("AAA", "BBB", "CCC");
// Printing array structure
print_r($names);
?>
</body>
</html>
```

Output



Here values get stored at corresponding index as follows -

```
$mylist[0]    =    10;
$mylist[1]    =    20;
$mylist[2]    =    30;
$mylist[3]    =    40;
$mylist[4]    =    50;
```

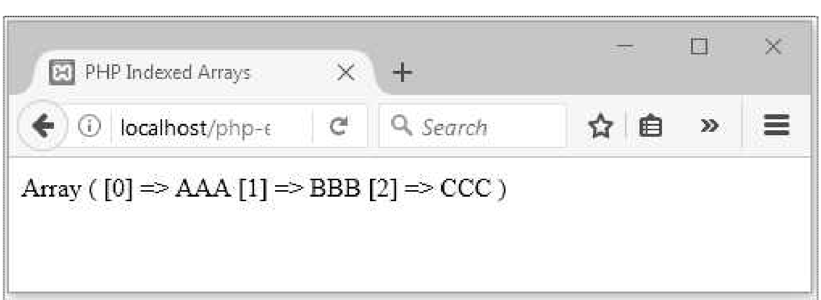
We can directly assign some value at specific index.

```
$mylist[5]    =    100;
```

2. Associated array : Associated arrays are the arrays with named keys. It is a kind of array with **name** and **value** pair. For example -

```
<html>
<head>
  <title>PHP Associative Array</title>
</head>
<body>
<?php
$city["AAA"] = "Pune";
$city["BBB"] = "Mumbai";
$city["CCC"] = "Chennai";

// Printing array structure
print_r($city);
?>
</body>
</html>
```

Output



3. Multidimensional Arrays

- PHP support for multidimensional arrays.
- We can store the elements in two dimensional array as

```
$Student = array
```

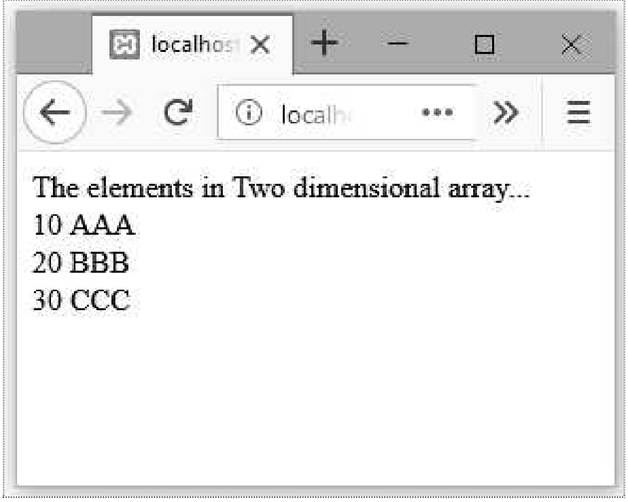
```
 (
  array(10,"AAA"),
  array(20,"BBB"),
  array(30,"CCC"),
  );
```

- The complete PHP program in which the multidimensional array is created and accessed is as follows -

PHP Document

```
<!DOCTYPE html>
<html>
<body>
<?php
$Student = array
 (
  array(10,"AAA"),
  array(20,"BBB"),
  array(30,"CCC"),
  );
echo "The elements in Two dimensional array...<br/>";
for ($row = 0; $row < 3; $row++) {
  for ($col = 0; $col < 2; $col++) {
    echo " ".$Student[$row][$col];
  }
  echo "<br/>";
}
?>
</body>
</html>
```

Output



5.8.3 Extracting Data from Arrays, Implode, Explode and Flip Functions

i) The extract function :

Using the extract() function the array keys becomes the variable name and the array values become the variable values.

Syntax :

```
int extract($input_array, $extract_rule, $prefix)
```

Where

$input_array : is the name of the array whose values need to be extracted. This is a required parameter.

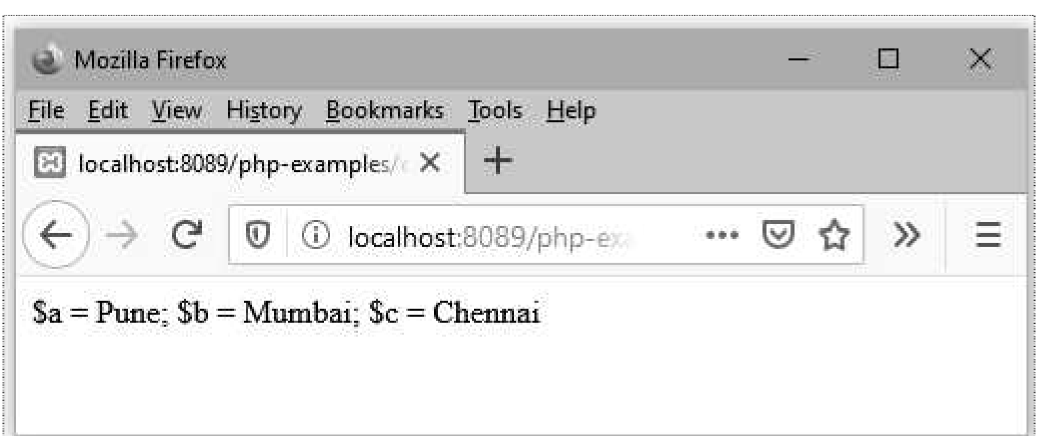
$extract_rule : This is an optional parameter. This parameter specifies how invalid variable names will be treated.

$prefix : The prefix is automatically separated from the array key by an underscore character. This is an optional parameter.

Example Program

```
<!DOCTYPE html>
<html>
<body>

<?php
$arr = array("a" => "Pune","b" => "Mumbai", "c" => "Chennai");
extract($arr);
echo "\$a = $a; \$b = $b; \$c = $c";
?>
</body>
</html>
```

Output



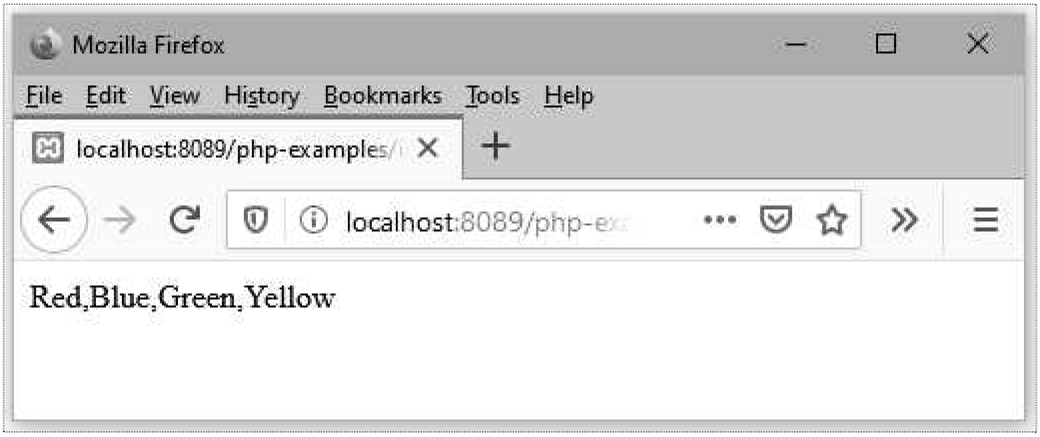
ii) The implode function :

The implode function converts array into the string. For example -

implodeDemo.php

```
<!DOCTYPE html>
<html>
<body>
<?php
$arr[0] = "Red";
$arr[1] = "Blue";
$arr[2] = "Green";
$arr[3] = "Yellow";
$text = implode(",",$arr);
echo $text;
?>
</body>
</html>
```

Output



iii) The explode function

The explode() function is to split a string.

Syntax :

```
explode(delimiter, string_name, limit)
```

Where

Delimiter : It sets the boundary string within the input string

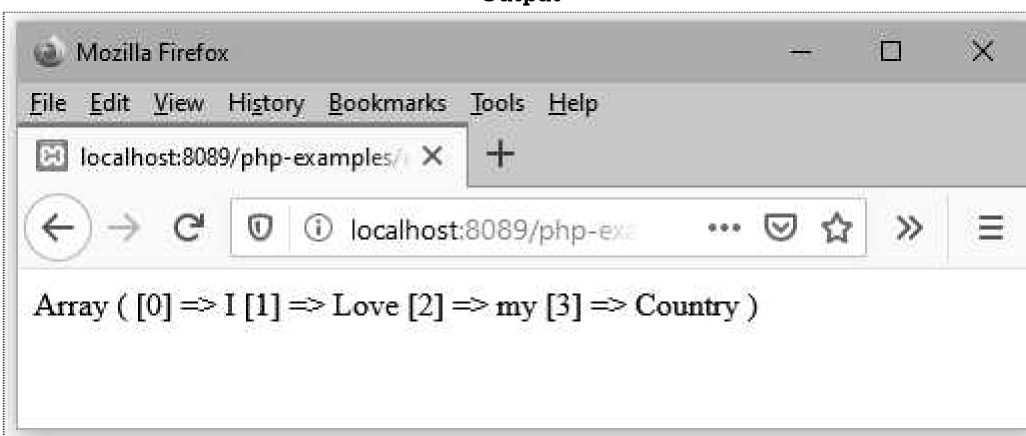
String_name : The name of the string to be split

Limit : It indicates the maximum number of elements in the output array if set to +ve value. If set to -ve value, all but the last element will be present in the output array.

Example Program

```
<!DOCTYPE html>
<html>
<body>

<?php
$str="I Love my Country";
$arr=explode(" ",$str);
print_r($arr);
?>
</body>
</html>
```

Output



iv) The flip function

The array_flip() function is used to exchange the keys with their associated values in array.

That means after applying the array_flip function we get keys from array become values and values from array become keys.

Syntax :

```
array_flip(name_of_array)
```

Example Program

```
<!DOCTYPE html>
<html>
<body>

<?php
$arr=array("Red"=>10,"Green"=>20,"Blue"=>30,"Black"=>40);
$result=array_flip($arr);
```

```
print_r($result);
?>

</body>
</html>
```

Output

5.8.4 Traversing Arrays

1. The current and next function

- The array element reference start at the first element and array maintains an internal pointer using which the next element can be easily accessible.
- This helps to access the array elements in sequential manner.
- The pointer **current** is used to point to the current element in the array. Using the **next** function the next subsequent element can be accessed. Following PHP code illustrates this idea -

PHP Document[ArrayFunDemo4.php]

```
<?php
$mylist = array("Hello", "PHP","You","Are","Wonderful!!!");
$myval=current($mylist);
print("The current value of the array is <b>$myval</b>");
print "<br/>";
$myval=next($mylist);
print("The next value of the array is <b>$myval</b>");
print "<br/>";
$myval=next($mylist);
print("The next value of the array is <b>$myval</b>");
```

```
print "<br/>";
$myval=next($mylist);
print("The next value of the array is <b>$myval</b>");
print "<br/>";
$myval=next($mylist);
print("The next value of the array is <b>$myval</b>");
?>
```

Output

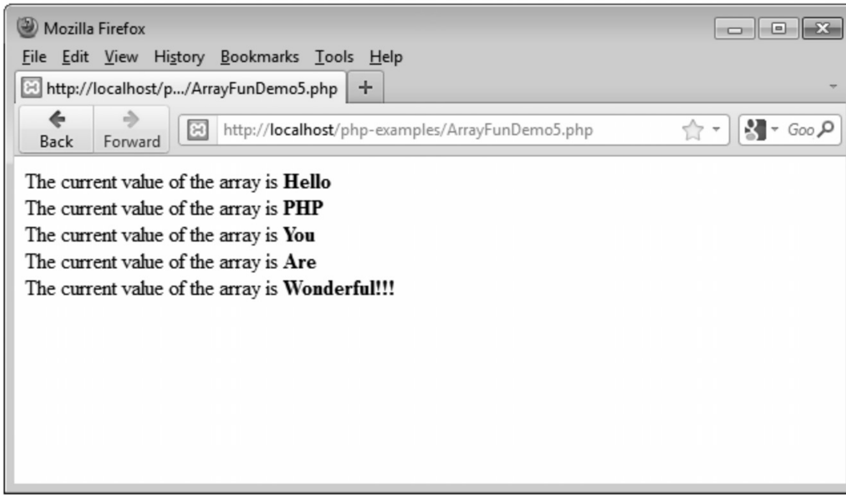
2. The each and foreach function

- Using **each** function we can iterate through the array elements.

PHP Document[ArrayFunDemo5.php]

```
<?php
$mylist = array("Hello", "PHP","You","Are","Wonderful!!!");
while($myval=each($mylist))
{
    $val=$myval["value"];
    print("The current value of the array is <b>$val</b>");
    print "<br/>";
}
?>
```

Output



- The **foreach** function is used to iterate through all the elements of the loop. The syntax of foreach statement is as follows -

```
foreach($array as $value)
{
  statements to be executed
}
```

- The above code can be modified and written as follows -

PHP Document[ArrayFunDemo6.php]

```
<?php
$mylist = array("Hello", "PHP","You","Are","Wonderful!!!");

foreach($mylist as $value)
{
    print("The current value of the array is <b>$value</b>");
    print "<br/>";
}
?>
```

The output will be the same as above.

Review Questions

1. *What is associative arrays in PHP ? Explain it with the help of simple PHP code.*
   SPPU : May-18,19, Marks 6
2. *What is multi-dimensional arrays in PHP ? Explain it with simple PHP code.*
   SPPU : Dec.-18, Marks 6

5.9 Functions

The functions in PHP are very much similar to the functions in C. Let us discuss it in detail.

5.9.1 General Characteristics of Functions

- The syntax of the function definition is as follows -

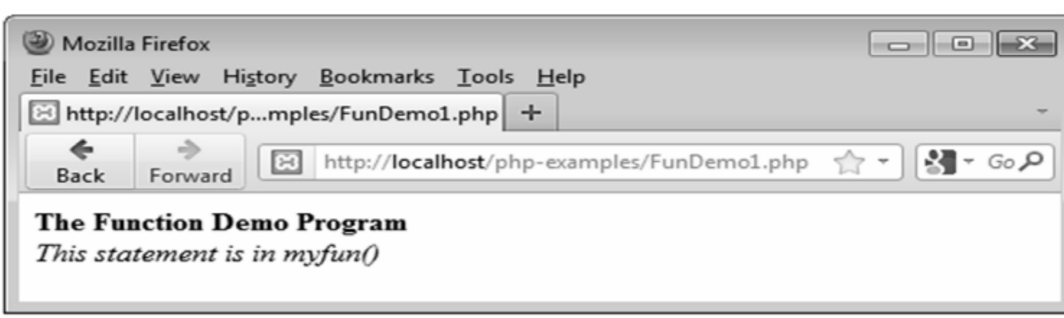
```
functionname_of_function(parameter list)
{
statements to be executed in function
 …
 …
 …}
```

- The function gets executed only after the call to that function. The call to the function can be from anywhere in the PHP code. For example -

PHPDocument[FunDemo1.php]

```
<?php
functionmyfun()
{
    print "<i>This statement is in myfun()</i>";
}
print "<b>The Function Demo Program</b>";
print "<br/>";
myfun();
?>
```

Output



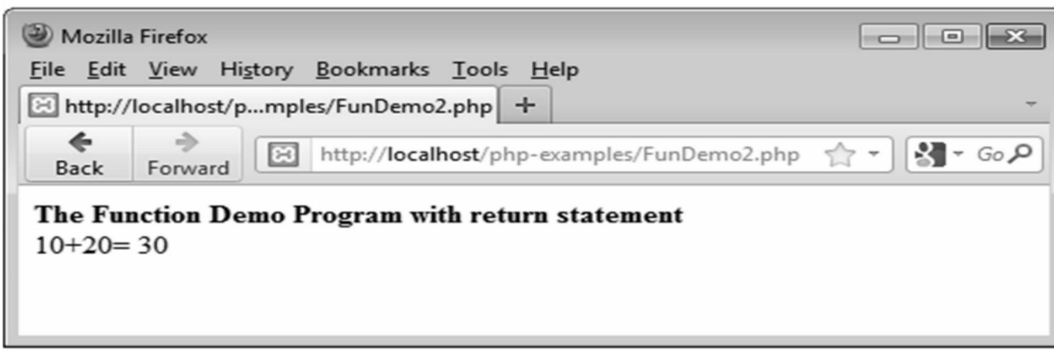
- The **return** statement is used for returning some value from the function body. Following PHP scripts shows this idea.

PHPDocument[FunDemo2.php]

```
<?php
function Addition()
```

```
{
    $a=10;
    $b=20;
    $c=$a+$b;
   return $c;
}
print "<b>The Function Demo Program with return statement</b>";
print "<br/>";
print "10+20= ".Addition();
?>
```

Output



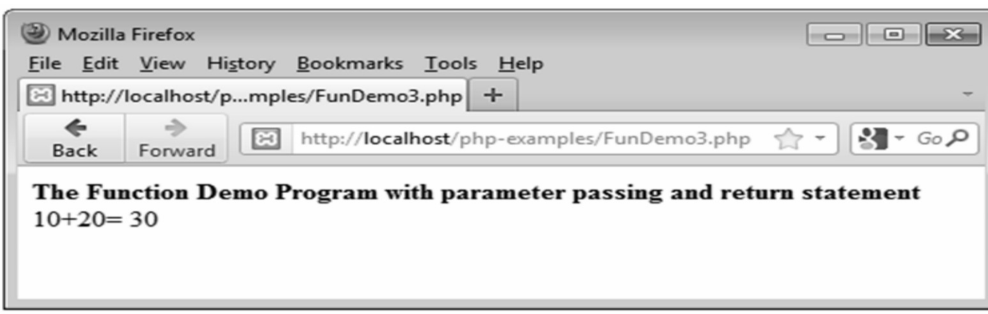
5.9.2 Parameters

- The parameter that we pass to the function during the call is called the **actual parameter.** These parameters are generally the expressions.
- The parameters that we pass to the function while defining it is called the **formal parameters.** These are generally the variables. It is not necessary that the number of actual parameters should match with the number of formal parameters.
- If there are few actual parameter and more formal parameters then the value of formal parameter will be some unbounded one.
- If there are many actual parameters and few formal parameters then the excess of actual parameters will be ignored.
- The default parameter passing technique in PHP is pass by value. The parameter passing by value means the values of actual parameters will be copied in the formal parameters. But the values of formal parameters will not be copied to the actual parameters.
- Following PHP script illustrates the functions with parameters.

PHP Document[FunDemo3.php]

```
<?php
function Addition($a,$b)
{
       $c=$a+$b;
      return $c;
}
print "<b>The Function Demo Program with parameter passing and return
statement</b>";
print "<br/>";
$x=10;
$y=20;
print "10+20= ".Addition($x,$y);
?>
```

Output



5.10 Built in Functions

PHP has number of predefined functions.

5.10.1 Functions based on Basic Operations

These function help in performing basic operations such floor, ceil, round and so on.

Function	Purpose
floor	The largest integer less than or equal to the parameter is returned. For example - floor(4.9) will return 4,
ceil	The smallest integer greater than or equal to the parameter is returned. For example - ceil(4.9) will return 5,
round	Nearest integer is returned.
abs	Returns the absolute value of the parameter.

min	It returns the smaller element.
max	It returns the larger element.

5.10.2 String Handling Functions

Various functions used for string handling are -

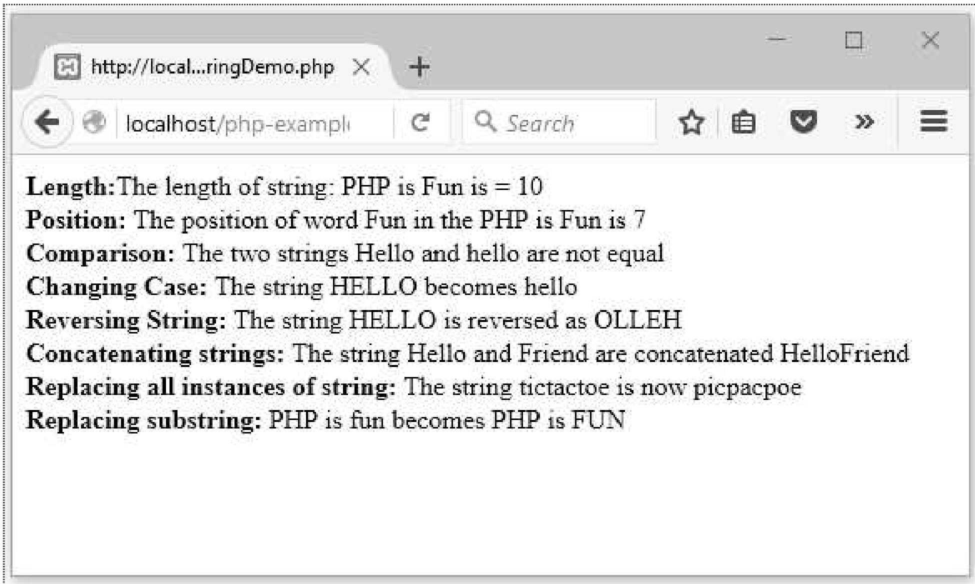
Function	Purpose	Sample PHP code	Output
strlen(string1)	It finds the total number of characters in the string	<?php $s="Friend"; echo $s <?php	6
strcmp (string1,string2)	It compares the two strings. It is case sensitive. If this function returns 0 then two strings are equal. If this function returns >0 then string1 is greater than string2. If this function returns <0 then string1 is less than string2.	<?php echo strcmp("PHP","PHP"); ?>	0
strtolower (string1)	This function converts the characters in string1 to lower case.	<?php echo strtolower("PHP."); ?>	php
strtoupper (string1)	This function converts the characters in string1 to upper case.	<?php echo strtolower("php"); ?>	PHP
trim(string1)	This function eliminates the white space from both the ends of the string.	<?php $str = " PHP "; echo " : $str "; echo " :".trim($str); echo " "; ?>	: PHP : PHP

Example 5.10.1 *Write a PHP program to do string manipulations.*

Solution : For this program we will apply various built in string manipulating functions to the string. The PHP code is as follows -

```
<?php
    $Str1="PHP is Fun";
    $length = strlen($Str1);
    echo "<b> Length:</b>The length of string: $Str1 is = $length";
    echo "<br/><b>Position: </b>The position of word Fun in the $Str1 is ".strpos($Str1,'Fun');
    $Str1="Hello";
    $Str2="hello";
    if(strcmp($Str1,$Str2))
        echo "<br/><b>Comparison: </b> The two strings $Str1 and $Str2 are not equal";
    else
        echo "<br/><b>Comparision: </b> The two strings $Str1 and $Str2 are equal";
    $Str1="HELLO";
    echo "<br/><b>Changing Case: </b> The string $Str1 becomes ".strtolower($Str1);
    echo "<br/><b>Reversing String: </b> The string $Str1 is reversed as ".strrev($Str1);
    $Str1="Hello";
    $Str2="Friend";
    echo "<br/><b>Concatenating strings: </b> The string $Str1 and $Str2 are concatenated
".$Str1.$Str2;
    echo "<br/><b>Replacing all instances of string: </b> The string tictactoe is now ";
    echo str_replace("t","p","tictactoe");
    $Str1="PHP is fun";
    $newstring=substr_replace($Str1,"FUN",7,9);
    echo "<br><b>Replacing substring: </b> $Str1 becomes $newstring";
?>
```

Output



5.10.3 Data Type Conversion Functions

Syntax :

```
Conversion_function($variable_name)
```

Example :

```
Intval($marks)
```

Syntax :

```
settype($variable_name,"datatype")
```

Example :

```
settype($marks,"integer");
```

- The **gettype** function is useful to obtain the data type of the variable.

5.10.4 Array Handling Functions

Function	Purpose
array()	Creates an array.
array_change_key_case(array,nameof case)	Changes the elements of array from lower to upper or from upper to lower case.
array_fill()	Fills the array with values.

array_keys()	Returns all the keys of the array.
array_merge()	Merges one or more arrays into one array.
array_search()	Searches array for given value and returns the keys.
array_value()	Returns the values of array.
count()	Returns the elements in array.
current()	Returns current element in an array.
each()	Returns current key and value pair from array.
end()	Returns last element value form array.
next()	Advances the internal pointer of the array to point to the next element.
sizeof()	Returns the total number of elements.
sort()	This function is for sorting the array.

5.11 Form Handling

- PHP is used for form handling. For that purpose the simple form can be designed in HTML and the values of the fields defined on the form can be transmitted to the PHP script using GET and POST methods.
- For forms that are submitted via "GET" method, we can obtain the form via the **$_GET** array variable.
- For forms that are submitted via "POST" method, we can obtain the form via the **$_POST** array variable.

Example 5.11.1 *Create HTML form with one textbox to get user's name. Also write PHP code to show length of entered name when, the HTML form is submitted.*

Solution : Step 1 : The HTML form can be created as follows.

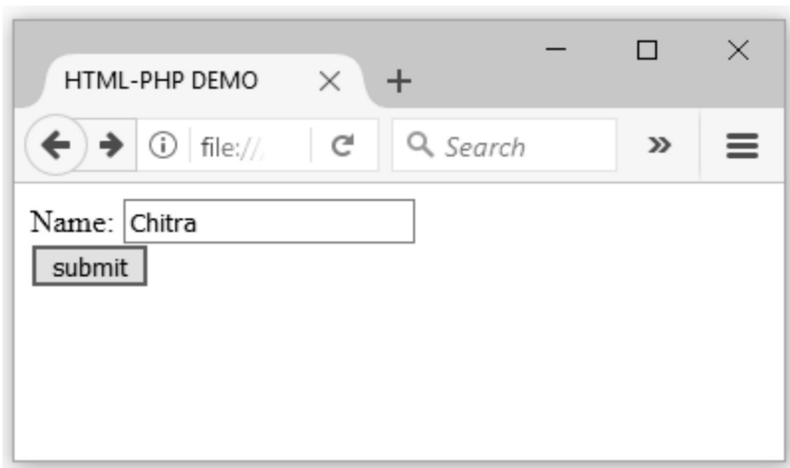
```
<!DOCTYPE html>
<html>
<head><title> HTML-PHP DEMO</title>
</head>
<body>
  <form method="post"
   action="http://localhost/getdata.php">
```

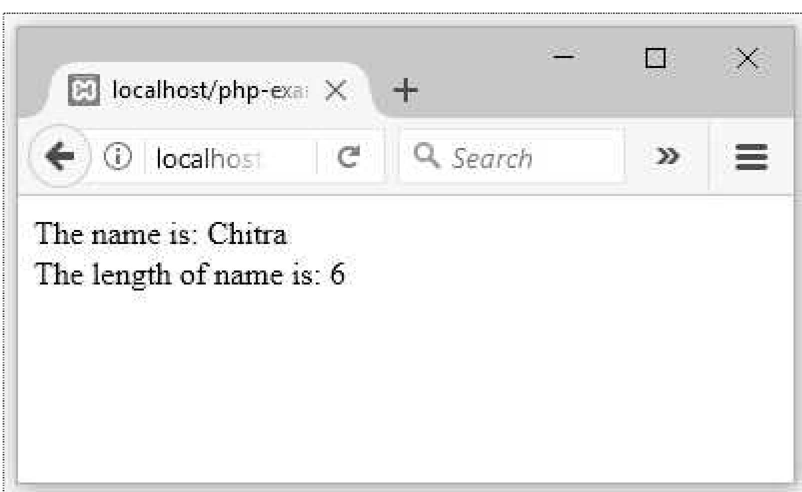
```
    Name: <input type="text" name="myname" size="20"/>
    <br/>
    <input type="submit" name="submit"
    value="submit"/>
  </form>
</body>
</html>
```

Step 2 : The PHP script to display the length of submitted name is as written below.

```
<?php
print "The name is: ";
print $_POST["myname"];
$len= strlen($_POST["myname"]);
print "<br/> The length of name is: ";
print $len;
?>
```

Output





Example 5.11.2 *Create HTML form to enter one number. Write PHP code to display the message about number is odd or even.*

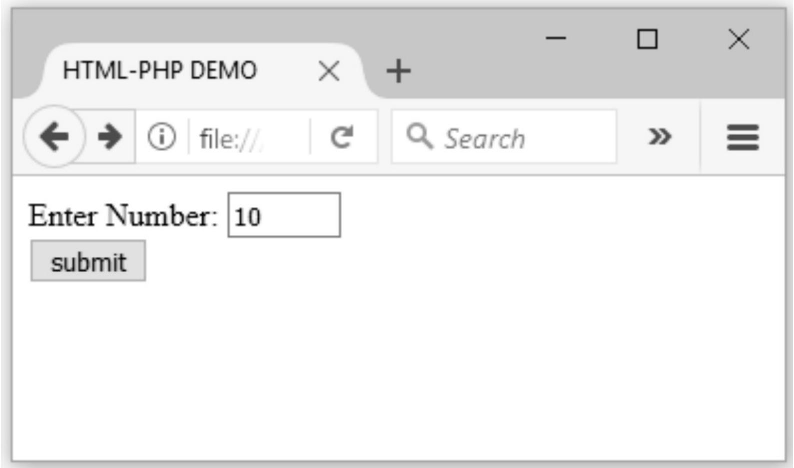
Solution : Step 1 : The HTML form for accepting number is created as below -

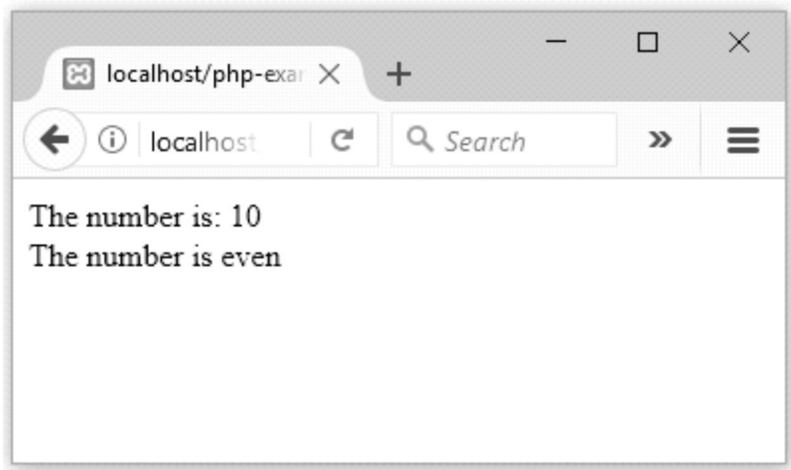
```
<!DOCTYPE html>
<html >
 <head><title> HTML-PHP DEMO</title>
</head>
<body>
   <form method="post" action="http://localhost/getdata.php">
     Enter Number: <input type="text" name="mynum"      size="5"/>
      <br/>
   <input type="submit" name="submit" value="submit"/>
   </form>

</body>
</html>
```

Step 2 : The PHP script deciding whether the number is even or odd is as given below -

```
<?php
print "The number is: ";
print $_POST["mynum"];
$a= $_POST["mynum"];
if($a%2==1)
 print "<br/> The number is odd ";
else
 print "<br/> The number is even ";
?>
```

Output





Example 5.11.3 *Create a form containing information sl.no, title of the book, publishers, quantity, price read the data from the form and display it using PHP script.*

Solution : Step 1 : Create an HTML page for inputting the data. Following is the code for HTML script .

HTML Document[input.html]

```
<!doctype html>
<html>
<head>
<title> Book Order Form </title>
</head>
<body>
<h3> Enter the Book Data </h3>
<form method="post" action="http://localhost/php-examples/formdemo.php">
<table>
<tr>
<td>Sr.No.</td>
<td><input type="text" name="SLNo"></td>
</tr>
<tr>
 <td>Book name</td>
 <td><input type="text" name="BName"></td>
</tr>
<tr>
 <td>Publisher</td>
 <td><input type="text" name="PUBName"></td>
</tr>
<tr>
```

```
 <td>Price</td>
 <td><input type="text" name="Price"></td>
</tr>
 <td>Quantity</td>
 <td><input type="text" name="Qty"></td>
</tr>
<tr>
 <td><input type="submit" value="Submit"></td>
 <td><input type="submit" value="Clear"></td>
</tr>
</table>
</form>
</body>
</html>
```

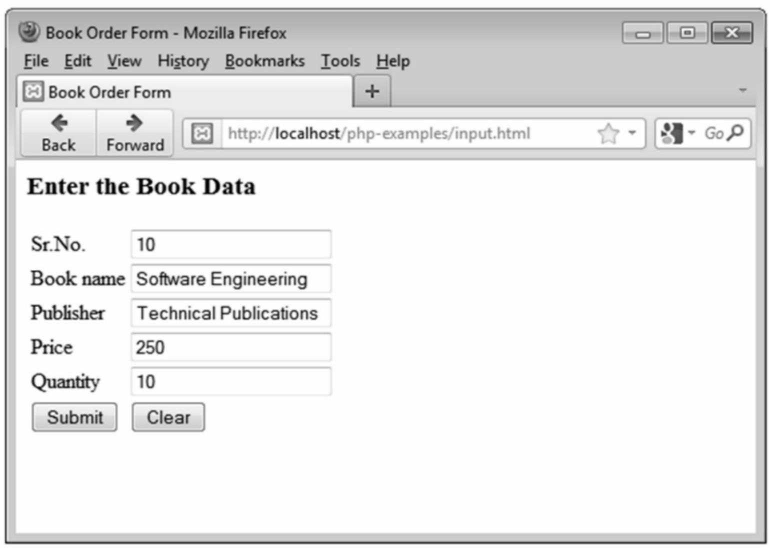
Step 2 : Create a PHP script which will read out the data entered by the user using HTML form. The code is as follows -

PHP Document[formdemo.php]

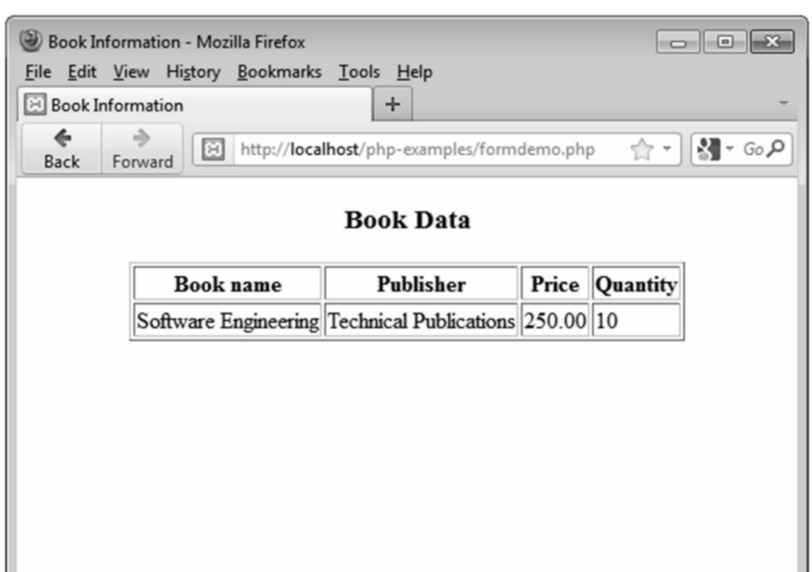
```
<html>
<head>
 <title>Book Information</title>
</head>
<body>
<?php
$BName=$_POST["BName"];
$PUBName=$_POST["PUBName"];
$Price=$_POST["Price"];
$Qty=$_POST["Qty"];
?>
<center>
<h3> Book Data </h3>
<table border=1>
<tr>
 <th>Book name</th>
 <th>Publisher</th>
 <th>Price</th>
 <th>Quantity</th>
</tr>
<tr>
 <td><?php print ("$BName"); ?></td>
 <td><?php print ("$PUBName"); ?></td>
 <td><?php printf("%3.2f",$Price); ?></td>
 <td><?php printf("%d",$Qty); ?></td>
```

```
</tr>
</table>
</center>
</body>
</html>
```

Step 3 : Open some suitable web browser and enter the address for the HTML file which you have created in step 1.



Now click on the Submit button and the PHP file will be invoked. The output will then be as follows -

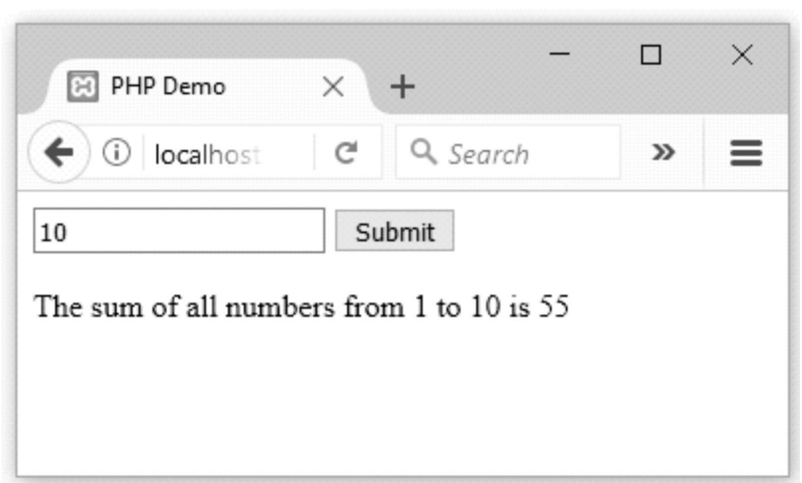


Example 5.11.4 *Write a PHP program to accept a positive integer 'N' through a HTML form and to display the sum of all the numbers from 1 to N.*

Solution :

```
<html>
    <head>
        <title> PHP Demo </title>
    </head>
    <body>
        <form method="post">
            <input type="text" name="num"/>
            <input type="submit" value="Submit"/>
        </form>
    </body>
</html>
<?php
    //getting values from HTML form
    $n = intval($_POST['num']);
    $sum=0;
    for($i=1;$i<=$n;$i++)
        $sum=$sum+$i;
    echo "The sum of all numbers from 1 to ".$n." is ".$sum;
?>
```

Output



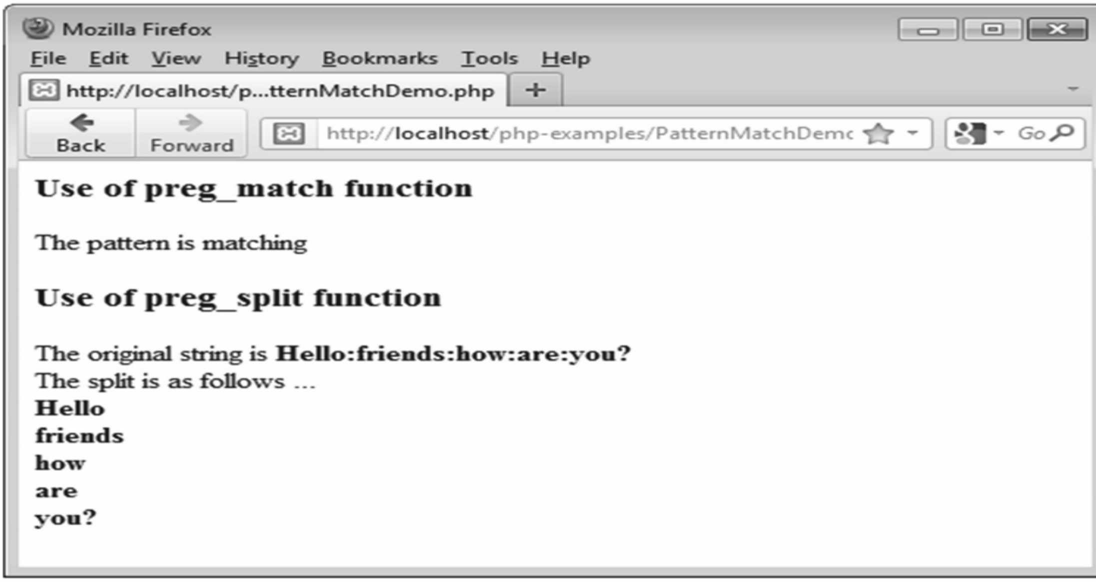
5.12 Pattern Matching

- The pattern matching technique which is used in PHP is of two types -The first one is based on **POSIX regular expression** and the another is based on **Perl regular expression.**
- There is a **preg_match** function which is used to check the matching of the pattern from the text. It takes two parameters the first parameter is the pattern to be matched and the second parameter is the text against which the pattern is to be matched.
- The **preg_split** is the function that splits the string into the chunks.
- Following is a PHP script in which both of these functions are discussed.

PHP Document[PatternMatchDemo.php]

```
<?php
print "<h3>Use of preg_match function</h3>";
if (preg_match("/gram/", "I like Programming in PHP!"))
      print "The pattern is matching <br />";
 else
      print "The pattern does not match <br />";
 print "<h3>Use of preg_split function</h3>";
 $text ="Hello:friends:how:are:you?";
 print "The original string is<b> $text</b>";
 print "<br/>";
 print "The split is as follows ...<br/>";
 $chunks=preg_split("/:/",$text);
 while($myval=each($chunks))
 {
     $val=$myval["value"];
     print("<b>$val</b>");
     print "<br/>";
 }
?>
```

Output



Example 5.12.1 *Explain the string comparison capability of PHP using regular expression with an example.*

Solution : The preg_match function is used to compare two strings using regular expression. The first parameter to this function is a regular expression and second parameter is the string which is used for matching with the regular expression.

Following PHP program illustrates, the matching of date pattern using regular expression.

Example program in PHP

```
<?php
$regex = "/[a-zA-Z]+ \d+/";
if (preg_match($regex, "May 16"))
{
   // The expression "[a-zA-Z]+ \d+" matches the date string
   echo "The date is matching with given regular expression!";
}
else
 {
   // If preg_match() returns false, then the regex does not
   // match the string
   echo "The regex pattern does not match.";
}
?>
```

Output

Example 5.12.2 *Write a PHP program that tests whether an email address is input correctly. Verify that the input begins with a series of characters followed by the @ character, another series of characters, a period '.' And a final series of characters. Test your program with both valid and invalid email address.*

Solution : Step 1 : The HTML form containing the text box for inputting the email address is created as follows -

```
<html>
<head>
    <title> Inserting Data into Database Table</title>
</head>
<body>
    <form method = "post" action = "http://localhost/php-examples/ValidateEmail.php ">
    <label>Email</label>
    <input type="text" name="email" />
    <br />
    <input type="submit" value="Submit">
    </form>
</body>
</html>
```

Step 2 : The PHP script that validates the given email address using regular expression is as follows -

ValidateEmail.php

```
<?php
$reg = '/^[_a-z0-9-]+(\.[_a-z0-9-]+)*@[a-z0-9-]+(\.[a-z0-9-]+)*(\.[a-z]{2,3})$/';
```

```
$email=$_POST["email"];
if (preg_match($reg, $email))
{
    echo "Valid Email ID";
}
else
{
    echo "Invalid Email ID.";
}
?>
```

Example 5.12.3 *Design application to send an email using PHP.*

Solution :

```
<?php
    $to = "xyz@mysite.com";
    $subject = "Hi!";
    $body = "Hi,\n\nHow are you?";
    if (mail($to, $subject, $body))
    {
        echo("<p>Email successfully sent!</p>");
    }
    else
    {
        echo("<p>Failure in Email delivery!!!</p>");
    }
?>
```

5.13 Files

SPPU : Dec.-18, Marks 6

PHP is known as a server-side scripting language. Hence file handling functions such as create, read, write, append are some file related operations that are supported by PHP.

5.13.1 Opening and Closing Files

- The first step in file handling is opening of the file.
- It takes two parameters - The first parameter of this function contains the name of the file to be opened and the second parameter specifies in which mode the file should be opened.

Modes	Description
r	Read only. Starts reading from the beginning of the file.
r+	Read/Write. Starts reading from the beginning of the file.

w	Write only. Opens and clears the contents of file; or creates a new file if it is not created
w+	Read/Write. Opens and clears the contents of file; or creates a new file if it is not created.
a	Append. Opens and writes to the end of the file or creates a new file if it is not created.
a+	Read/Append. Preserves file content by writing to the end of the file.

For example :

```
$my_file = 'file.txt';
$file_handle = fopen($my_file, 'a') or die('Cannot open file:  '.$my_file);
```

The **fopen** function returns TRUE if the required file is opened.

5.13.2 Reading from File

- The **fread** is the function which is used to read the file.
- It takes two parameters. The first parameter is the handle to the file and the second parameter is the number of bytes to be read. The **filesize** is the function which takes the filename as the parameter. For example -

```
$mystring = fread($file_handle, filesize(“file.txt”));
```

- There is another function named **file_get_contents** using which the contents of the file can be obtained.
- The **fgets()** function is used to read a single line from the file. For example, following code displays the contents of the file line by line.

```
while(!feof($file_handle))
 {
    echo fgets($file_handle). "<br />";
  }
```

5.13.3 Writing to a File

- The **fwrite** is the function which is used to write the contents to the file.
- It takes two parameters - the first parameter is the handle to the file and the second parameter is the number of bytes to be written. For example -

```
$Written_string = fread($file_handle, $my_data);
```

Example 5.13.1 *Create a form containing information sl.no, title of the book, publishers, quantity, price read the data from the form and write onto the file using PHP script.*

Solution : Step 1 : We will create an input form for the user to enter the data. The HTML code is as follows. (It is the same HTML form which we have discussed in the last section)

HTML Document[input.html]

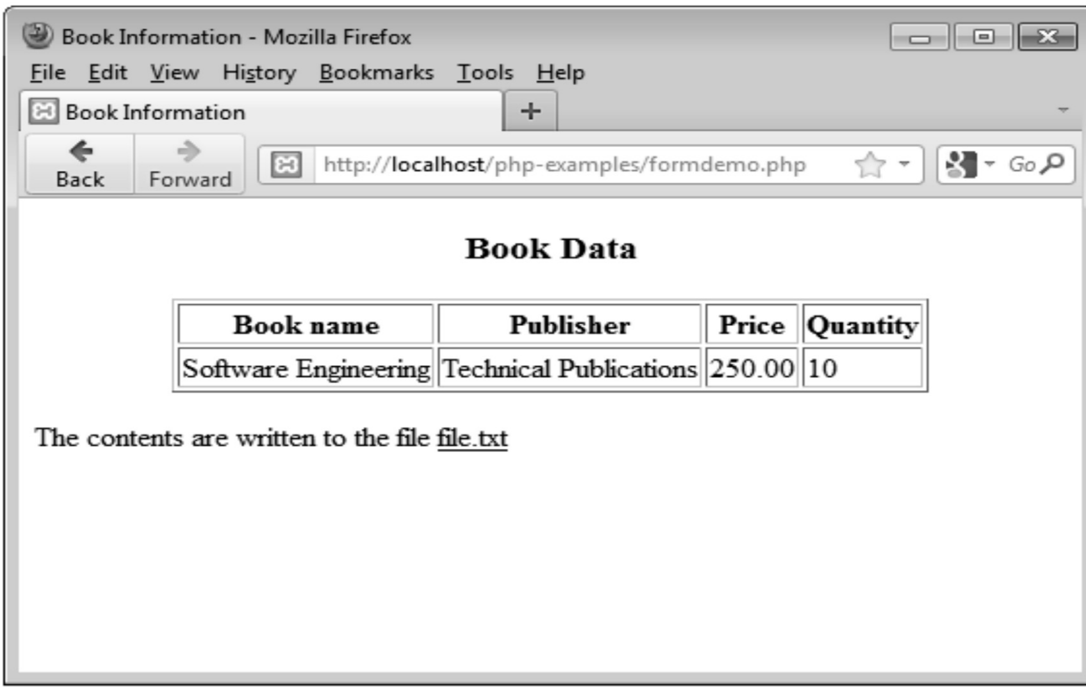
```
<!doctype html public "-//w3c//dtd html 4.0 transitional//en">
<html>
<head>
<title> Book Order Form </title>
</head>
<body>
<h3> Enter the Book Data </h3>
<form method="post" action="http://localhost/php-examples/formdemo.php">
<table>
<tr>
<td>Sr.No.</td>
<td><input type="text" name="SLNo"></td>
</tr>
<tr>
 <td>Book name</td>
 <td><input type="text" name="BName"></td>
</tr>
<tr>
 <td>Publisher</td>
 <td><input type="text" name="PUBName"></td>
</tr>
<tr>
 <td>Price</td>
 <td><input type="text" name="Price"></td>
</tr>
 <td>Quantity</td>
 <td><input type="text" name="Qty"></td>
</tr>
<tr>
 <td><input type="submit" value="Submit"></td>
 <td><input type="submit" value="Clear"></td>
</tr>
</table>
</form>
</body>
</html>
```

Step 2 : Following is a simple PHP script which displays the data and then writes it to the file.

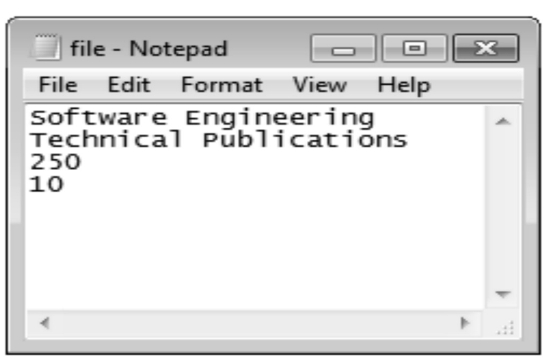
PHP Document

```
<html>
<head>
 <title>Book Information</title>
</head>
<body>
<?php
$BName=$_POST["BName"];
$PUBName=$_POST["PUBName"];
$Price=$_POST["Price"];
$Qty=$_POST["Qty"];
?>
<center>
<h3> Book Data </h3>
<table border=1>
<tr>
 <th>Book name</th>
 <th>Publisher</th>
 <th>Price</th>
 <th>Quantity</th>
</tr>
<tr>
 <td><?php print ("$BName"); ?></td>
 <td><?php print ("$PUBName"); ?></td>
 <td><?php printf("%3.2f",$Price); ?></td>
 <td><?php printf("%d",$Qty); ?></td>
</tr>
</table>
</center>
<?php
$my_file = 'file.txt';
$file_handle = fopen($my_file, 'a') or die('Cannot open file:  '.$my_file);
fwrite($file_handle, $BName);
fwrite($file_handle, "\r\n");
fwrite($file_handle, $PUBName);
fwrite($file_handle, "\r\n");
fwrite($file_handle, $Price);
fwrite($file_handle, "\r\n");
fwrite($file_handle, $Qty);
fclose($file_handle);
?>
<p>The contents are written to the file <u>file.txt<u></p>
</body>
</html>
```

Step 3 : Now open the web browser and type the address of the html file which you have created in step 1. On clicking the submit button on the this input form the data will be displayed as follows -



Step 4 : Open the current directory folder (The folder in which your html and PHP scripts are present). You will obtain a text file file.txt. Just Open the file and see the contents. I have got the following contents -



Review Question

1. *Explain server side include in PHP with sample code.*

SPPU : Dec.-18, Marks 6

5.14 Cookies

In this section we will discuss how to create and read cookies.

5.14.1 Introduction to Cookies

- Cookie is a small file that server embeds in the user's machine.
- Cookies are used to identify the users.
- A cookie consists of **a name** and **a textual value**. A cookie is created by some software system on the server.
- In every HTTP communication between browser and server a header is included. The header stores the information about the message.
- The header part of http contains the cookies.
- There can be one or more cookies in browser and server communication.

5.14.2 PHP Support for Cookies

- PHP can be used to create and retrieve the cookies.
- The cookie can be set in PHP using the function called setcookie()
- The syntax for the cookie is -

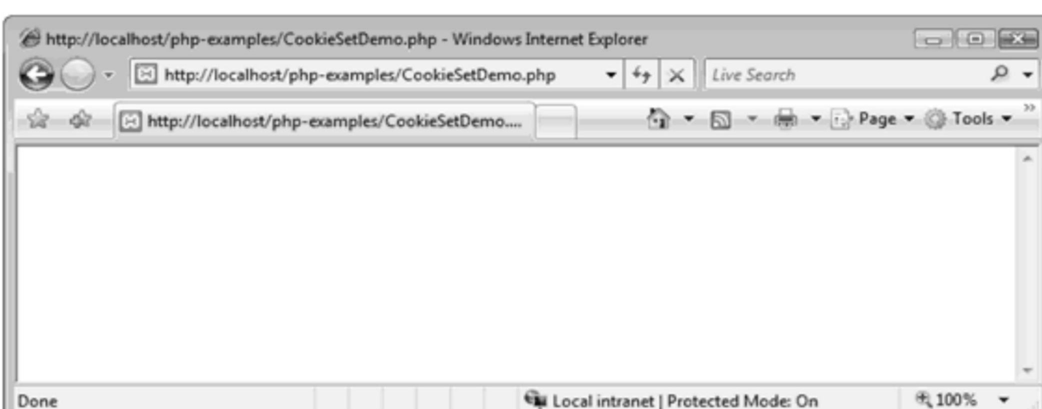
```
setcookie(name,value,expire_period,path,domain)
```

- Following is a simple PHP document which illustrates how to set the cookies -

PHP Document[CookieSetDemo.php]

```
<?php
$Cookie_period=time()+60*60*24*30;
setcookie("Myname", "Monika", $Cookie_period);
?>
```

Output



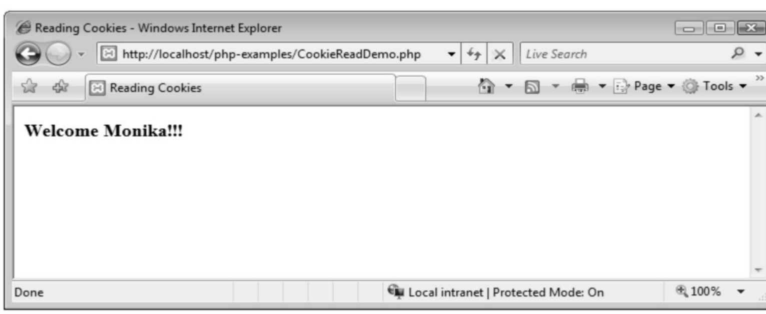
Note that you have got the blank screen it indicates that the cookie is set. In above PHP document we have set the PHP script for one month. Just observer the third parameter of the setcookie function.

Now you can retrieve the cookie and read the value to ensure whether or the cookie is set.

PHP Document[CookieReadDemo.php]

```
<html>
 <head><title>Reading Cookies</title>
<body>
 <?php
 if (isset($_COOKIE["Myname"]))
   echo "<h3>Welcome " . $_COOKIE["Myname"]."!!!</h3>";
 else
   echo "<h3>Welcome guest!</h3>";
 ?>
</body>
</html>
```

Output



Program Explanation

The isset function is used for checking whether or not the cookie is set. Then using the $_COOKIE the value of the cookie can be retrieved.

5.15 Session Tracking

SPPU : Dec.-19, Marks 8

- When you open some application, use it for some time and then close it. This entire scenario is named as **session.**
- **Session state** is a server-based state mechanism that lets web applications store and retrieve objects of any type for each **unique user session.** That is, each browser session has its own session state stored as a serialized file on the server, which is deserialized and loaded into memory as needed for each request. Refer Fig. 5.15.1.

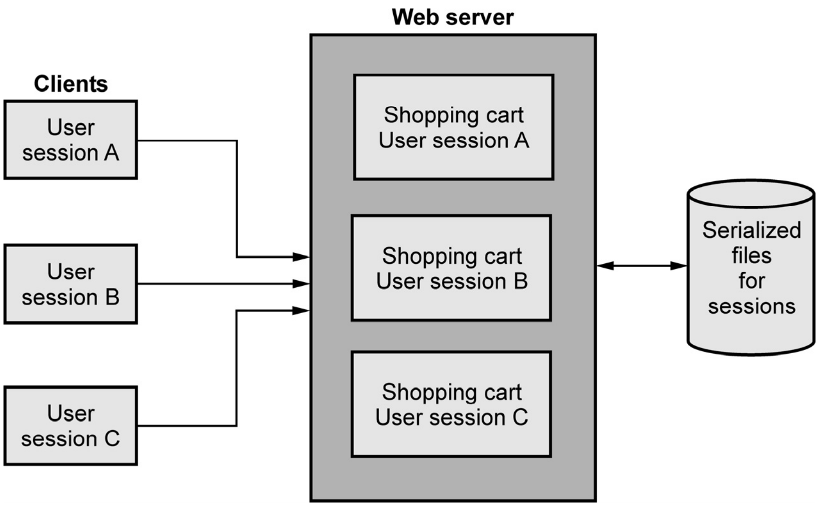


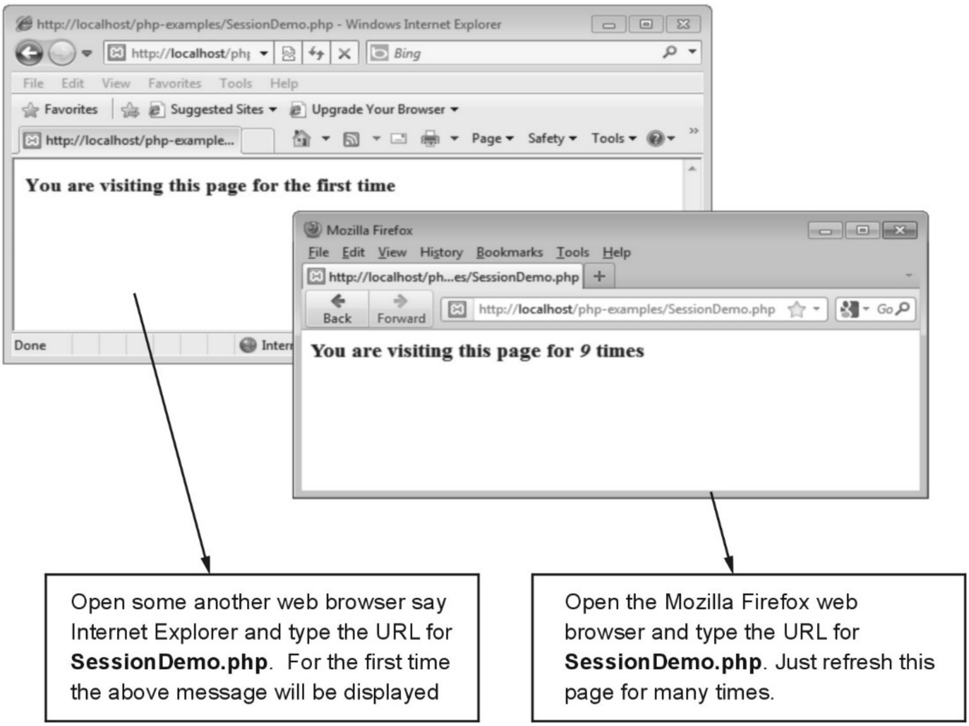
Fig. 5.15.1 Session state

- Because server storage is a finite resource, objects loaded into memory are released when the request completes, making room for other requests and their session objects. This means there can be more active sessions on disk than in memory at any one time.
- Sometimes the information about the session is required by the server. This information can be collected during the session. This process is called **session tracking.**
- In PHP, session state is available to the developer using the $_SESSION variable. The unique ID for the sessions can be stored in superglobal array $_SESSION.
- PHP keeps track of session by using a function called **session_start().** Due to the call to **session_start()** function the session ID is created and recorded.
- Following is a simple PHP script in which the information about session is tracked.

PHP Document[SessionDemo.php]

```
<?php
session_start();
if(isset($_SESSION['pgvisit']))
{
    $_SESSION['pgvisit']=$_SESSION['pgvisit']+1;
echo "<h3>You are visiting this page for <i>". $_SESSION['pgvisit']."</i> times</h3>";
}
else
{
    $_SESSION['pgvisit']=1;
```

```
echo "<h3>You are visiting this page for the first time</h3>";
}
?>
```



Program Explanation :

- In above program, we have started the session by using **session_start()** function.
- The **isset()** function checks if the **pgvisit** variable has already been set. If **pgvisit** has been set, we can increment our counter. If **pgvisit** is not set, then we create a set it to 1.
- The value of **pgvisit** is displayed on the screen.

Review Question

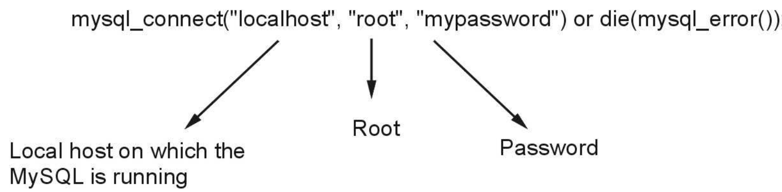
1. *Explain how session and cookies are used for session management in PHP.*
   SPPU : Dec.-19, Marks 8

5.16 Using MySQL with PHP

SPPU : May-18, Marks 6, Dec.-19, Marks 8

5.16.1 Connecting to Database

The PHP function **mysql_connect** connects to the MySQL server. There are three parameters that can be passed to this function. For example -



For example

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost:3306/mydb","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
else
{
print"connected";
}
mysql_close($conn); //closing the database
?>
```

The database can be closed using the function **mysql_close**

5.16.2 Creating Database

We can create a database using the function **mysql_query**. The **mysql_error()** function is used to obtain the error messages if any command gets failed.

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost ","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
//Create a database
if(mysql_query("CREATE DATABASE mydb",$conn))
    {
    print "Database created";
    }
    else
    {
     print "Error creating database: " . mysql_error();
    }
```

```
mysql_close($conn); //closing the database
?>
```

5.16.3 Selecting Database

The database can be selected using the function **mysql_select_db().**

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost:3306/mydb ","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
//Select a database
mysql_select_db("mydb", $conn);
mysql_close($conn); //closing the database
?>
```

5.16.4 Listing Database

There are various databases present in the MySQL which can be displayed using the function **mysql_list_db().**

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost:3306/mydb ","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}

//Show databases
$db_lists=mysql_list_db($conn)
while($i=$mysql_fetch_object($db_list))
{
echo $i->Database. "\n"; //Present Databases will be displayed
}
mysql_close($conn); //closing the database
?>
```

5.16.5 Listing Table Names

The tables are present inside the databases. The function **mysql_list_tables** function is used to display the tables present in the database or we can use **mysql_query**

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost ","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
$table_list=mysql_query("SHOW TABLES FROM mydb",$con)
if(!$table_list)
{
 print "Error!!! " . mysql_error();
}
while($i=$mysql_fetch_row($table_list))
{
echo $i[0]. "\n"; //Listing the tables
}
mysql_close($conn); //closing the database
?>
```

5.16.6 Creating a Table

Before creating the table a database must be created and within which the table can be created. Note that before creating a table the desired database must be selected.

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost ","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}

//Create a database
if(mysql_query("CREATE DATABASE mydb",$con))
    {
    print "Database created";
    }
    else
    {
```

```
     print "Error creating database: " . mysql_error();
     }
mysql_select_db("mydb",$conn); //before creating table select the database
$query="CREATE TABLE my_table
        (
            id INT(4),
            name VARCHAR(20)
         )";
mysql_query($query,$conn);//Execution of Query

mysql_close($conn); //closing the database
?>
```

5.16.7 Inserting Data

Using PHP we can insert the data in the database created in the MySQL.

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("mydb",$conn); //select the database
$query=" INSERT INTO my_table (id,name)  VALUES(1,'SHILPA')";
mysql_query($query,$conn);//Execution of Query
$query=" INSERT INTO my_table (id,name)  VALUES(2,'MONIKA')";
mysql_query($query,$conn);//Execution of Query

mysql_close($conn); //closing the database
?>
```

Sometimes values that can be inserted in the table can be obtained from some another script and these values might be present in the variables. Insertion of such data can be done using $_POST variables. It is as shown below -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("mydb",$conn); //select the database
```

```
$query=" INSERT INTO my_table (id,name)
    VALUES('$_POST[MyID]','$_POST[MyName]')";
mysql_query($query,$conn);//Execution of Query

mysql_close($conn); //closing the database
?>
```

We can display these records using the SELECT query. For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost ","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("mydb",$conn); //select the database
//Execution of Query for displaying the data
$result=mysql_query("SELECT * FROM my_table");
while($row = mysql_fetch_array($result))
       {
            echo $row['id'] . " " . $row['name'];          // Each record will be displayed line by line

            echo "<br />";
        }
mysql_close($conn); //closing the database
?>
```

5.16.8 Altering Table

The ALTER table command is useful for various reasons such as to add some columns in the existing table, to delete some column from the table, for changing the size of some field or to change the name of some column. In the following example we are adding one column named **addr** to the existing table **my_table**

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("mydb",$conn); //select the database
$query=" ALTER TABLE my_table ADD COLUMN addrVARCHAR(30)";
mysql_query($query,$conn);//Execution of Query
mysql_close($conn); //closing the database
?>
```

5.16.9 Deleting Database

We can delete the values from the database using the DELETE query.

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("mydb",$conn); //select the database
$query=" DELETE FROM my_table WHERE id=1";
mysql_query($query,$conn);//Execution of Query
mysql_close($conn); //closing the database
?>
```

Similarly we can delete a database using the query DROP.

For example -

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","mypassword");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("mydb",$conn); //select the database
$query=" DROP DATABASE mydb";
mysql_query($query,$conn);//Execution of Query
mysql_close($conn); //closing the database
?>
```

Examples

Example 5.16.1 *Write a PHP script to create a new database table with 4 fields of your choice and perform following database operations.*

1. Insert 2. Delete 3. Update 4. Display

Solution : We will create a table in the database test. The name of the table ismytable. Then we will insert the record into the table using the INSERT command, update particular field of the record using the command UPDATE and delete the record using the command DELETE.

The PHP script is as follows -

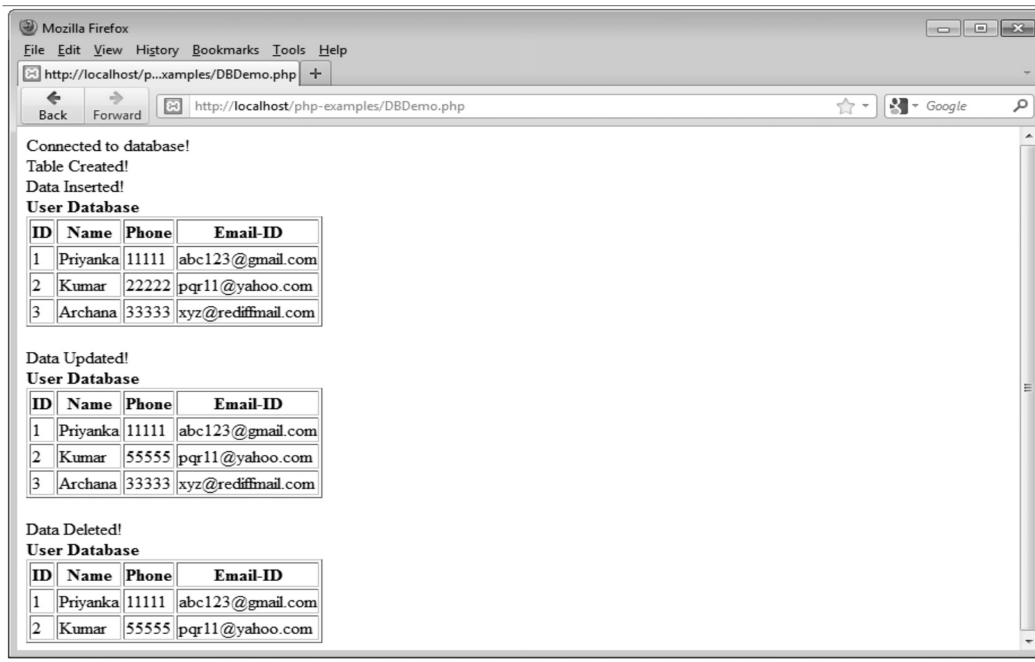
PHPDocument[DBDemo.php]

```
<?php
// Make a MySQL Connection
mysql_connect("localhost", "root", "mypassword") or die(mysql_error());
mysql_select_db("test") or die(mysql_error());
echo "Connected to database!";
mysql_query("CREATE TABLE mytable(id INT NOT NULL AUTO_INCREMENT, PRIMARY
KEY(id), name VARCHAR(30),
phoneINT,emailIdVARCHAR(30))")
or die(mysql_error());
print "<br/>";
echo "Table Created!";

// Insert a row of information into the table
mysql_query("INSERT INTO mytable
(name, phone,emailId) VALUES('Priyanka', '11111','abc123@gmail.com' ) ")
or die(mysql_error());

mysql_query("INSERT INTO mytable
(name, phone,emailId) VALUES('Kumar', '22222','pqr11@yahoo.com' ) ")
or die(mysql_error());

mysql_query("INSERT INTO mytable
(name, phone,emailId) VALUES('Archana', '33333','xyz@rediffmail.com' ) ")
or die(mysql_error());
print "<br/>";
echo "Data Inserted!";
$result =mysql_query("SELECT * FROM mytable")
or die(mysql_error());
print "<br/>";
print "<b>User Database</b>";
echo "<table border='1'>";
echo "<tr><th>ID</th><th>Name</th><th>Phone</th><th>Email-ID</th>
</tr>";
while($row = mysql_fetch_array( $result ))
    {
    // Print out the contents of each row into a table
    echo "<tr><td>";
    echo $row['id'];
    echo "</td><td>";
    echo $row['name'];
    echo "</td><td>";
```

```
    echo $row['phone'];
    echo "</td><td>";
    echo $row['emailId'];
    echo "</td></tr>";
}
echo "</table>";
$result = mysql_query("UPDATE mytable SET phone='55555' WHERE phone='22222'")
or die(mysql_error());

print "<br/>";
echo "Data Updated!";
$result =mysql_query("SELECT * FROM mytable")
or die(mysql_error());
print "<br/>";
print "<b>User Database</b>";
echo "<table border='1'>";
echo "<tr><th>ID</th><th>Name</th><th>Phone</th><th>Email-ID</th>
</tr>";
while($row = mysql_fetch_array( $result ))
{
// Print out the contents of each row into a table
echo "<tr><td>";
echo $row['id'];
echo "</td><td>";
echo $row['name'];
echo "</td><td>";
echo $row['phone'];
echo "</td><td>";
echo $row['emailId'];
echo "</td></tr>";
}
echo "</table>";
$result = mysql_query("DELETE FROM mytable WHERE phone='33333'")
or die(mysql_error());
print "<br/>";
echo "Data Deleted!";
$result =mysql_query("SELECT * FROM mytable")
or die(mysql_error());
print "<br/>";
print "<b>User Database</b>";
echo "<table border='1'>";
echo "<tr><th>ID</th><th>Name</th><th>Phone</th><th>Email-ID</th>
</tr>";
while($row = mysql_fetch_array( $result ))
```

```
    {
    // Print out the contents of each row into a table
    echo "<tr><td>";
    echo $row['id'];
    echo "</td><td>";
    echo $row['name'];
    echo "</td><td>";
    echo $row['phone'];
    echo "</td><td>";
    echo $row['emailId'];
    echo "</td></tr>";
}
echo "</table>";
?>
```

Output



Example 5.16.2 *Create a HTML form "result.html" with a text box and a submit button to accept registration number of the student. Write a "result.php" code to check the status of the result from the table to display whether the student has "PASS" or "FAIL" status. Assume that the MYSQL database "my_db" has the table "result_table" with two columns REG_NO and STATUS.*

Solution : Step 1 : Create a database named my_db. Create a table result_table for this database and insert the values in this table. The table is created as follows -

Sr. No.	REG_NO	STATUS
1	101	PASS
2	102	FAIL
3	103	PASS
4	104	FAIL
5	105	PASS

Table 5.16.1

Step 2 : Create an HTML form to accept the registration number; the HTML document is as follows -

```
result.html
<!DOCTYPE html>
<html>
    <head>
        <title> STUDENT RESULT </title>
    </head>
    <body>
        <form name="myform" method="post" action="http://localhost/php-
examples/result.php">
            <input type="text" name="reg_no"/>
            <input type="submit" value="Submit"/>
        </form>
    </body>
</html>
```

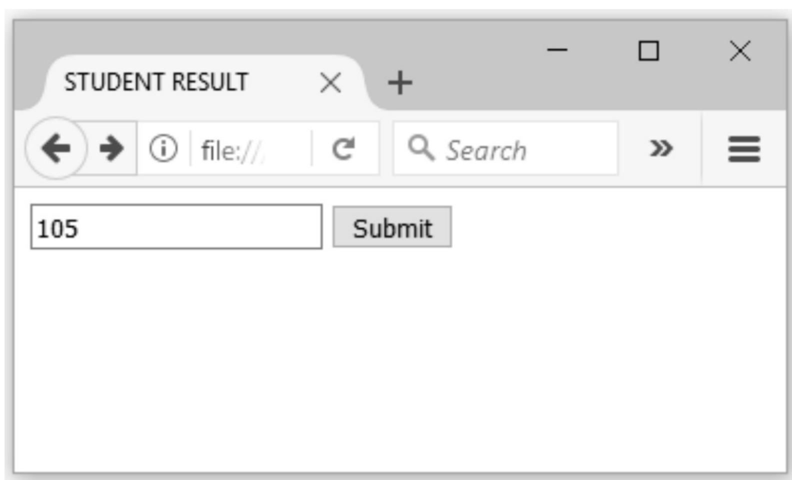
Step 3 : Create a PHP script to accept the registration number. This php script will connect to MYSQL database and the status (PASS or FAIL) of the corresponding registration number will be displayed.

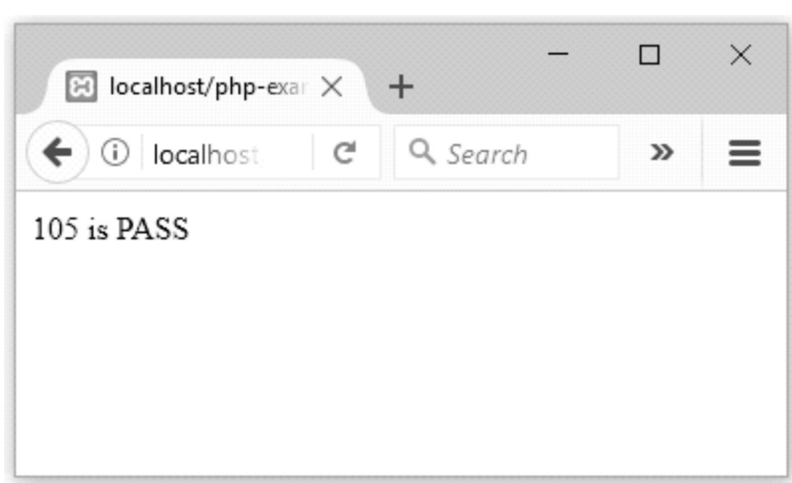
result.php

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("my_db",$conn); //select the database
//Execution of Query for displaying the data
```

```
$reg_no = intval($_POST['reg_no']);

$result=mysql_query("SELECT REG_NO,STATUS FROM result_table
where REG_NO=$reg_no ");
 while($row = mysql_fetch_array($result))
  {
   echo $row['REG_NO'] . " is " . $row['STATUS'];
   echo "<br />";
  }
mysql_close($conn); //closing the database
?>
```

Step 4 : Load the HTML form created in Step 2 and click the submit button by entering some registration number.





Example 5.16.3 *Write a PHP program to delete a record from result_table. (Refer example 5.16.2)*

Solution :

```
<?php
// Make a MySQL Connection
$conn=mysql_connect("localhost","root","");
if(!$conn)
{
die('error in connection'.mysql_error());
}
mysql_select_db("my_db",$conn); //select the database
//Execution of Query for displaying the data
$reg_no = intval($_POST['reg_no']);

$result=mysql_query("DELETE FROM result_table where REG_NO=$reg_no");
 while($row = mysql_fetch_array($result))
  {
   echo $row['REG_NO'] . " is " . $row['STATUS'];          //Each record will be displayed line by line

   echo "<br />";
  }

mysql_close($conn); //closing the database
?>
```

Example 5.16.4 *Write a user defined function 'CalculateInterest' using PHP to find the simple interest to be paid for a loan amount. Read the loan amount, the number of years and rate of interest from a database table called LOANDETAILS having three fields AMT, YEARS, and RATE, and calculate the interest using the user defined function.*

Solution : Step 1 : Create a database table named LOANDETAILS having three fields AMT, YEARS and RATE. Insert the values in it. The sample table will be as follows -

AMT	YEARS	RATE
10000	5	6.9

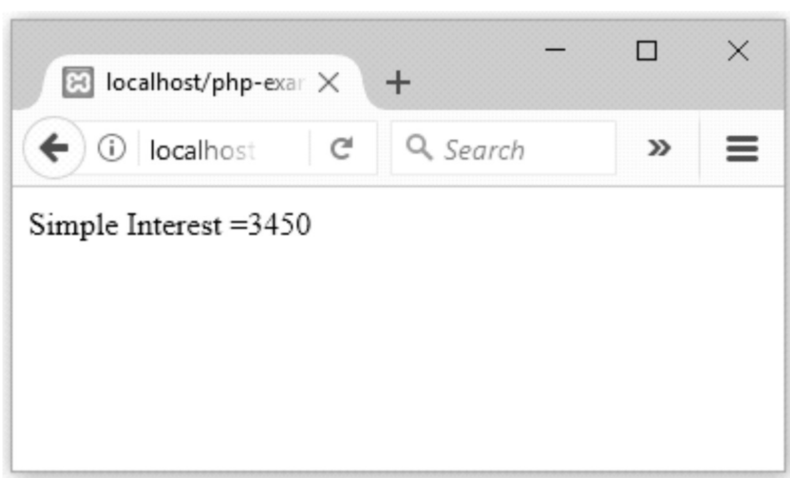
Table 5.16.2

Step 2 : The PHP code for calculating the simple interest the above values in a function will be as follows -

Interest.php

```
<?php
function CalculateInterest()
{
     // Make a MySQL Connection
     $conn=mysql_connect("localhost","root","");
     if(!$conn)
     {
         die('error in connection'.mysql_error());
     }
     mysql_select_db("my_db",$conn); //select the database
     //Execution of Query for displaying the data
     $result=mysql_query("SELECT * FROM LOANDETAILS");
     $row = mysql_fetch_array($result);
     $amount= $row['AMT'];
     $rate= $row['RATE'];
     $years= $row['YEARS'];
      $interest=($amount * $rate *$years)/100;
     mysql_close($conn); //closing the database
     return $interest;
}
print "Simple Interest =".CalculateInterest();
?>
```

Output



Review Question

1. *Identify and explain steps involved in connecting to MySQL with PHP.*
SPPU : May-18, Marks 6, Dec.-19, Marks 8

5.17 WAP

- Wireless Application Protocol (WAP) is a set of standards designed to extend Internet services to mobile phones and Personal Digital Assistants (PDA).
- A WAP browser is a web browser for mobile devices such as mobile phones that uses the protocol.
- The mobile network gets connected to the Internet through **WAP Gateway**. The function of WAP gateway is to convert the **WAP request** to a **WEB request** or a WEB request to a WAP request.
- The mobile phones act as a client when gets connected to the internet. The demanded web pages can be displayed on the mobile phones if it is WAP enabled.
- The Web pages are normally written in HTML language but the web pages written in HTML gets loaded very slowly on the mobile phones. Hence the web pages for mobile phones are written in the language called **Wireless Markup Language (WML)**
- The WAP model follows the OSI model and is represented by Fig. 5.17.1.

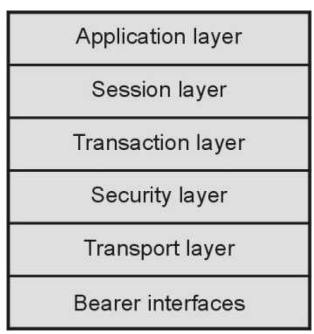


Fig. 5.17.1 WAP model

Application Layer

- o The WAP's application layer is the Wireless Application Environment (WAE).
- o The WAE supports WAP application development using the Wireless Markup Language (WML).
- o The WAE also includes the Wireless Telephony Application Interface (WTA) which provides the programming interface to telephones. This allows WAP technology to support for sending messages, initiating calls and other networking capability.

Session Layer

- The session layer makes use of **Wireless Session Protocol** (WSP).
- This protocol is similar to HTTP. But HTTP is not efficient for displaying the web pages on mobile devices, WAP makes uses of WSP.
- The WSP conserves the bandwidth on the wireless links.
- The HTTP works with textual data whereas the WSP work with binary data.
- It supports for both **connection-oriented** and **connectionless** sessions.

Transaction, Security and Transport Layer

There are three different protocols used in these layers.

- **Wireless Transaction Protocol (WTP)** provides transaction level services. These services are used for both reliable and unreliable transports. If packets gets dropped then WTP performs the retransmission of packets. Similarly it avoids transmission of duplicate copies of packets.
- **Wireless Transaction Layer Security (WTLS)** is just similar to Secured Socket Layer (SSL). It performs the authentication and encryption functionality.
- **Wireless Datagram Protocol (WDP)** is similar to User Datagram Protocol (UDP). WDP offers to the upper layers an invisible interface independent of the underlying network technology used.
- At the lowest layer of the stack, there are **bearer** services. WAP supports dial-up networking using IP and Point to Point Protocol (PPP) as a bearer service. It also supports for Short Message Service (SMS) and General Packet Radio Service (GPRS).

5.18 WML

- The topmost layer in the WAP architecture is made up of WAE (Wireless Application Environment), which consists of WML and WML scripting language.
- WML stands for Wireless Markup Language.
- The role of WML is the same as that of HTML in web applications. WAP sites are written in WML, while web sites are written in HTML.
- WML is basically the web language for making sites on mobile phones.
- WML is an application of XML, which is defined in a document-type definition.

How to write WML ?

- The WML is a scripting language just similar to HTML. The web script can be written in Notepad and executed on WAP simulator.
- The extension to the WML file is .wml.
- I have used WinWap for executing the WML scripts. The WinWAP is a web browser for WAP made by Winwap Technologies available for Microsoft Windows. Get it downloaded from internet using the site www.winwap.com, if you wish to use WinWap browser for executing the WML scripts.

5.18.1 Cards and Decks

- In HTML, we can navigate from one page to another. But in case of WML you can not move from page to page. The WML contains the very basic unit called **card.**
- WML file contains multiple cards. The multiple cards in a WML file forms a **deck**. User can move from one card to another using the same deck.
- We can put text, images, links, form components and so on in the card.

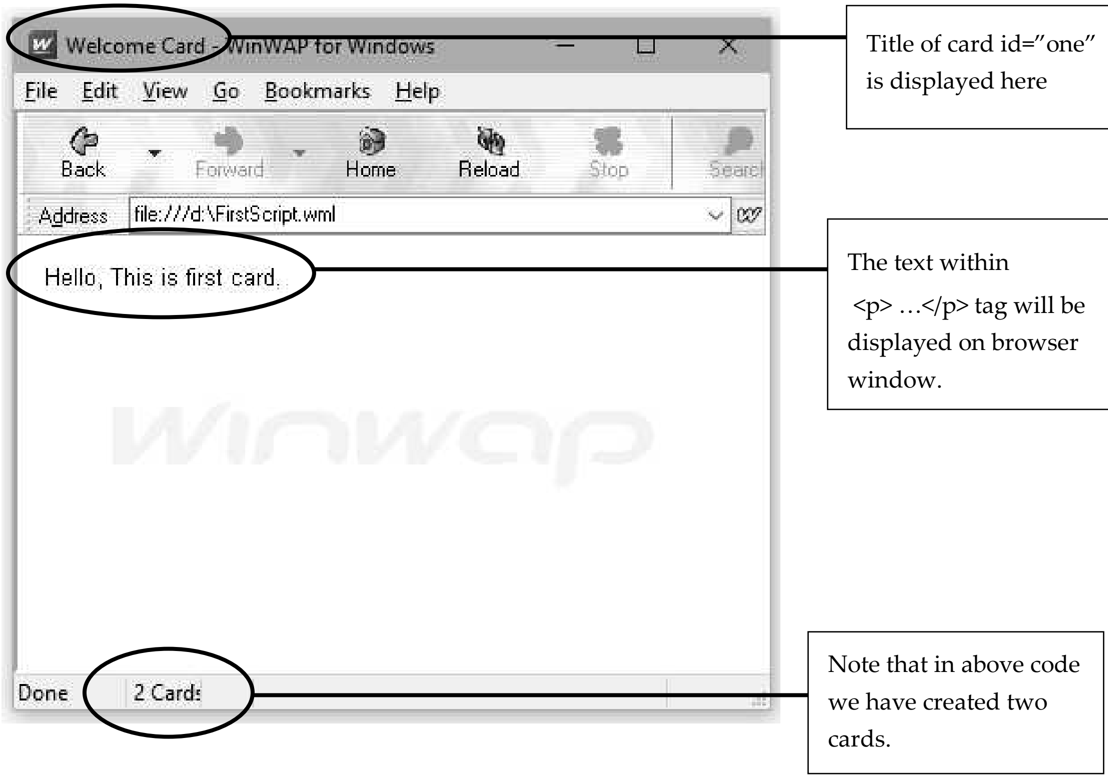
5.18.2 Sample Program

Let us write the first simple WML script. Open the Notepad and type the following code.

WML Document[FirstScript.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card id="one" title="Welcome Card">
<p>
Hello, This is first card.
</p>
</card>
<card id="two" title="GoodBye Card">
<p>
Good bye, this is second card.
</p>
</card>
</wml>
```

Now open the WinWap browser and execute the above code. It will be as follows



Script Explanation :

1) The first line of the script says that it is XML document.
2) The second line indicates that document type definition (DTD). These are the standard lines written in every WML script.
3) Then within <wml> …</wml> tag the entire code is written.
4) The two cards are created with their own ids. The title tag is used to display the content on the title bard. Within the <p>…</p> i.e. paragraph tag the text which is to be displayed on the browser window is displayed.

```
<card id="one" title="Welcome Card">

<p>
Hello, This is first card.
</p>
</card>
```

For displaying on title bar for first card

For displaying on browser window for first card

```
<card id="two" title="GoodBye Card">

<p>
Good bye, this is second card.
</p>
</card>
```

For displaying on title bar for second card

For displaying on browser window for second card

Instead of having pages, WAP sites have cards. These are what is displayed on the screen at one time, just like a page. More than one card can be inserted in each WML document. To declare a card, insert the following :

```
<card id="some id" title="some text">
```

Note that : Only one card will be displayed at a time on the web browser.

Closing Tag

Unlike HTML it is important to close every opened tag. Otherwise WML page will not get executed. The card and wml tags are closed as follows

```
</card>
</wml>
```

5.18.3 Comments

Comments are placed inside <!-- --> in WML. This is the same as HTML. For example, the following lines are comments. WAP browsers ignore all comments.

```
<!-- This is a comment in WML -->
<!-- This is a multi-line
   comment -->
```

5.18.4 Paragraph

The <p> tag defines the paragraph of text. The paragraph in WML is always displayed on a new line. The attributes of paragraph are as follows -

Attribute	Value	Purpose
align	Left right center	The text is displayed using left, right or center alignment.
mode	Wrap nowrap	The paragraph lines can be wrapped or not wrapped.
Id	Some user defined ID	The unique ID of the element is specified.

Example

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card id="one" title="Welcome Card">
<p align="left">
Hello, This is first card.
</p>
</card>
<card id="two" title="description Card">
<p align="center">
And, this is second card.
</p>
</card>
<card id="three" title="GoodBye Card">
<p align="right">
Good bye, this is third card.
</p>
</card>
</wml>
```

5.18.5 Line Break

The line break in WML is specified using <br/> tag. Note that it is important to close the tag.

Example

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card id="one" title="Welcome Card">
<p>
Hello, This is<br/> first card.
This is <br/> another line with line break
</p>
</card>
</wml>
```

Output

5.18.6 Table

The table tags are just similar to HTML tags. For example - using <table> … </table> the table can be created. The <tr> </tr>tag is used for creating rows and <td> </td> is used for creating columns. The <table> element is placed within <p> </p> tag.

Various attributes used along with the <table> tag are

Attribute	Value	Purpose
align	L C or R	The text is displayed using left (L), right (R) or center (C) alignment. If we specify LCR then that means, first column is left aligned, Second column is center aligned and third column is right aligned.
Id	Some user defined ID	The unique ID of the element is specified.

WML Document[TableDemo.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying Tables">
<p>
<table columns="3" align="LCR">
```

```
    <tr>
      <td>One</td>
      <td>Two</td>
      <td>Three</td>
    </tr>
    <tr>
      <td>Four</td>
      <td>Five</td>
      <td>Six</td>
    </tr>
</table>
</p>
</card>
</wml>
```

5.18.7 Image

- The <img> element is used to include an image in a WAP card. WAP-enabled wireless devices only supported the Wireless Bitmap (WBMP) image format.
- WBMP images can only contain two colors : black and white.
- The file extension of WBMP is ".wbmp"
- There are free tools available on internet to convert the jpg or GIF images to .wbmp format. For example Nokia Mobile Internet Toolkit, ImageMagic and so on. Hence to display image on the browser using wml, we must convert our image file to .wbmp format first.
- **Example**

WML Document[ImageDemo.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying Image">
<p>
Good Morning!!!
<img src="flower.wbmp" alt="flower image"/>
</p>
</card>
</wml>
```

Various commonly used attributes of <img> tag are -

Attribute	Value	Purpose
align	Top, middle or bottom	Image is aligned at the top, middle or bottom.
src	Image file path	
alt	Alternative text	The alternative text will be displayed if the image is not getting loaded on the page.
width	% or pixels	The number of pixels or % is specified to mention the width of the image.
height	% or pixels	The number of pixels or % is specified to mention the height of the image.
Id	Some user defined ID	The unique ID of the element is specified.

5.18.8 Hyperlinks

The hyperlink can be used in WML using the anchor tag.

In WML we can use <anchor>, <a> tag to create anchor link.

Various attributes of **<a> tag** are as given below.

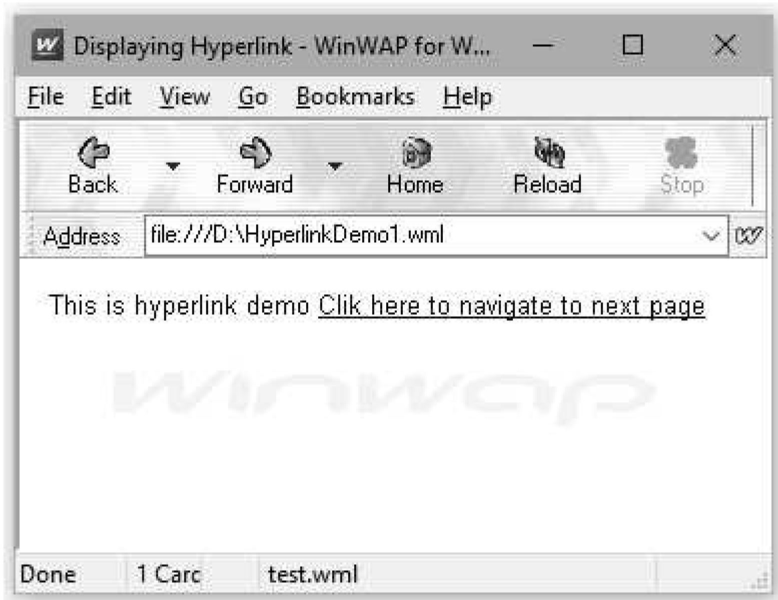
Attribute	Value	Purpose
href	URL or address of the file	The specified URL is linked with page and can be opened up on clicking the link.
title	String	The text for identifying the link is specified
Id	Some user defined ID	The unique ID of the element is specified.

Example

WML Document[HyperlinkDemo1.wm]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying Hyperlink">
  <p> This is hyperlink demo
    <a href="test.wml">Clik here to navigate to next page</a>
  </p>
</card>
</wml>
```

Output



- Similarly we can use **<anchor> tag** for using hyperlink in WML document. This tag is basically used along with <go/>, <refresh> or <prev/> tag.
- **Example**

WML Document[HyperlinkDemo2.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying Hyperlink">
<p> This is hyperlink demo
    <anchor>
        <go href="test.wml"/>
    </anchor>
</p>
<p>
<anchor>
<prev/>
</anchor>
</card>
</wml>
```

Using <prev/> tag we can go to Back page

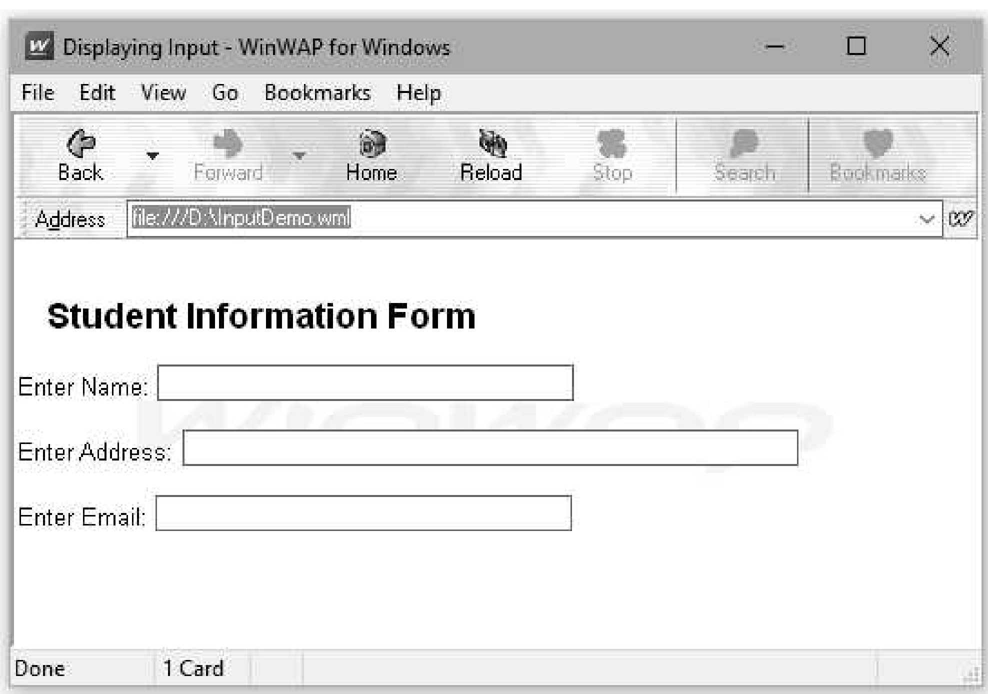
5.18.9 Inputs

The **<input/>** tag allows the user to enter alphanumeric characters such as name, age, address, phone number, email address and so on.

For example -

WML Document[InputDemo.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying Input">
<p>
    <h2>Student Information Form</h2>
    Enter Name: <input name="name" size="20"/><br/><br/>
    Enter Address: <input name="address" size="30"/><br/><br/>
    Enter Email: <input name="email" size="20"/><br/>
</p>
</card>
</wml>
```



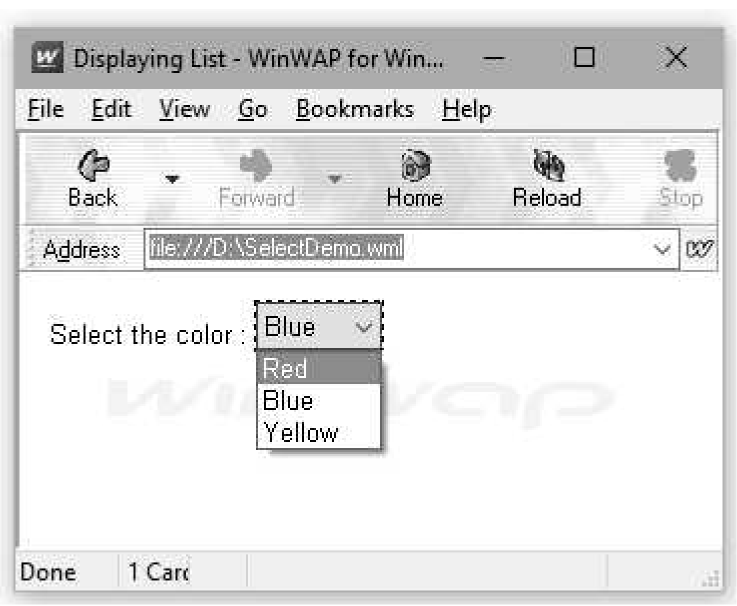
5.18.10 Select

The <select>...</select> WML elements are used to define a selection list and the <option>...</option> tags are used to define an item in a selection list.

WML Document[Select.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying List">
<p> Select the color :
 <select>
  <option value="red">Red</option>
  <option value="blue">Blue</option>
  <option value="yellow">Yellow</option>
 </select>
</p>
</card>
</wml>
```

Output



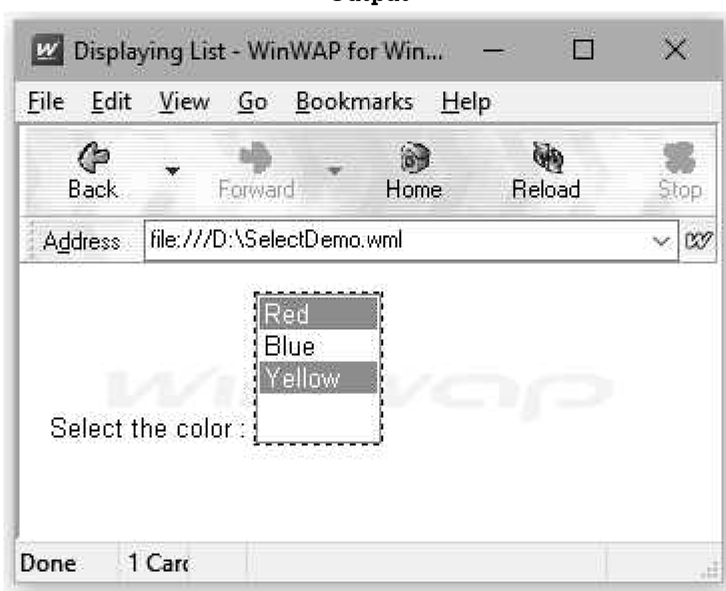
You can select multiple value if the attribute **"multiple"** is set to true. For example

WML Document[SelectDemo.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
```

```
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card title="Displaying List">
<p> Select the color :
 <select multiple="true">
  <option value="red">Red</option>
  <option value="blue">Blue</option>
  <option value="yellow">Yellow</option>
 </select>
</p>
</card>
</wml>
```

Output



5.18.11 Event

- The event is supported in WML using <onevent></onevent> tag. We can specify the action to be taken whenever an event occurs.
- There are four types of events that are supported in WML -

  (1) ontimer : This event occurs when a timer expires.

  (2) onenterbackward : This event occurs when a user goes back to a card through the WAP browser's URL history.

  (3) onenterforward : This event occurs when a user goes to a card in the forward direction. For example, if a user goes to a card by entering its URL directly in the WAP browser, the *onenterforward* event will be triggered.

(4) **onpick :** This event occurs when an item of a selection list is selected or deselected.

- The syntax of specifying the event tag is as follows -

```
<onevent type="event_type">
   Task to be performed when event occurs
</onevent>
```

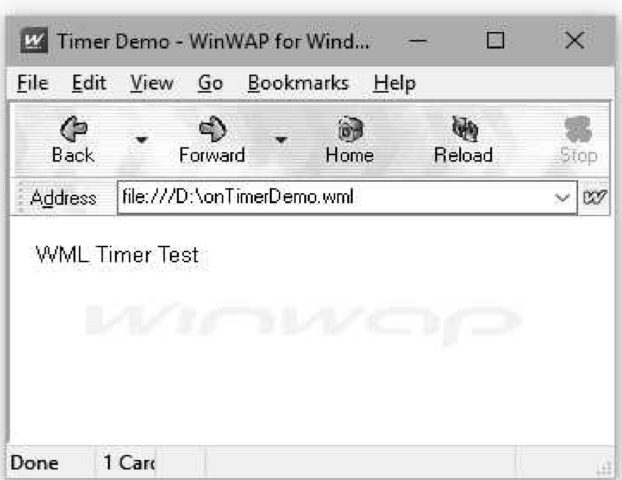
Example using ontimer Event :

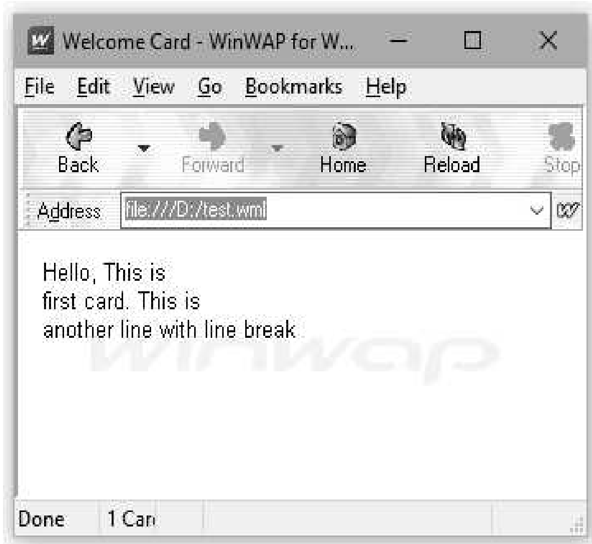
Using ontimer event, we can load the specific web page after some desired amount of time automatically. User need not have to click any hyper link in this case. Following script loads the **test.wml** web page after 3 second. The timer value is specified using **value** attribute and the time unit is 1/10 seconds.

WML Document[onTimerDemo.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.3//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
  <card id="card1" title="Timer Demo">
    <onevent type="ontimer">
      <go href="test.wml"/>
    </onevent>
    <timer value="30"/>
    <p>
      WML Timer Test
    </p>
  </card>
</wml>
```

Output





After 3 seconds

Example using onenterbackward Event :

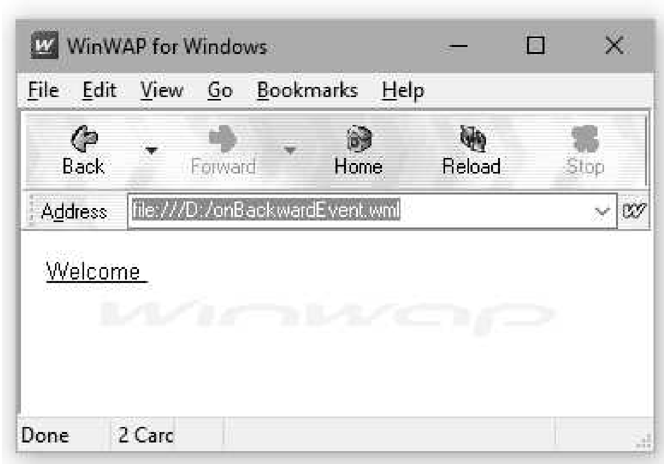
The onenterbackward event is triggered if a user goes back to a previous card through the WAP browser's URL history, and the WML code placed in the event handler will be executed.

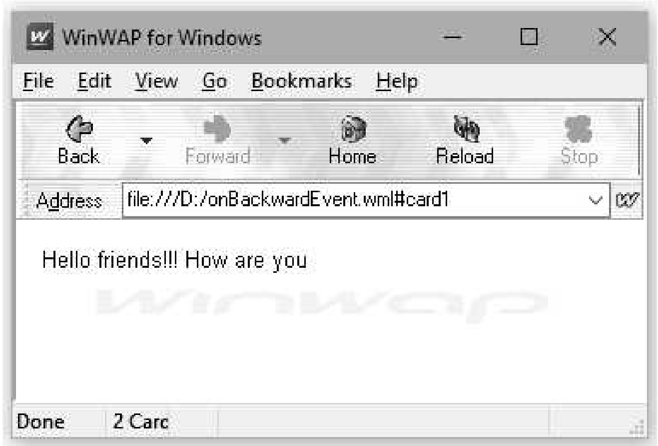
WML Document[onenterBackward.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.3//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<template>
 <onevent type="onenterbackward">
  <go href="#card1"/>
 </onevent>
</template>
<card id="card2">
<p>
<anchor>
<prev/>
 Welcome
</anchor>
</p>
</card>
<card id="card1">
<p>
 Hello friends!!! How are you
```

```
</p>
</card>
</wml>
```

Output





Example using onenterforward

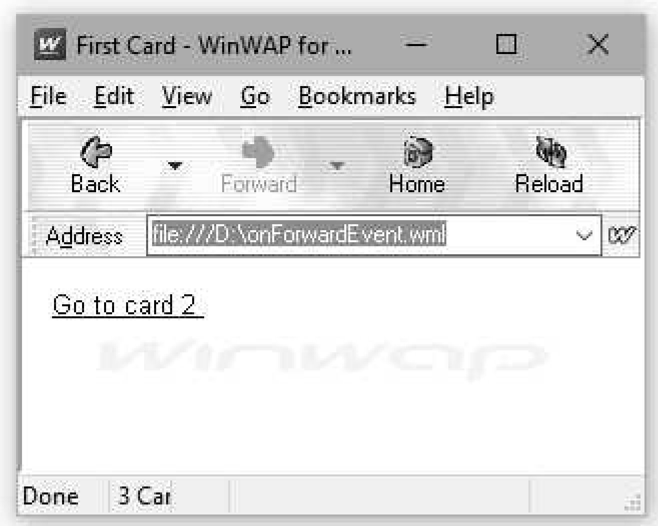
The onenterforward event is triggered when a user goes to a card in the forward direction. For example, if you go to a card by entering the URL directly or by following an anchor link of which the action is <go>, the onenterforward event will be triggered and the WML code associated with the event will be executed.

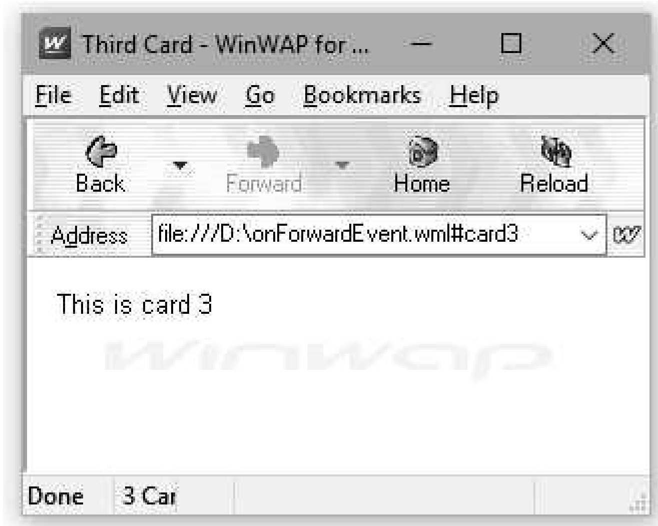
WML Document[onForwardEvent.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
```

```
<card id="card1" title="First Card">
<p>
 <anchor>
   <go href="#card2"/>
   Go to card 2
 </anchor>
</p>
</card>
<card id="card2" title="Second Card">
<onevent type="onenterforward">
 <go href="#card3"/>
</onevent>
<p>
 This is card 2
</p>
</card>
<card id="card3" title="Third Card">
<p>
This is card 3
</p>
</card>
</wml>
```

Output





Note that on clicking the link Go to card2 the control will move directly to card3, because in card2 the **onenterforward** event will occur and control will be transferred directly to card3.

Example using onpick

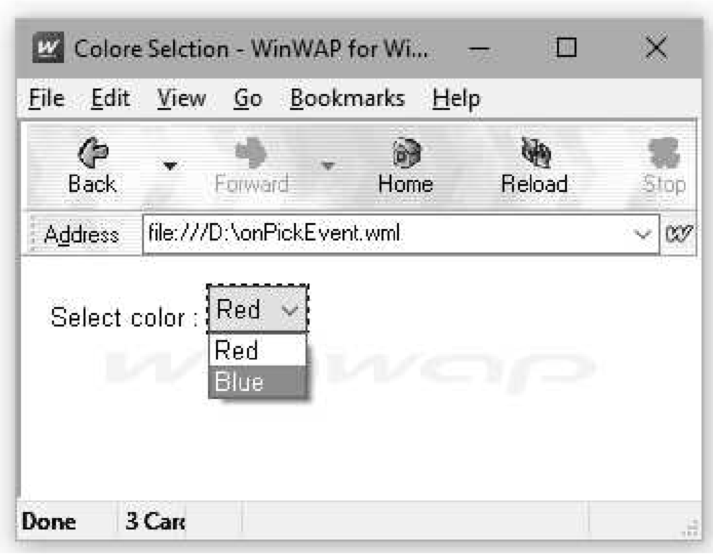
When an item from the selection list is selected or deselected then this even occurs.

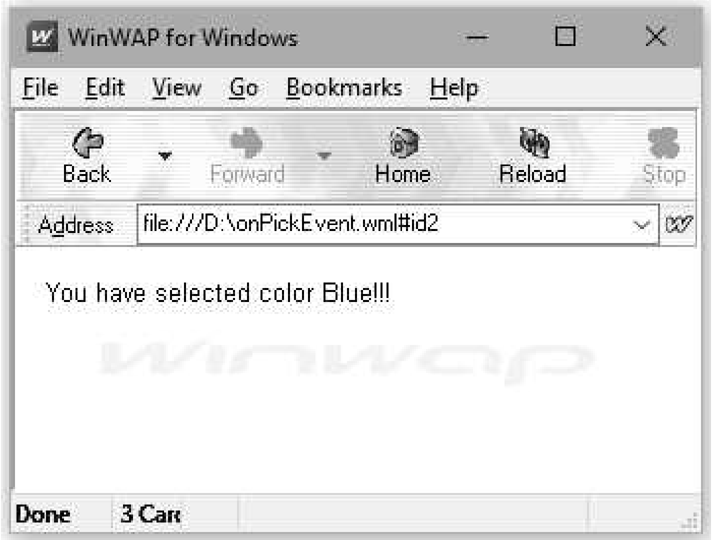
WML Document[onPickEvent.wml]

```
<?xml version="1.0"?>
<!DOCTYPE wml PUBLIC "-//WAPFORUM//DTD WML 1.2//EN"
"http://www.wapforum.org/DTD/wml12.dtd">
<wml>
<card id="MainCard" title="Color Selection">
<p>
 Select color :
 <select>
   <option onpick="#id1">Red</option>
   <option onpick="#id2">Blue</option>
 </select>
</p>
</card>
<card id="id1">
<p>
You have selected color RED!!!
</p>
</card>
<card id="id2">
```

```
<p>
You have selected color Blue!!!
</p>
</card>
</wml>
```

Output





Part - II : ASP.NET

5.19 Introduction to ASP.NET

Active Server Pages (ASP) is a server side scripting language. It is a part of **Internet Information Server** (IIS). This is basically a product of Microsoft.

The ASP.NET is a server side scripting technology offered by Microsoft. It is a powerful scripting language for creating dynamic and interactive web pages. The ASP.NET is next generation ASP but it is entirely new technology. The ASP.NET runs under IIS server.

5.19.1 How does ASP.NET Works ?

Step 1 : When user demands for the ASP.NET file, the web browser of the client's PC sends this request to IIS server.

Step 2 : The IIS server passes this request to .NET Engine.

Step 3 : The .NET engine reads the file line by line and generates the script in a file.

Step 4 : The requested .NET file is returned to the web browser.

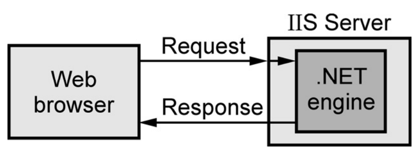


Fig. 5.19.1

5.19.2 Features of ASP.NET

- ASP.NET is built on the Common Language Runtime (CLR), allowing programmers to write ASP.NET code using any supported .NET language.
- ASP.NET pages are called **web forms.** These web forms have the extensions **.aspx.**
- These files contain static XHTML markup.
- The static and dynamic content can be inserted in the ASP.NET code.
- The code for appearance can be separated out from the code which is responsible for responding the events. Microsoft treats the dynamic code using the **code behind the model.** In this model the dynamic code is collected and placed in a separate file. Such a file typically has the names like xyz.aspx.cs. This separates out the **contents** from **presentation.**
- ASP.NET supports various types of controls. It also allows the programmer to build the **custom controls.** These controls can be built in a .DLL file and can be loaded whenever required.

- ASP.NET supports some advanced techniques such as cookies, sessions handling, query strings.

5.20 Overview of the .NET Framework

- The .NET framework is a software development platform developed by Microsoft. It is designed for building and running windows applications.
- The first version of .NET framework was released in the year 2002. This version was known as .NET framework 1.0.
- The .NET framework is used to create both form based and web based applications.
- The .NET framework is also useful in building the web services.
- It supports various programming languages like C#, VB.NET, Visual Basic and many more. Developers can choose and select the language to develop the required application.

Architecture of .NET Framework

The .NET framework architecture consists of three major components -

1. CLR
2. Library
3. Language

Let us discuss them in detail.

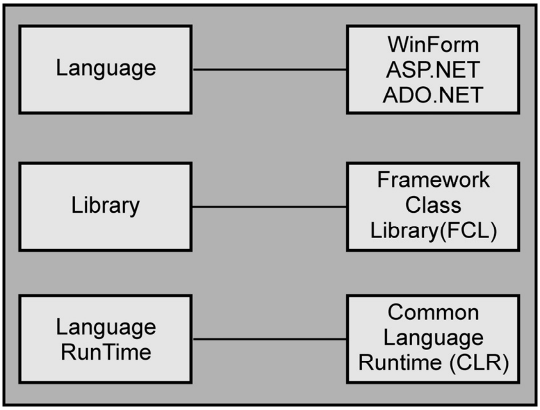


Fig. 5.20.1 The .NET framework

1. CLR

- The CLR stands for Common Language Runtime. It is a platform on which the .NET programs are executed. Any .NET programming language can be executed by this platform.

- Just in Time (JIT) is a part of CLR. It converts the source code into native intermediate code and then into machine instructions.
- The CLR is also responsible for performing some other tasks such as exception handling, memory management and garbage collection.
- The important features of CLR are interoperability, security and portability.

2. Framework Class Library (FCL)

- It is a standard library which is actually a collection of various classes, interfaces, and namespaces that can be used to build the application.
- It is the key component of .NET framework that provides the core functionalities of .NET.
- Most of the methods of classes are split either into System.* or Microsoft.* namespaces. A namespace is a logical separation of methods.

3. Language

- The .NET supports many languages. Developer can choose the desired language for development of required application.
- Following are the categories of various types of applications that can be developed using .NET
    - **WinForms :** This is useful for developing form based applications.
    - **ASP.NET :** The web based applications can be developed using ASP.NET. These applications will run on the web browsers such as Google Chrome, Mozilla FireFox, Internet Explorer and so on. The IIS server is responsible for executing the ASP.NET application which is a part of Microsoft component.
    - **ADO.NET :** This technology is used to interact with the databases like Microsoft SQL servers, Oracle and so on. The most commonly used languages for this category is C# and VB.NET.

5.21 Overview of C#

- The C# is a modern, **object oriented programming** language developed by Microsoft.
- C# was developed by Anders Hejlsberg during the development of .NET framework.

- C# is designed for **Common Language Infrastructure (CLI),** which consists of the executable code and runtime environment that allows use of various high-level languages to be used on different computer platforms and architectures.
- Various **features** of C# are
  1. Object Oriented
  2. Standard Library
  3. Automatic Garbage Collection
  4. Boolean Condition
  5. Properties and Events
  6. Simple Multithreading
  7. Integration with Windows.

Required Environment

For executing the C# programs we need Visual Studio 2019. Install the Microsoft Visual Studio 2019 edition.

5.21.1 Parts of C# Program

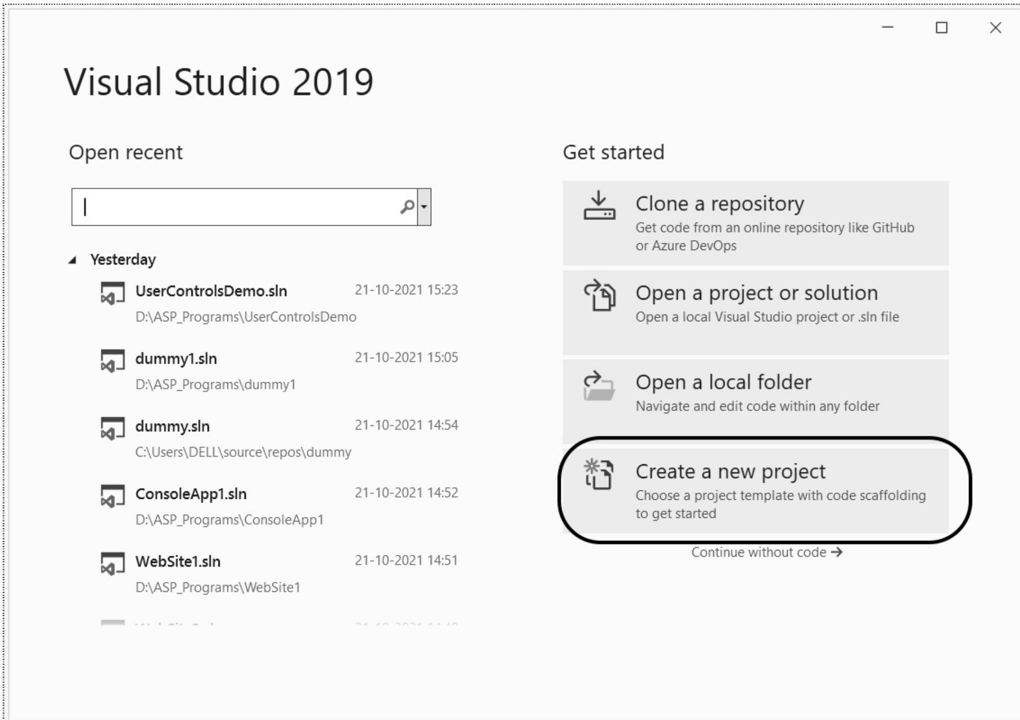
A C# program basically consists of the following parts :

- Namespace declaration
- A class
- Class methods
- Class attributes
- A Main method
- Statements and Expressions
- Comments

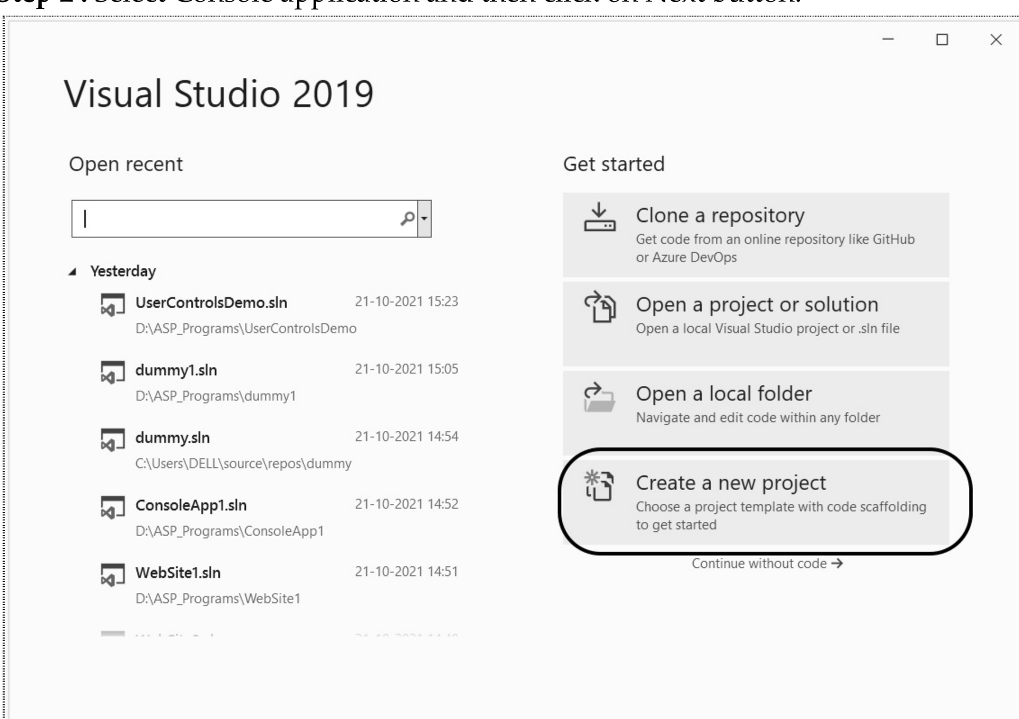
5.21.2 Creating First C# Program

In this section we will write and execute the first C# program.

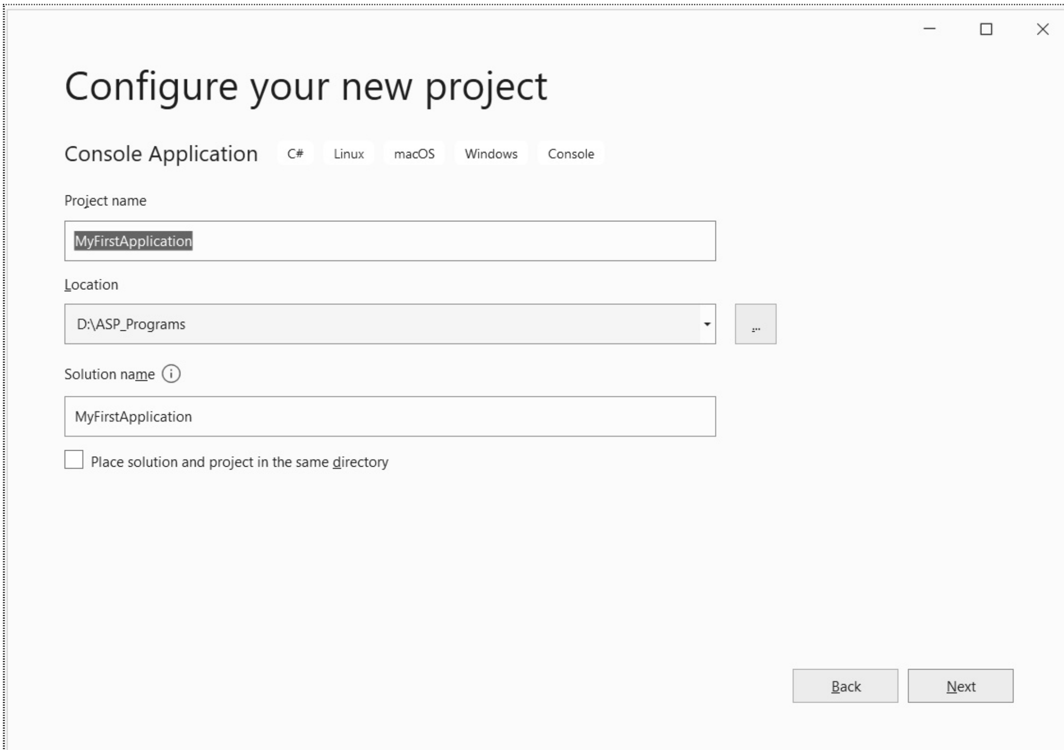
Step 1 : Open Visual Studio. Click on 'Create a new Project'



Step 2 : Select Console application and then click on Next button.



Step 3 : Give any suitable name to your project. I have given the name as **MyFirstApplication.**

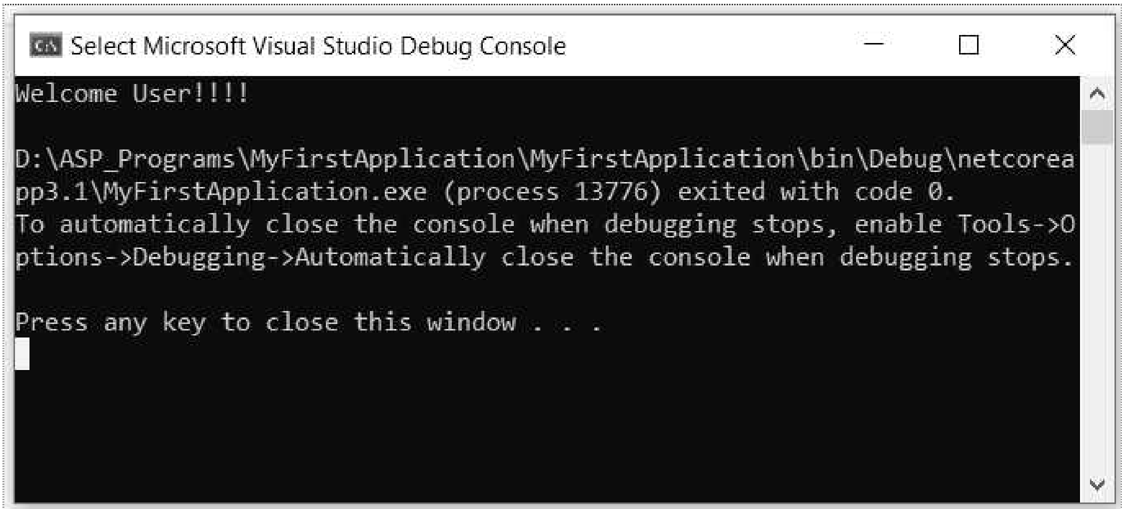


Step 4 : The source code window will appear now. This window contains the default C# code. We can write our first C# code as follows -

```
using System;

namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("Welcome User!!!!");
      }
   }
}
```

Step 5 : Click on

The console window will appear and display the output of the above program.



Program Explanation :

1. The first line of the program is **using System;** the word **using** is a keyword that is used to include the **System** namespace in the program. The namespace is a collection of classes.
2. On the next line the namespace is declared by the name of the program. This namespace contains the class definition named **Program** in which the method Method() is defined.
3. This is the only method defined in the class. This is the **entry point** for all C# programs. The **Main** method states what the class will do when executed.
4. Inside the Main method we have written **Console.writeln** statement. The Console is the class and we are invoking the **writeln** method of this class. By this statement we can display the desired message on the output screen.

Comment Statement

The comments in C# are similar to C++ comment statements. The // and /* … */ are used as comments in C#.

Points to Remember

1. C# is case sensitive
2. The execution of the program starts at Main method.
3. Every statement in C# must be terminated by semicolon.
4. The file name can be different from the class name.

5.21.3 Variables

Variables are used for storing values. The syntax for variables is as follows -

```
Data type variable = value
```

For example -

```
int age = 20;
```

5.21.4 Data Types

The data types are used for specifying the type of the variable.

Following table shows various data types used -

Data Types	Memory Size
char	1 byte
signed char	1 byte
unsigned char	1 byte
short	2 byte
signed short	2 byte
unsigned short	2 byte
int	4 byte
signed int	4 byte
unsigned int	4 byte
long	8 byte
signed long	8 byte
unsigned long	8 byte
float	4 byte
double	8 byte
decimal	16 byte

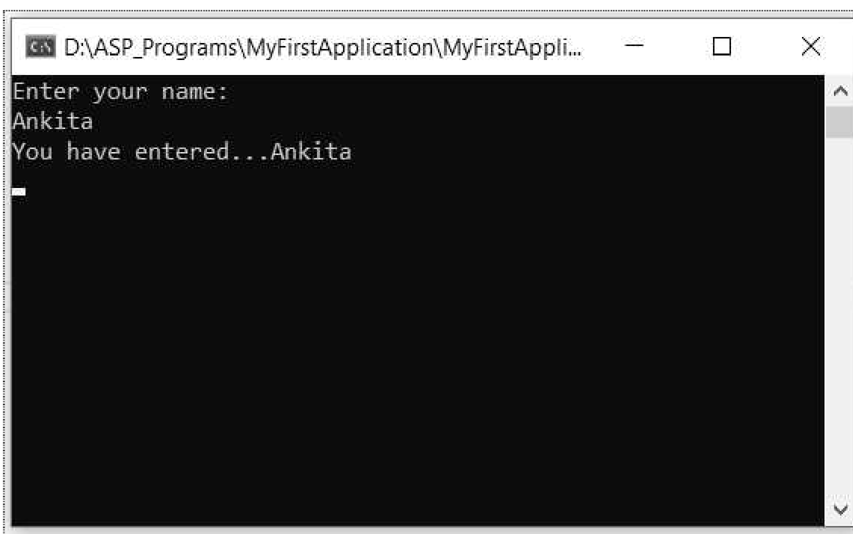
5.21.5 Input and Output

The user input is given by **Console.ReadLine()** and the output is displayed using **Console.WriteLine()** method. For example :

```
using System;
using System.Collections.Generic;
```

```
using System.Linq;
using System.Text;

namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("Enter your name: ");
         String name = Console.ReadLine();
         Console.WriteLine("You have entered..."+name);
         Console.ReadKey();
      }
   }
}
```

Output



5.21.6 Operators

The operators are the symbols that are used to perform various operations.

Various operators that are used in Java are enlisted in following table -

Type	Operator	Meaning	Example
Arithmetic	+	Addition or unary plus	c=a+b
	–	Subtraction or unary minus	d= – a
	*	Multiplication	c=a*b

	/	Division	c=a/b
	%	Mod	c=a%b
Relational	<	Less than	a<4
	>	Greater than	b>10
	<=	Less than equal to	b<=10
	>=	Greater than equal to	a>=5
	==	Equal to	x==100
	!=	Not equal to	m!=8
Logical	&&	And operator	0&&1
	||	Or operator	0 ||1
Assignment	=	is assigned to	a=5
Increment	++	Increment by one	++i or i++
Decrement	– –	Decrement by one	– – k or k– –

Conditional operator

The conditional operator is "?" The syntax of conditional operator is

```
Condition?expression1:expression2
```

where expression1 denotes the true condition and expression2 denotes false condition.

For example :

```
a>b?true:false
```

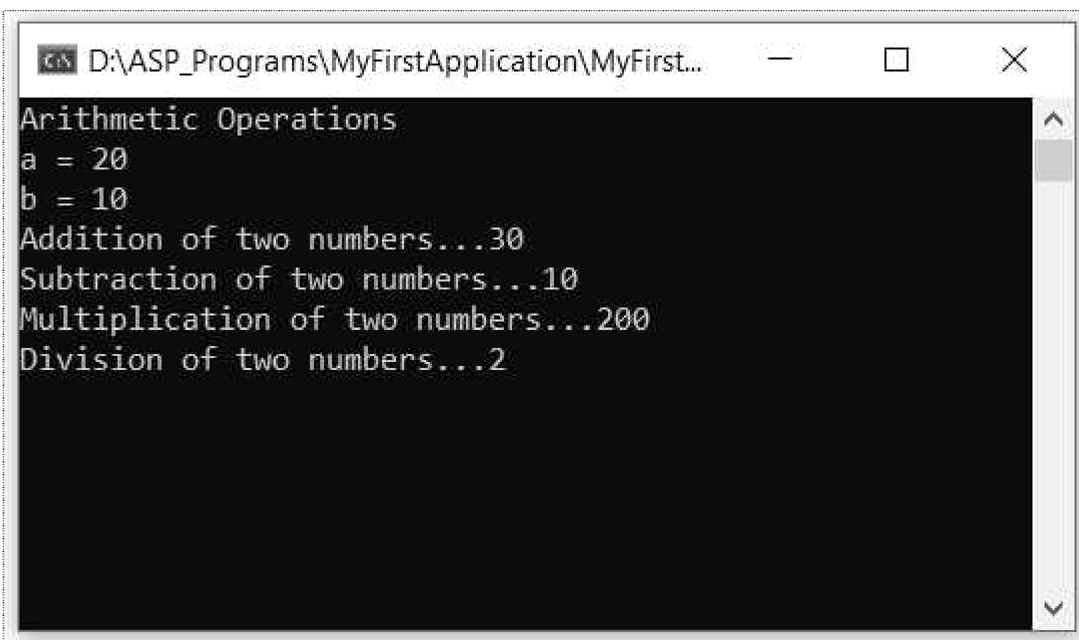
This means that if a is greater than b then the expression will return the value true otherwise it will return false.

Program for Demonstrating Arithmetic Operations

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
```

```
            Console.WriteLine("Arithmetic Operations");
            int a = 20;
            int b = 10;
            int c;
            Console.WriteLine("a = " + a);
            Console.WriteLine("b = " + b);
            c = a + b;
            Console.WriteLine("Addition of two numbers..."+c);
            c = a - b;
            Console.WriteLine("Subtraction of two numbers..." + c);
            c = a * b;
            Console.WriteLine("Multiplication of two numbers..." + c);
            c = a / b;
            Console.WriteLine("Division of two numbers..." + c);

            Console.ReadKey();
        }
    }
}
```

Output

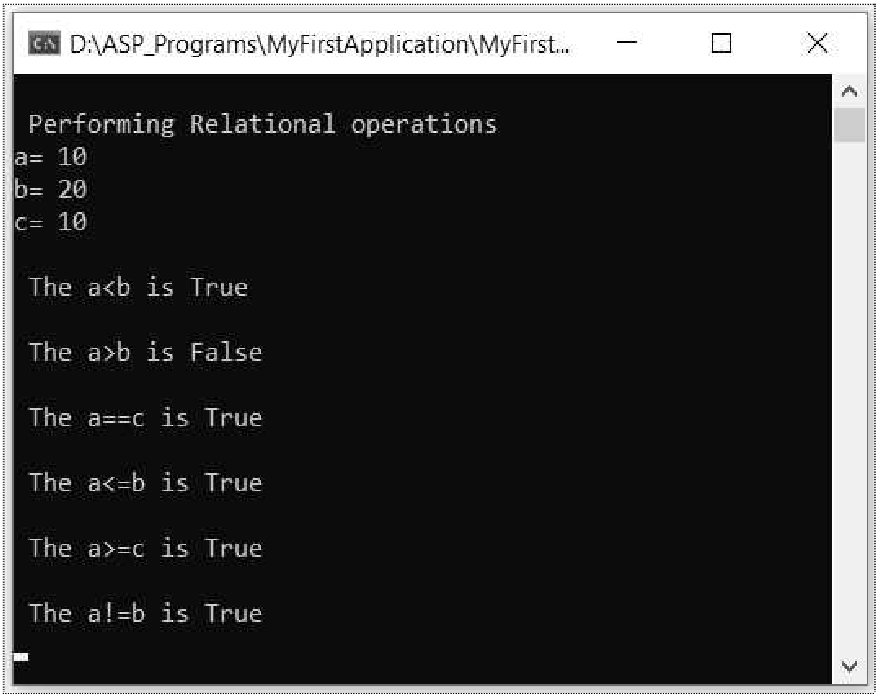


Program Demonstrating Relational Operators

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
```

```
namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("\n Performing Relational operations ");
         int a = 10, b = 20, c = 10;
         Console.WriteLine("a= " + a);
         Console.WriteLine("b= " + b);
         Console.WriteLine("c= " + c);
         Console.WriteLine("\n The a<b is " + (a < b));
         Console.WriteLine("\n The a>b is " + (a > b));
         Console.WriteLine("\n The a==c is " + (a == c));
         Console.WriteLine("\n The a<=b is " + (a <= b));
         Console.WriteLine("\n The a>=c is " + (a >= c));
         Console.WriteLine("\n The a!=b is " + (a != b));
         Console.ReadKey();
      }
   }
}
```

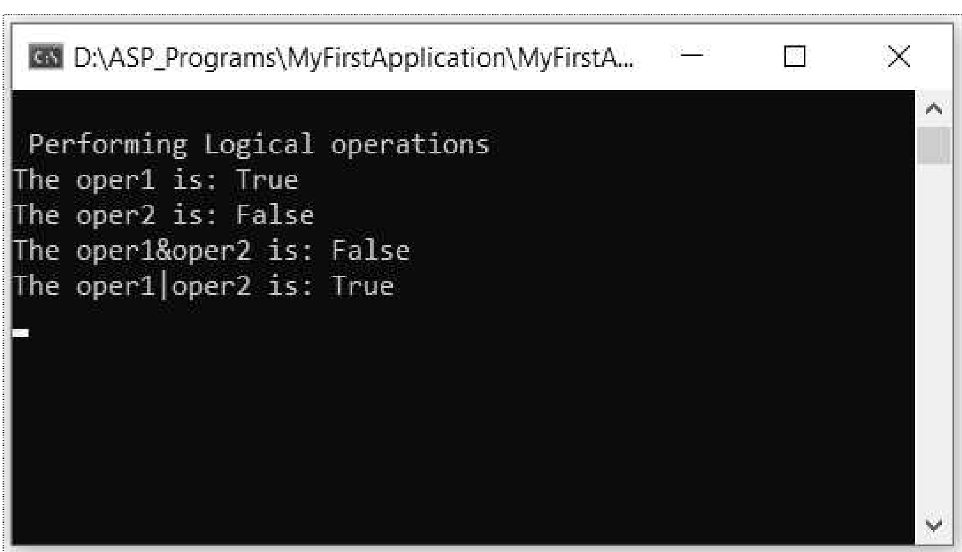
Output



Program Demonstrating Logical Operators

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("\n Performing Logical operations ");
         Boolean oper1, oper2;
         oper1 = true;
         oper2 = false;
         Boolean ans1, ans2;
         ans1 = oper1 & oper2;
         ans2 = oper1 | oper2;
         Console.WriteLine("The oper1 is: " + oper1);
         Console.WriteLine("The oper2 is: " + oper2);
         Console.WriteLine("The oper1&oper2 is: " + ans1);
         Console.WriteLine("The oper1|oper2 is: " + ans2);
         Console.ReadKey();
      }
   }
}
```

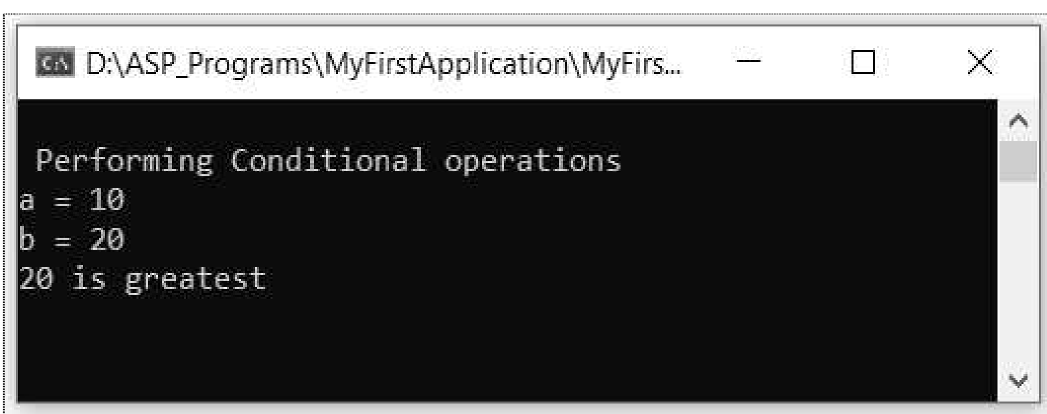
Output



Program Demonstrating Conditional Operator

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("\n Performing Conditional operations ");
         int a, b, c;
         a = 10;
         b = 20;
         Console.WriteLine("a = " + a);
         Console.WriteLine("b = " + b);
         c = a > b ? a : b;
         Console.WriteLine(c + " is greatest");
         Console.ReadKey();
      }
   }
}
```

Output



5.21.7 Control Structure

Various control structures are -

1. if statement
2. while statement
3. do-while statement
4. switch case statement
5. for loop

Let us discuss these control structures with the help of simple examples.

1. If statement

- There are two types of if statements - simple if statement and compound if statement. The simple if statement is a kind of if statement which is followed by single statement. The compound if statement is a kind of if statement for which a group of statements are followed. These groups of statements are enclosed within the curly brackets.
- If statement can also be accompanied by the else part.
- Following table illustrates various forms of if statement.

Type of statement	Syntax	Example
simple if	if(condition) statement	if(a<b) Console.WriteLine(”a is smaller than b”);

compound if	if(condition)
	{
	statement 1;
	…
	…
	}
		if(a<b)
		{
		Console.WriteLine(”a is smaller than b”);
		Console.WriteLine(”b is larger than a”);
		}

if…else	if(condition)
	statement;
	else
	statement;
		if(a<b)
		Console.WriteLine(”a is smaller than b”);
		else
		Console.WriteLine(”a is larger than b”);

compound if…else	if(condition)
	{
	statement 1;
	…
	…
	}
	else
	{
	statement 1;
	…
	…
	}
		if(a<b)
		{
		Console.WriteLine(“a is smaller than b”);
		Console.WriteLine(”b is larger than a”);
		}
		else
		{
		Console.WriteLine(”a is larger than b”);
		Console.WriteLine(“b is smaller than a”);
		}

if…else if	if(condition)
	{
	statement 1;
	…
	…
	}
	else if(condition)
	{
	statement 1;
	…
	…
	}
	else
	{
	statement 1;
	…
	…
	}
		if(a<b)
		Console.WriteLine("a is smaller than b");
		else if(a<c)
		Console.WriteLine("a is smaller than b");
		else
		Console.WriteLine("a is larger than b and c");

2. while statement

The while statement is executing repeatedly until the condition is false. The while statement can be simple while or compound while. Following table illustrates the forms of while statements -

Type of statement	Syntax	Example
simple while	while(condition) statement	while(a<10) Console.WriteLine("a is smaller than 10");

compound while	while(condition)
	{
	statement 1;
	…
	…
	}
		while(a<10)
		{
		Console.WriteLine("a is less than b");
		a++;
		}

3. do..while

- The do…while statement is used for repeated execution. The difference between do while and while statement is that, in while statement the condition is checked before executing any statement whereas in do while statements the statement is executed first and then the condition is tested. There is at least one execution of statements in case of do…while. Following table shows the use of do while statement.

Type of statement	Syntax	Example

do…while	do
	{
	statement 1;
	…
	…
	}while(condition);
		do
		{
		Console.WriteLine(”a is less than b”);
		a++;
		} while(a<10);

Note that the while condition is terminated by a semicolon.

4. Switch statement

From multiple cases if only one case is to be executed at a time then switch case statement is executed. Following table shows the use of switch case statements.

Type of statement	Syntax	Example

switch …case	switch(condition)
	{
	case caseno:statements
	break;
	….
	default: statements
	}
		Console.WriteLine(”\n Enter choice”);
		cin>>choice;
		switch(choice)
		{
		case 1: Console.WriteLine(”You have selected 1”);
		break;
		case 2: Console.WriteLine(”You have selected 2”);
		break;
		case 3: Console.WriteLine(”You have selected 3”);
		break;
		default: Console.WriteLine(”good bye”);
		}

5. for loop

The for loop is a not statement it is a loop, using which the repeated execution of statements occurs. Following table illustrates the use of for loop -

Loop	Syntax	Example
Simple for loop	for(initialization; termination; step count) statement	for(i=0;i<10;i++) c[i]=a[i]+b[i];

Compound for loop	for(initialization; termination; step count)
	{
	statement 1;
	statement 2;
	…
	}
		for(i=0;i<10;i++)
		{
		for(j=0;j<10;j++)
		c[i][j]=a[i][j]+b[i][j];
		}

5.21.8 The Break and Continue Statements

The break and continue statements are used to alter the normal control flow of the program.

Break statement : The break statement is used to terminate the loops such as for, while, do…while and switch.

Syntax :

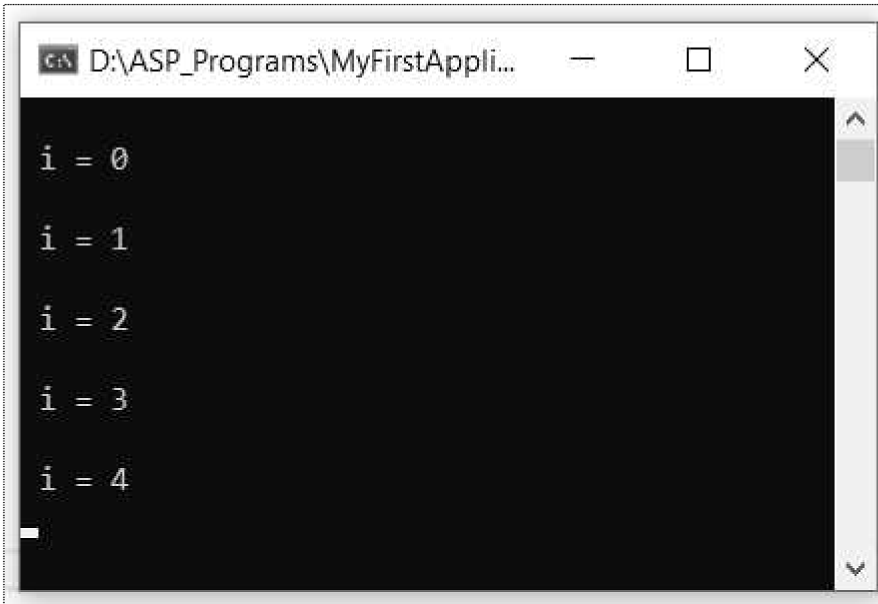
```
break;
```

Example Program :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace MyFirstApplication
{
   class Program
   {
      static void Main(string[] args)
      {
         int i, count = 5;
         for (i = 0; i < 10; i++)
         {
            if (i == count)
            {
               break;
            }
            Console.WriteLine("\n i = "+i);
         }
```

```
            Console.ReadKey();
        }
    }
}
```

Output



Continue statement : Sometime we need to skip certain test condition within a loop. At that time continue statement is used.

Syntax :

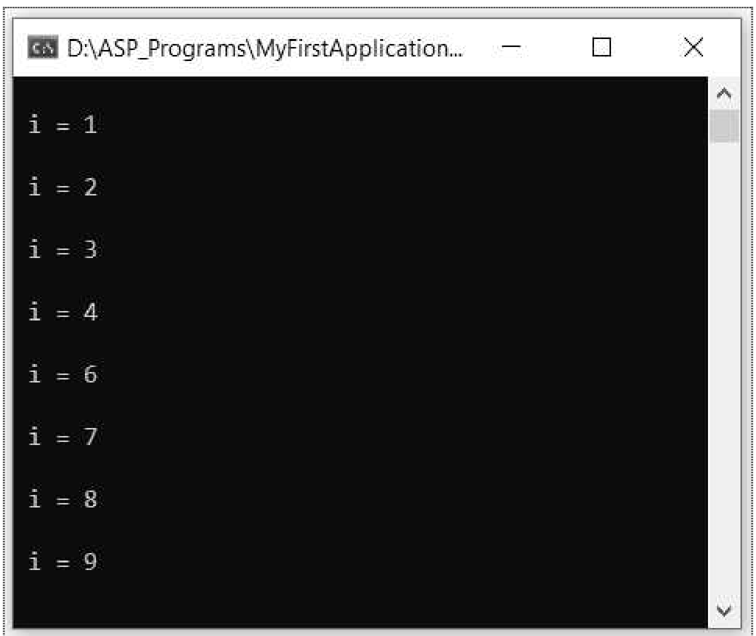
```
continue;
```

Example Program

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace MyFirstApplication
{
    class Program
    {
        static void Main(string[] args)
        {
            int i, count = 5;
            for (i = 0; i < 10; i++)
```

```
        {
            if (i == count)
            {
                continue;
            }
            Console.WriteLine("\n i = " + i);
        }
        Console.ReadKey();
    }
  }
}
```

Output



5.21.9 Class and Object

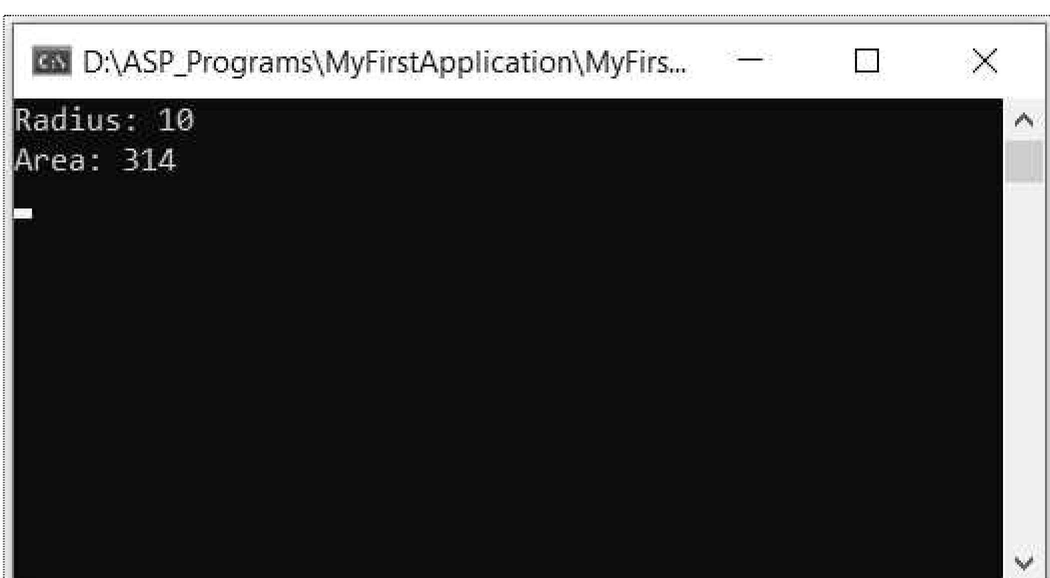
The programming style of C# is just similar to C++. Just similar to C++, you can write the class in C# and can access the methods of that class using the object of the class. Following example illustrates this concept.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;

namespace ConsoleApplication2
{
  class Circle
```

```
    {
        double radius;
        double pi=3.14;
        double a;
        public void getdata()
        {
            radius = 10;
        }
        public void area()
        {
           a = pi * radius * radius;
          Console.WriteLine("Radius: {0}", radius);
          Console.WriteLine("Area: {0}", a);
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            Circle obj = new Circle();
            obj.getdata();
            obj.area();
            Console.ReadKey();
        }
    }
}
```

Note that while displaying the variables they are displayed using the index{0},{1} and so on.
These two statements can be combined into one statement as follows -
Console.WriteLine("Radius: {0},Area: {1}", radius, a);

Output



5.22 Introduction to ASP.NET

- ASP.NET is a platform for developing web applications. It works on **HTTP protocol** and makes use of HTTP commands.
- The ASP.NET application programs can be written in one of the following languages -

  1. C#
  2. Visual Basic .NET
  3. Jscript
  4. J#

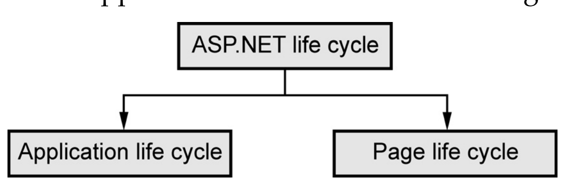
- ASP.NET consists of large number of controls such as textboxes, buttons, labels and checkboxes and so on for developing the web pages. Basically the HTML tags are used to manipulate the code.

5.22.1 ASP.NET Component Model

- The ASP.NET component model provides various building blocks of the ASP.NET pages. It describes the -
  1. Server side **pages** that uses the HTML tags
  2. Server **controls** that are used for building complex web applications.
- In ASP.NET programming the **.aspx page** uses the HTML tags. During execution, the .aspx page is transformed into an **instance of a class** which inherits from the base class Page of the .Net framework.
- In this technology **each ASP.Net page is an object** and all its components i.e., the server-side controls are also objects.

5.22.2 Life Cycle of ASP.NET Application

The life cycle of ASP.NET application is divided into two categories -



Application life cycle

There are mainly two phases in the application life cycle -

1. Request phase

a) User makes a request for accessing particular resource.

b) Browser sends this request to the web server.

c) An object of **ApplicationManager** class is created.

d) An object of the **HostingEnvironment** class is created to provide information regarding the resources.

e) Required items in the application are compiled.

2. Response Phase

a) Response objects are created.

b) There are some application objects of the class **HttpContext, HttpRequest** and **HttpResponse.** These objects are initialized.

c) The request is processed by the **HttpApplication** class. Hence an instance of the **HttpApplication** object is created and assigned to the request.

d) Different events are raised by this **HttpApplication** class for processing the request.

3. Page life cycle

Various stages of page life cycle are -

1. **Page request :** The page request occurs before the page life cycle begins. When the page is requested by a user, ASP.NET determines whether the page needs to be parsed and compiled or whether a cached version of the page can be sent in response without running the page.
2. **Start :** In stage the page properties such as **Request** and **Response** are set. At this stage, the page determines whether the request is postback or it is a new request. It then sets the **IsPostBack** property. At this stage the **UICulture** property is also set.
3. **Initialization :** In this stage controls on the page are available. Each control's **UniqueID** property is set. A master page and themes are set. For a new request postback data is loaded and the control properties are restored to the view-state values.
4. **Load :** At this stage, the control properties are loaded with information obtained from view state and control state.
5. **Postback event handling :** If the request is a postback, then control event handlers are called. Then to validate the controls the **Validate** method is called.
6. **Rendering :** At this stage, the page calls the **Render** method for each control. The output of rendering is written to the **OutputStream** class of the Page's Response property.

7. **Unloading :** The Unload event is raised after the page is fully rendered. At this point, page properties such as Response and Request are unloaded and cleanup is performed.

Page life cycle events

Within each stage of the life cycle of a page, the page raises events. The following table lists the page life-cycle events that are used most commonly.

Page event	Purpose
PreInit	This event is raised after start stage is complete and before initialization stage. It Checks the **IsPostBack** property to determine whether this is the first time the page is being processed. It sets the Master page and theme property dynamically.
Init	This event is raised after all controls have been initialized.
InitComplete	It is raised at the end of the page's initialization stage
PreLoad	It is raised after the page loads view state.
Load	The **Page** object calls the **OnLoad** method. This events is raised for the page loading and then recursively for all the child controls.
LoadComplete	It is raised at the end of the event-handling stage. This event can be handled by overloading the **OnLoadComplete** method
PreRender	When all the controls for the page are created then this event is raised. The **Page** object raises the **PreRender** event on the **Page** object, and then recursively does the same for each child control.
PreRenderComplete	After completion of preRendering stage this event is raised.
Render	This is not an event; instead, at this stage of processing, the **Page** object calls this method on each control that writes out the control's markup to send to the browser
Unload	Raised for each control and then for the page. During the unload stage, the page and its controls have been rendered so further changes to response stream can not be done.
Control Events	These events are used to handle specific control events. For instance Button **control** has **click** event.

5.23 ASP.NET Web Form Model

- The ASP.NET web forms make use of events-driven techniques for the interaction of the web application. The browser actually submits the web form to the web server and then server returns the HTML page in response.
- The client side activities are forwarded to the server and the server sends the response to its client. The HTTP protocol plays an important role in this communication.
- The HTTP is a stateless protocol. So ASP.Net helps in storing the information regarding the state of the application. It consists of :
  1. Page state
  2. Session state

The page state is the state of the **client**. That means at this state, the client has the web application page having various input fields on the web form. On the other hand the **session state** is the state in which the information is collected from various pages the user visited or worked with. ASP.Net session state and server side infrastructure keeps track of the information collected globally over a session.

5.24 ASP.NET Controls

Various controls that can be used by the web forms are -

1. Button controls
2. Text boxes and labels
3. Check boxes and Radio buttons
4. List controls
5. List items

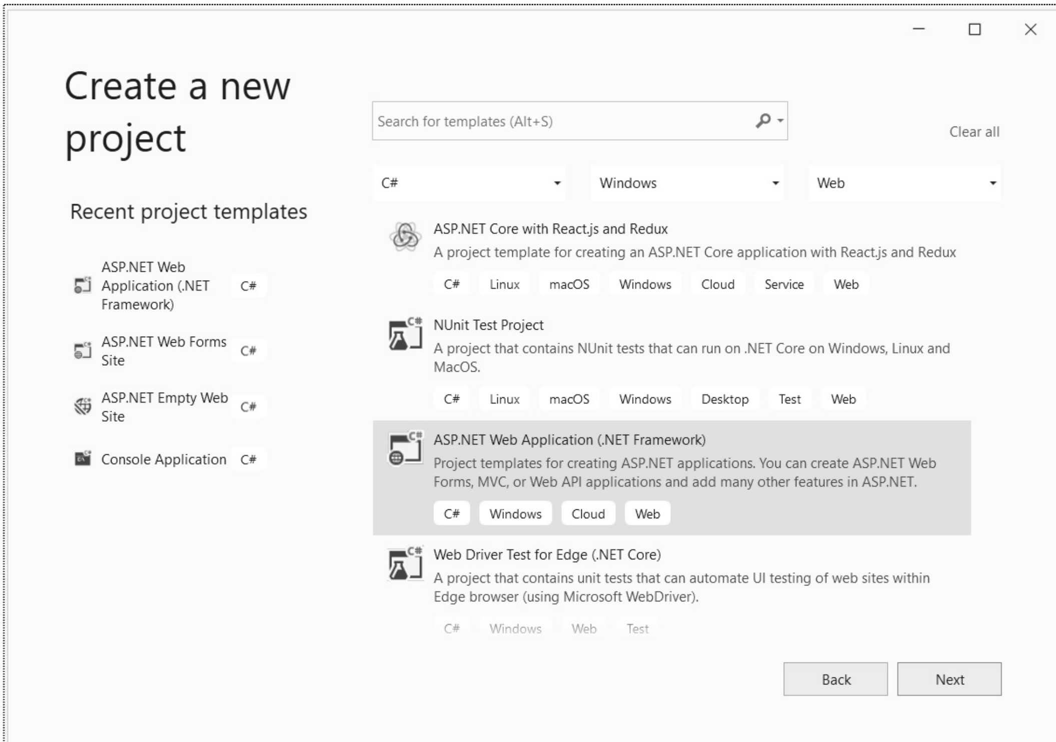
   and so on.

We will discuss some most commonly used controls for designing the web forms.

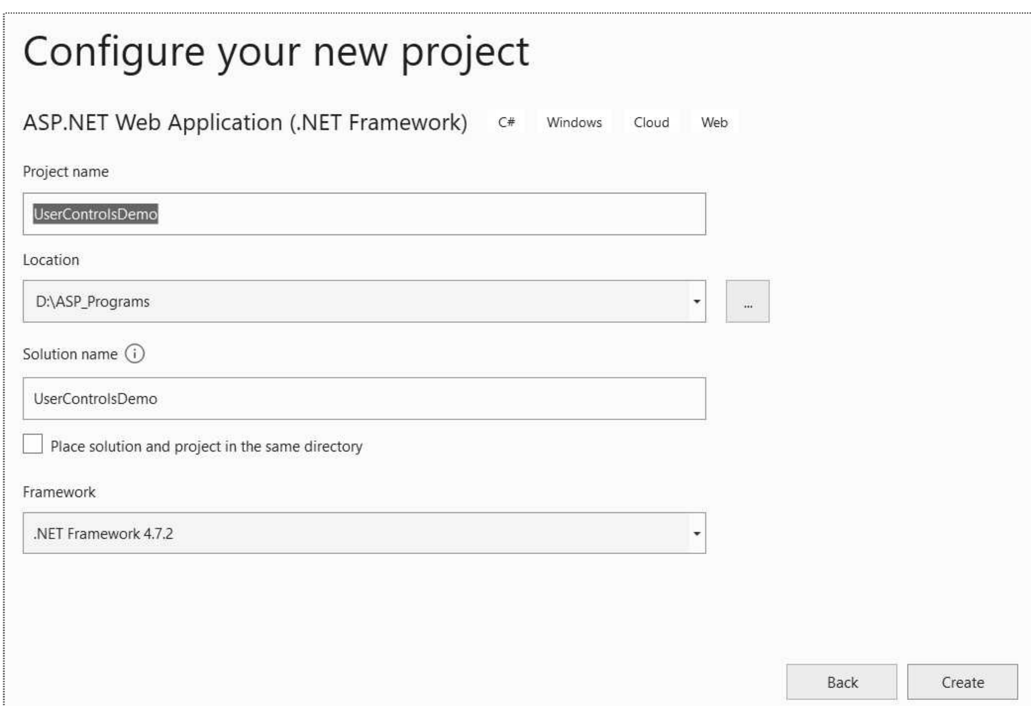
5.24.1 How to Create Web Form Application ?

Following are the steps to be taken to create and build the web forms

Step 1 : Start an instance of Visual Studio 2019 and create a new Web Application Project by clicking **Create a New Project.** Locate **ASP.NET Web Application (.NET Framework)**

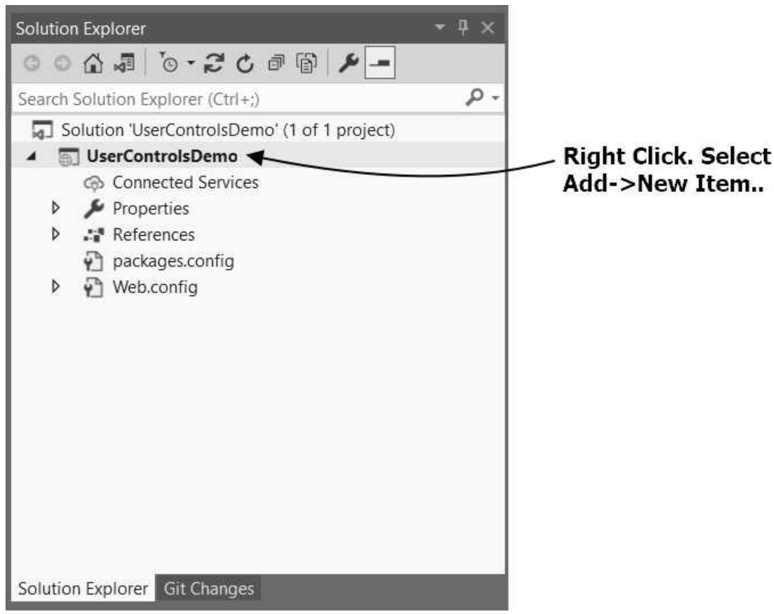


Click on Next button. Give the suitable project name. I have given project name as **UserControlsDemo.**

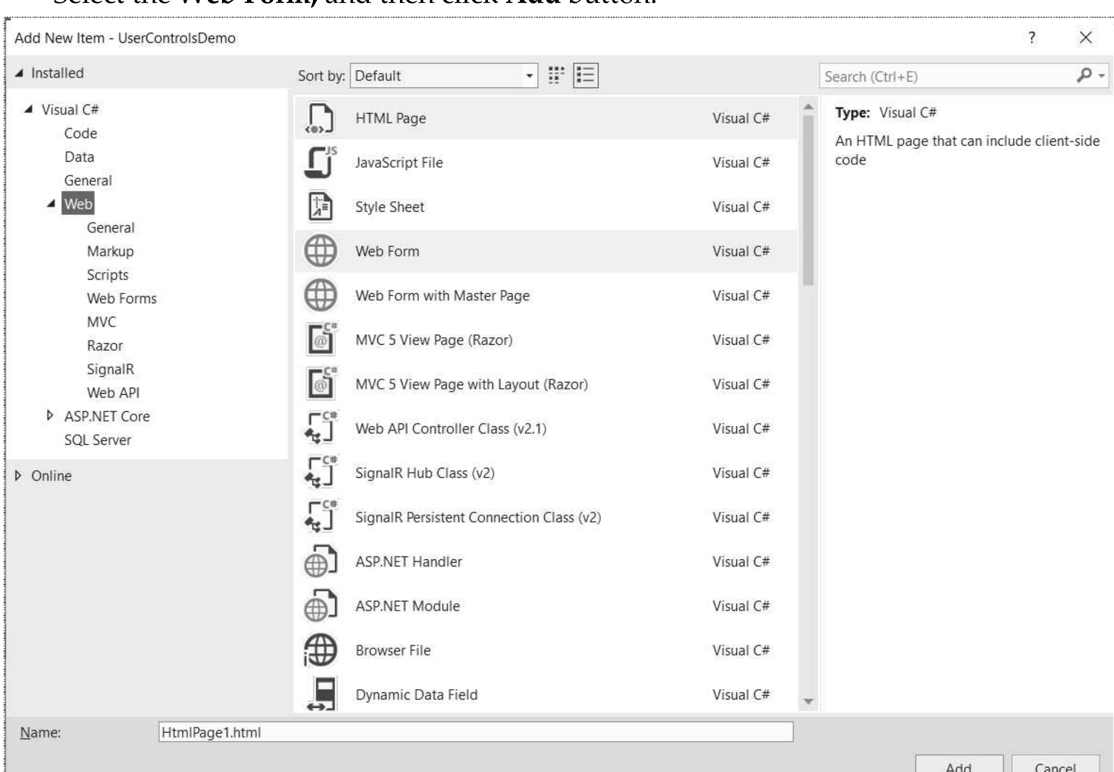


Click on **Create** button. Then Select **Empty** project.

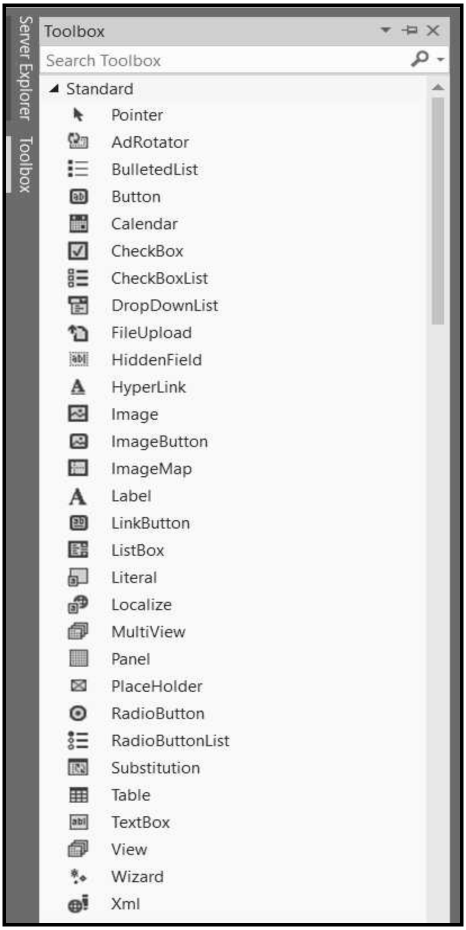
Step 2 : Now in the Solution Explorer window, right click on the project name



Select the **Web Form,** and then click **Add** button.



Step 3 : The default code for **WebForm1.aspx** will be displayed in the **Source** code window. We can add the user controls using **Toolbox.**

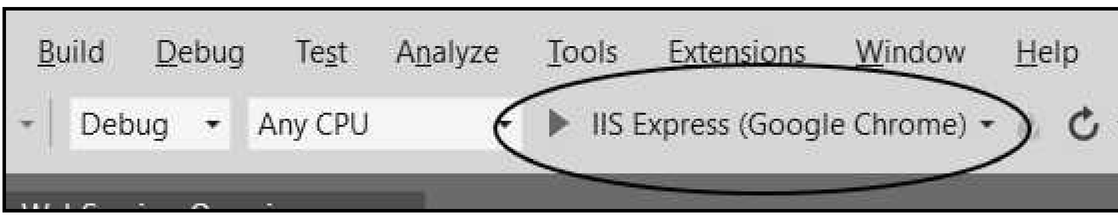


If we double click on some control, then the code for the user control will appear in the source code window. For instance if we want to insert a text box then,

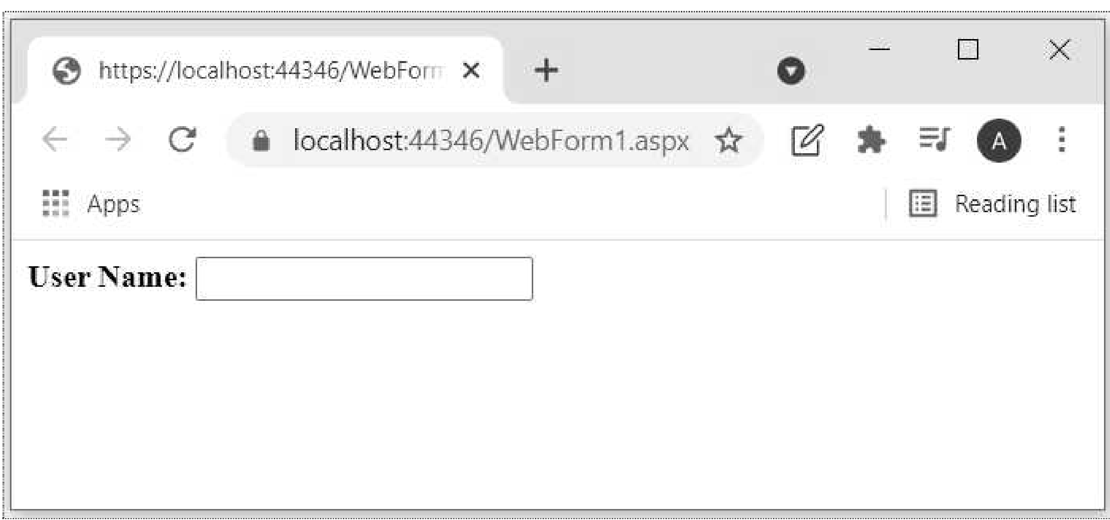
```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="UserControlsDemo.WebForm1" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
  <title></title>
</head>
<body>
  <form id="form1" runat="server">
```

```
      <div>
         <b> User Name:</b>
         <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
</div>
   </form>
</body>
</html>
```

Step 4 : The above web from can be executed on **IIS Express** server. Click on the IIS Express option present below the menu bar.



The output of the web service will be displayed on the web browser as follows -



Step 5 : The default code for WebForm1.aspx is displayed in the Code window. Just below that the **Design, Split** and **Source** options are located. Click on **Design.** The form window will get opened up. Now we can select the controls from the ToolBox under the **Standard** option. Click and drag the desired controls on the form. The most commonly used controls are Button, TextBoxes, Labels, Checkboxes, ListBoxes and so on.

Step 6 : Set the properties of the controls using the **Properties Window.** If you can not locate this window just press F4. You can set the properties like ID, Text, ForeColor, Font and so on.

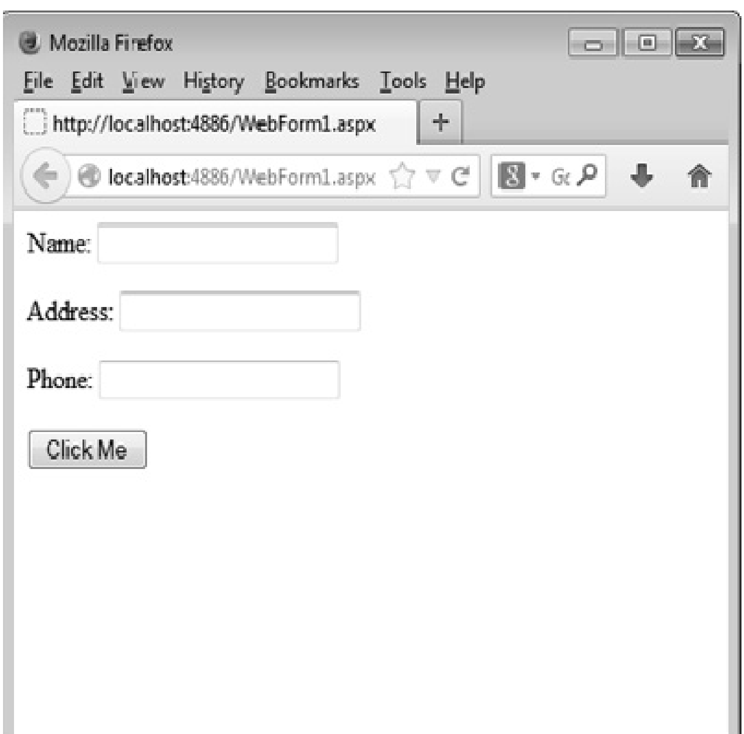
Step 7 : A sample form is designed as follows by simply dragging the controls.

Automatically the code will get generated. This code can be viewed by clicking the **Source.**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="WebApplication3.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <div>

      <asp:Label ID="Label1" runat="server" Text="Name:"></asp:Label>
```

```
        <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>

    </div>
    <p>
        <asp:Label ID="Label2" runat="server" Text="Address:"></asp:Label>
        <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
    </p>
    <p>
        <asp:Label ID="Label3" runat="server" Text="Phone:"></asp:Label>
        <asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>
    </p>
    <asp:Button ID="Button1" runat="server" Text="Click Me" />
    </form>
</body>
</html>
```

Step 8 : The output will be displayed on the web browser.



Examples on Web Form

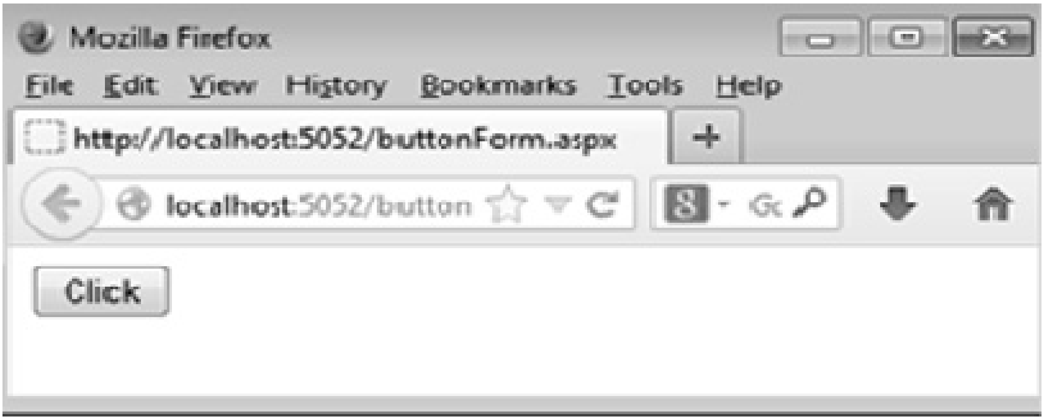
Example 5.24.1 *Write ASP.NET code to demonstrate the button control.*

Solution :

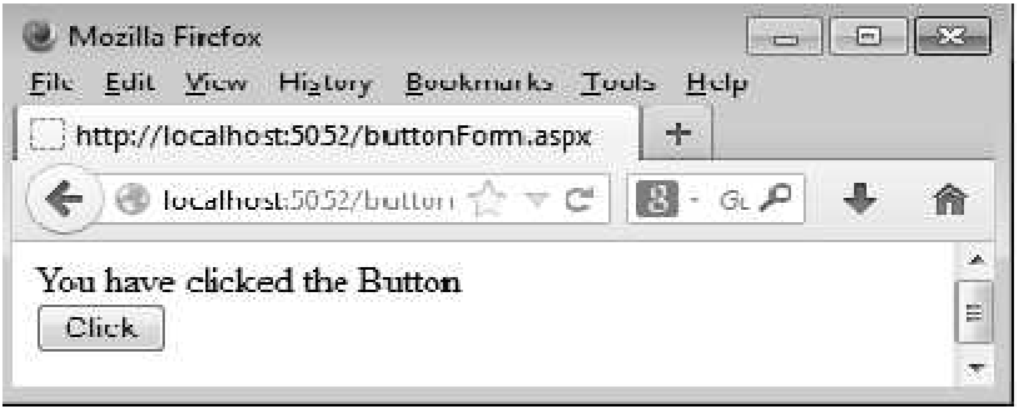
```
<%@ Page Language="C#"  %>
<script  runat="server">
void Button1_Click(object sender, EventArgs e)
{
   Label1.Text = "You have clicked the Button";
}
</script>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
   <div>
      <asp:Label ID="Label1" runat="server"></asp:Label>
   </div>
   <div>
        <asp:Button   ID="Button1"   runat="server"   Type="submit"   onclick="Button1_Click"
Text="Click" />
    </div>
   </form>
</body>
</html>
```

In above code, on the Button click the function **Button_Click** is called. In this function we are just assigning the Button clicked message as a Label control's text.

The output will be



just click on the Click button and you will get



Example 5.24.2 *Write ASP.NET code to demonstrate the TextBox control.*

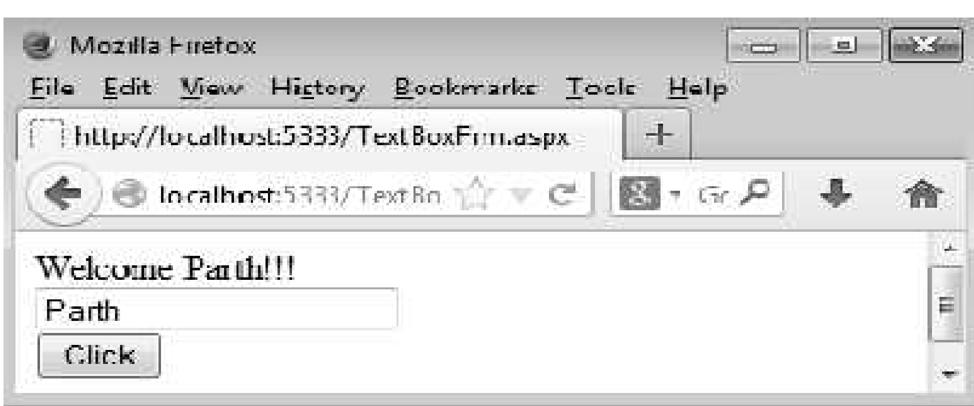
Solution :

```
<%@ Page Language="C#" %>
<script  runat="server">
void Button1_Click(object sender, EventArgs e)
{
   Label1.Text ="Welcome "+ TextBox1.Text+ "!!!";
}
</script>



<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
```

```
</head>
<body>
   <form id="form1" runat="server">
   <div>
     <asp:Label ID="Label1" runat="server" Text="Enter Some Text in the TextBox
"></asp:Label>
   </div>
   <div>
   <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
   </div>
   <div>
   <asp:Button ID="Button1" runat="server" onclick="Button1_Click" Text="Click" />
   </div>
   </form>
</body>
</html>
```

Output

Just type some name in TextBox. And then click the button.



Example 5.24.3 *Write ASP.NET code to demonstrate the CheckBox and ListBox control.*

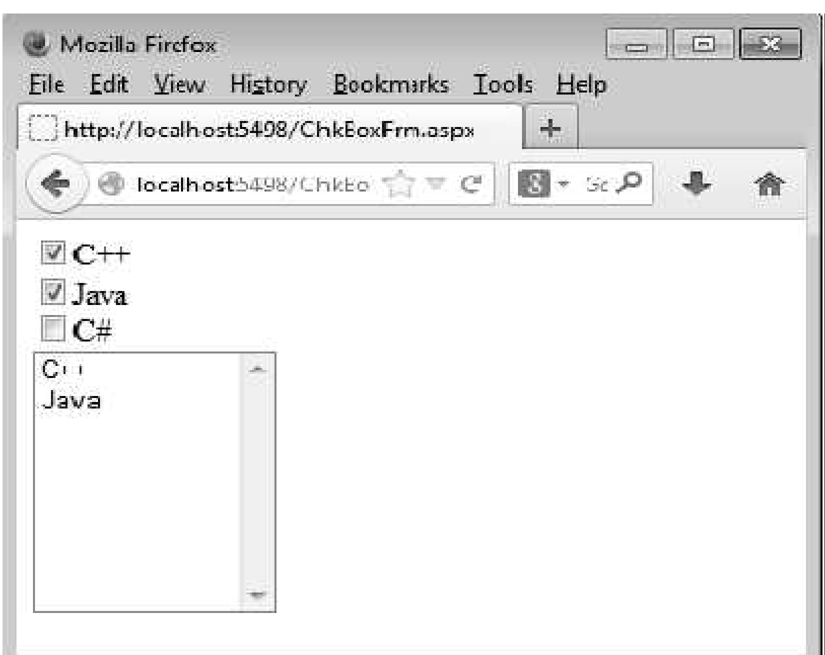
Solution :

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="ChkBoxFrm.aspx.cs"
Inherits="CheckBoxDemo.ChkBoxFrm" %>

<script runat="server">
void CheckBox1_CheckedChanged(object sender, EventArgs e)
{
   if (CheckBox1 .Checked==true)
   {
      ListBox1.Items.Add(CheckBox1.Text);
   }
   else
   {
      ListBox1.Items.Remove(ListBox1.Items.FindByText(CheckBox1.Text));
   }
}
void CheckBox2_CheckedChanged(object sender, EventArgs e)
{
   if (CheckBox2.Checked == true)
   {
      ListBox1.Items.Add(CheckBox2.Text);
   }
   else
   {
      ListBox1.Items.Remove(ListBox1.Items.FindByText(CheckBox2.Text));
   }
}
void CheckBox3_CheckedChanged(object sender, EventArgs e)
{
   if (CheckBox3.Checked == true)
   {
      ListBox1.Items.Add(CheckBox3.Text);
   }
   else
   {
      ListBox1.Items.Remove(ListBox1.Items.FindByText(CheckBox3.Text));
   }
}
</script>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
```

```
<body>
   <form id="form1" runat="server">
   <div>
      <asp:CheckBox ID="CheckBox1" runat="server" AutoPostBack="True"
        oncheckedchanged="CheckBox1_CheckedChanged" Text="C++" />

   </div>
   <div>
     <asp:CheckBox ID="CheckBox2" runat="server" AutoPostBack="True"
        oncheckedchanged="CheckBox2_CheckedChanged" Text="Java" />
   </div>
   <div>
     <asp:CheckBox ID="CheckBox3" runat="server" AutoPostBack="True"
        oncheckedchanged="CheckBox3_CheckedChanged" Text="C#" />
   </div>
   <div>
   <asp:ListBox ID="ListBox1" runat="server"  Height="140px" Width="120px">
   </asp:ListBox>
   </div>
   </form>
</body>
</html>
```

Output

Just check or uncheck the checkboxes to experience the effect of this control.



5.25 Web Services

- Web service is a software program using which the information can be exchanged. That means using web service one application can invoke the method another application. These applications can be on the same computer or different computers.
- The web services make use of the standard protocols such as HTTP, XML and SOAP. As these are open and standard protocols the applications can exchange the information on any platform with the help of web services. For instance - Web service built using .NET application can be consumed using JSON or Java application.

Features

1) Web service is a language independent.
2) It is platform independent.
3) It is based on XML.
4) It is discoverable. That means, applications and developers can search for and locate the desired web services through registries.
5) It is based on a stateless service architecture.

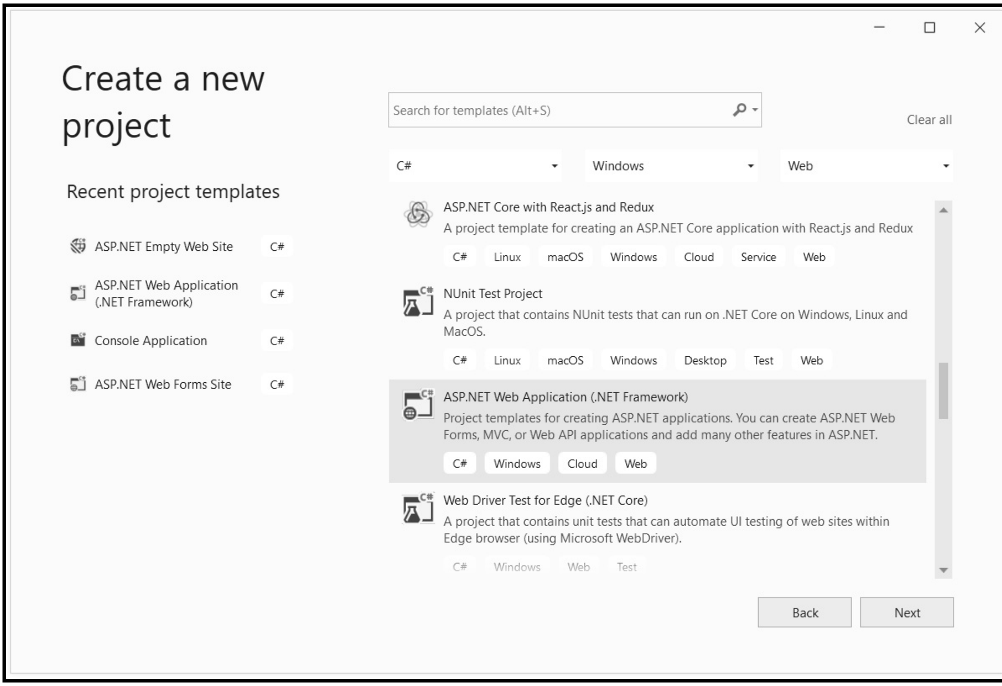
Basic terminologies used in web services

1) **HTTP :** The HTTP is a standard protocol which is widely used by the web services to send and receive messages.
2) **SOAP :** The SOAP (Simple Object Access Protocol) provides the mechanism between web services and applications. SOAP messages are in XML format.
3) **XML :** XML is a scripting languages intended for describing the data.
4) **WSDL :** It stands for Web Services Description Language. It is basically XML notation for describing the interface provided by web service provider.
5) **Extension .asmx :** In .NET the web services have the extension ASMX.

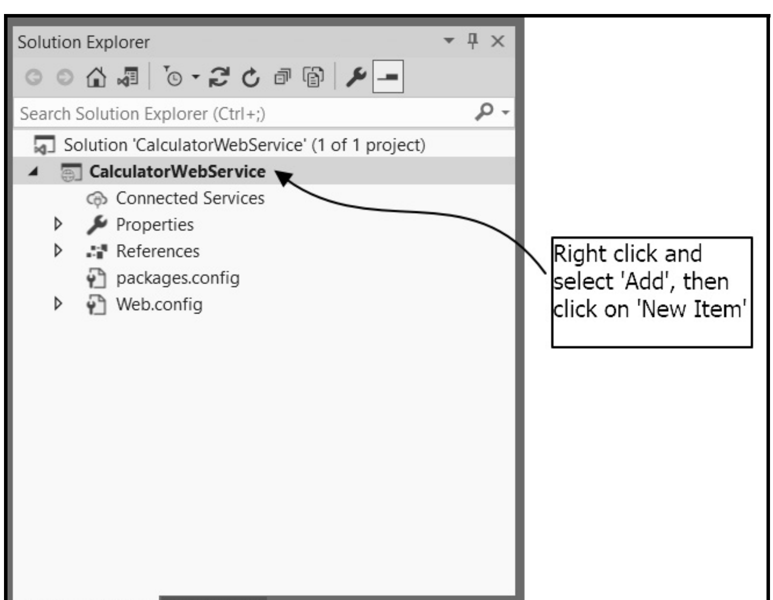
5.25.1 Writing Web Service

- In this section we will write the **web service method** in the web application program. For writing the web services, ASP.NET Web application (.NET Framework) is used.

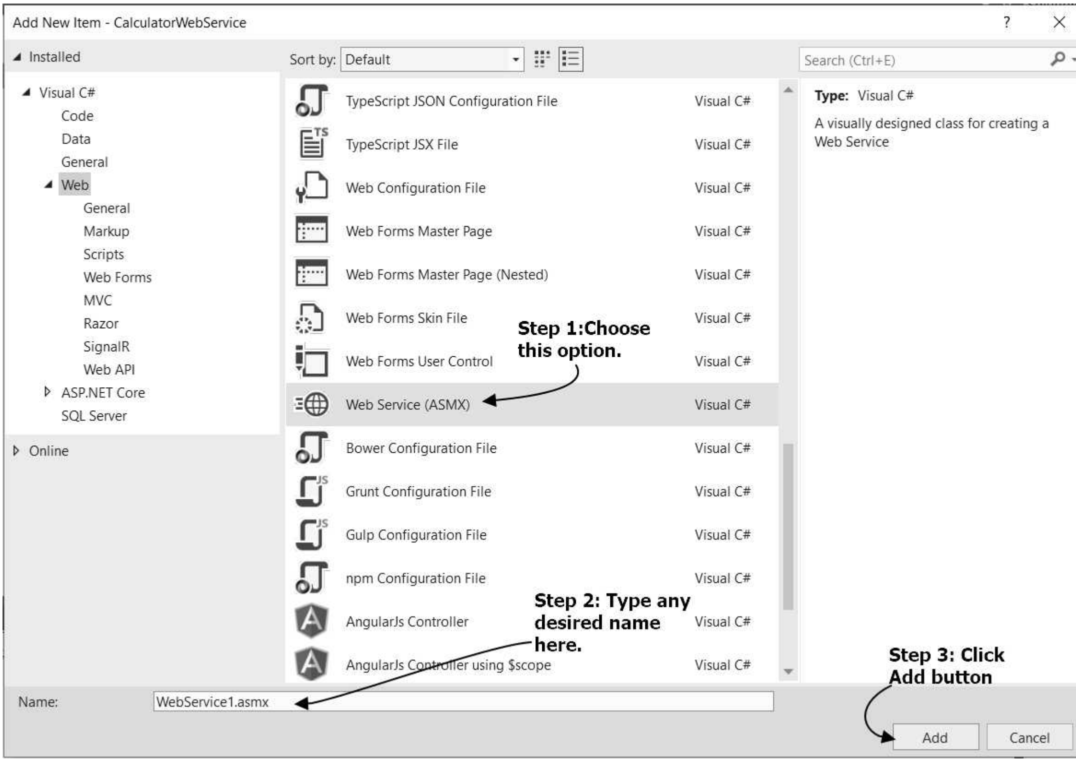
Step 1 : Open Visual Studio. Click on **Create a New Project**. Select **ASP.NET Web Application (.NET Framework)**. Give the name of the application. I have given the name as **CalculatorWebService**. Then select **Empty** project template.



Step 2 : In the Solution Explorer window locate the name of the project. Right click on project name.



Locate Web service option. Following screenshot illustrates it.



Step 3 : The source code window appears. Initially the default source code will be present in the source code window. Inside the webservice class, there exists a default **[WebMethod]**. Replace default **WebMethod** by any desired method for which you want to develop a web service. I want to write a web service for displaying maximum number out of two numbers. I have written a method with the name **ShowMaxValue**. To this method two integer parameters are passed as arguments.

- The code for this web service is as follows -

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Services;
namespace CalculatorWebService
{
   /// <summary>
   /// Summary description for WebService1
   /// </summary>
   [WebService(Namespace = "http://tempuri.org/")]
   [WebServiceBinding(ConformsTo = WsiProfiles.BasicProfile1_1)]
   [System.ComponentModel.ToolboxItem(false)]
```

```
    // To allow this Web Service to be called from script, using ASP.NET AJAX, uncomment the following line.
    // [System.Web.Script.Services.ScriptService]
    public class WebService1 : System.Web.Services.WebService
    {
        [WebMethod]
        public int ShowMaxValue(int a,int b)
        {
            if (a > b)
                return a;
            else
                return b;
        }
    }
}
```

Program Explanation :

The web service application is written in some namespace. In above program we have saved the web service by the name of the namespace as **CalculatorWebService.**

1) Web service is a class which is declared with **[WebService]** attribute. It is inherited from **System.Web.Services.WebService** base class. The attribute [WebService] tells that this class contains the code for web services.

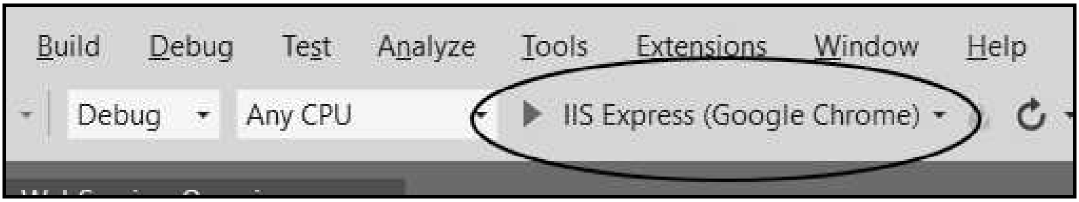
WebService namespace is used to uniquely identify our web service among all other web services present on the internet. It is generally written as follows -

```
[WebService(Namespace = "http://pragimtech.com/webservices")]
```

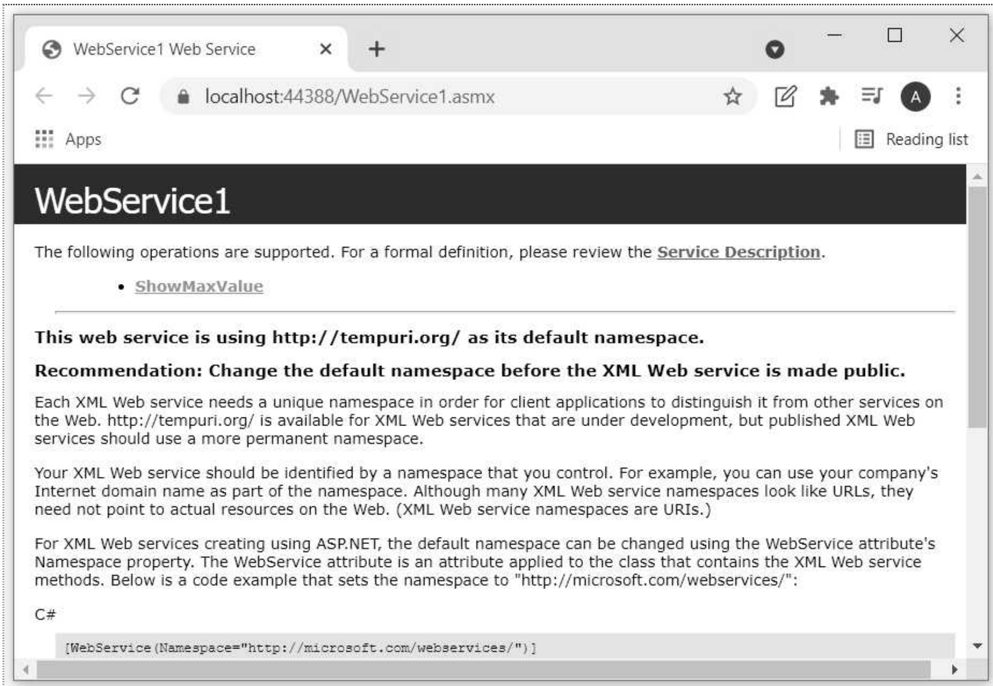
In above program, we have created a web service using the method **ShowMaxValue.** Before declaring this method we use the attribute **[WebMethod]**. If you want a method name to be displayed on the web browser as a part of the web service, then it must be declared as **public** and it must be decorated with [WebMethod] attribute.

We have passed two arguments to **ShowMaxValue** method. This method finds the largest among the two numbers and return this largest number.

Step 4 : Here the web service is created. The above web service can be executed on IIS Express server. Click on the **IIS Express** option present below the menu bar.

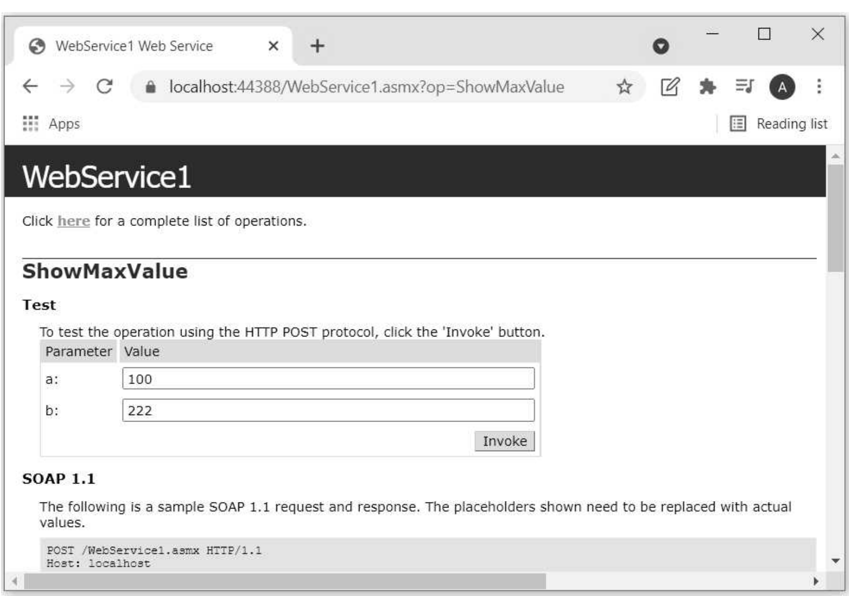


The output of the web service will be displayed on the web browser as follows -



Step 5 : From above window, just click on **<u>Service Description</u>** link. In the address bar we get the URL as **https://localhost:44388/WebService1.asmx?WSDL**. Just copy this link and save it to notepad.

If we click on **<u>ShowMaxValue</u>** we get following screen.

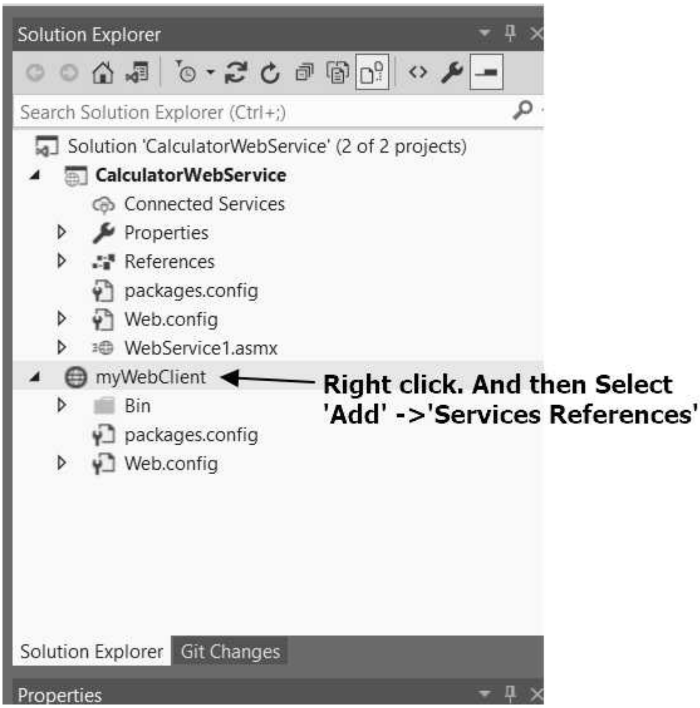


The two different values can be entered in the text boxes. On hitting the **Invoke** button, we get the result in the XML document. It is as follows. Note that the above layout is a default layout provided by .NET framework.

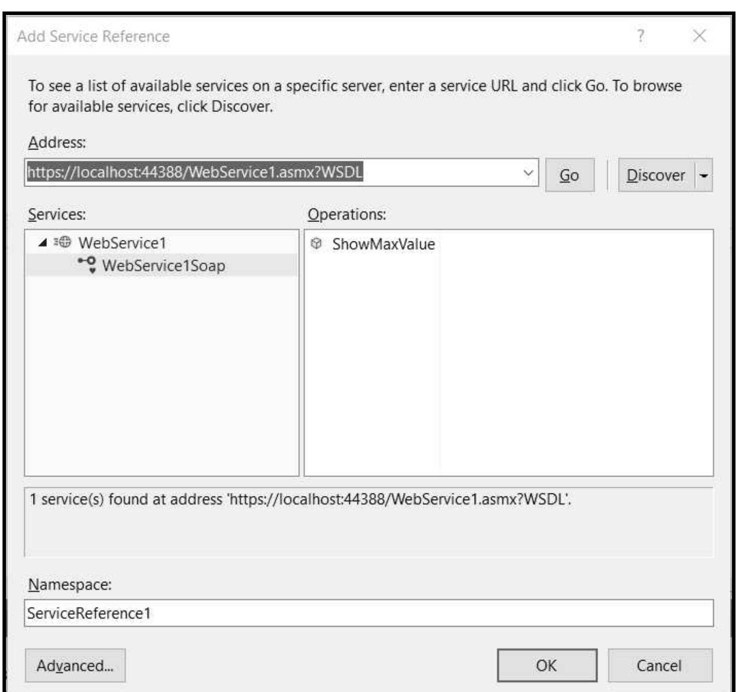
5.25.2 Writing Web service Client

- The above created web service can be invoked using the proxy web client. In this section we will create web service client by creating a HTML form. On the button click event of the form, the above created web service method **ShowMaxValue** will be invoked. The return value of this method is displayed as a result on the web browser.

Step 1 : In above section we have created a web service for displaying the maximum out of two numbers. Just right click on the name of the above web project which is displayed in the solution explorer window. Select **Add New Project->ASP.NET Empty Web Site**. Give suitable name to your web service client. I have given the name **myWebClient**.



In the step 5 of section **, we have copied a link as https://localhost:44388/WebService1.asmx?WSDL for WSDL in notepad. Just paste this link as a service reference and then click on **Go** button.



Click **OK** button. Thus the web service reference by the namespace **ServiceReference1** is created for the web service client.

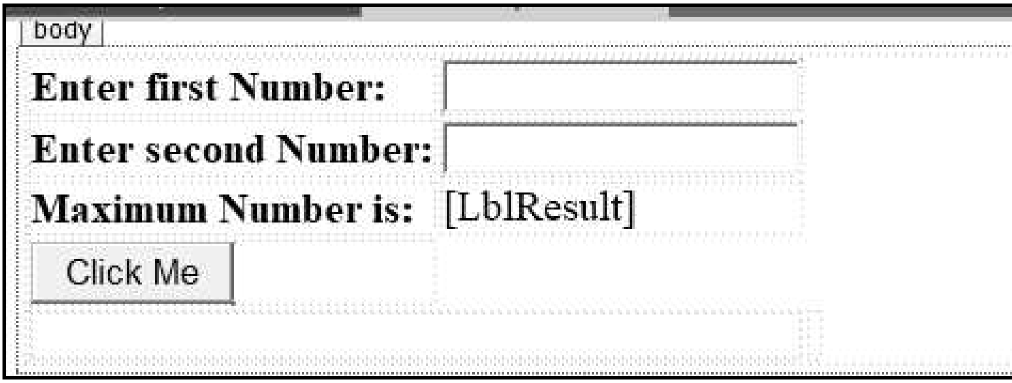
Step 2 : In the solution explorer window, right click on **myWebClient** and add a form.

Write following HTML code for creating two text boxes for inputting two numbers, one label for displaying the result and one button to invoke the web service for showing the maximum out of two numbers.

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Default.aspx.cs"
Inherits="_Default" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
      <table>
         <tr><td><b>Enter first Number:</b></td>
            <td> <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox></td>
         </tr>
         <tr><td><b>Enter second Number:</b></td>
            <td> <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox></td>
         </tr>
         <tr><td><b>Maximum Number is:</b></td>
            <td><asp:Label ID="LblResult" runat="server" Text=""></asp:Label> </td>
         </tr>
         <tr><td>
            <asp:Button ID="btnResult" runat="server" Text="Click Me" OnClick="btn_Click"/>
         </tr>
          <tr><td colspan="2">
            <td> </td>
         </tr>

      </table>
   </form>
</body>
</html>
```

Script Explanation : In above given HTML document we have created two text boxes for inputting two numbers. One label control for displaying the maximum of the two numbers. On button click, the web service method **ShowMaxValue** is called.

Step 3 : Click on **Design** tab, the form design will be as follows –



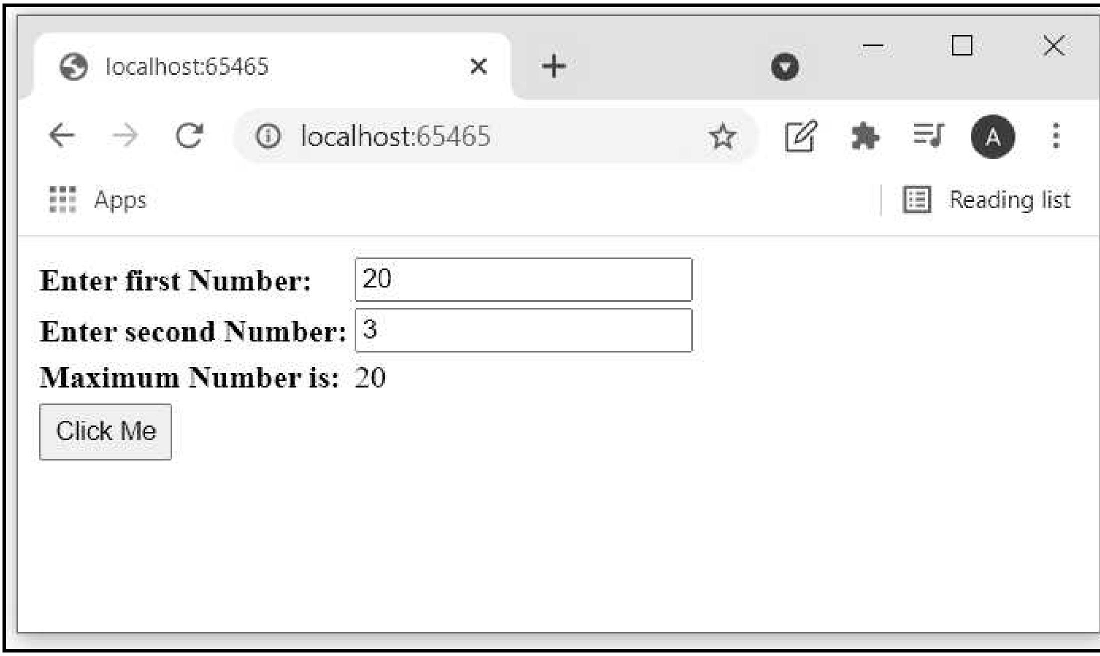
- Double click on 'Click Me' button. Write the code for button click event as follows –

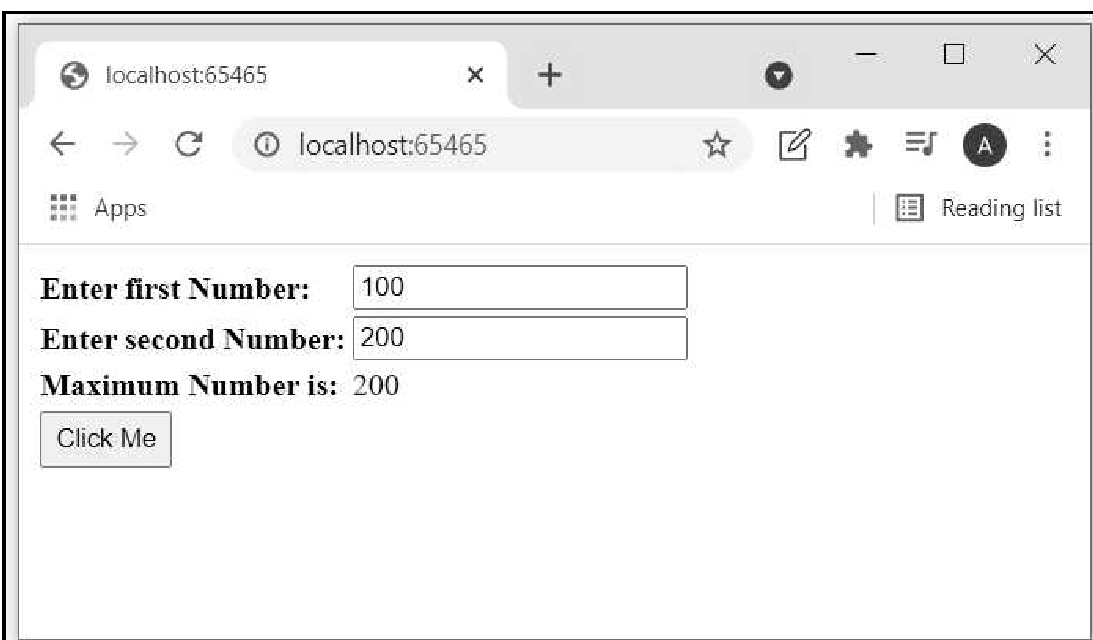
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
public partial class _Default : System.Web.UI.Page
{
   protected void Page_Load(object sender, EventArgs e)
   {
   }
   protected void btn_Click(object sender, EventArgs e)
   {
         ServiceReference1.WebService1SoapClient client =
         new ServiceReference1.WebService1SoapClient();
         int result = client.ShowMaxValue(
Convert.ToInt32(TextBox1.Text), Convert.ToInt32(TextBox2.Text));
         LblResult.Text = result.ToString();
   }
}
```

Web Service method named **(ShowMaxValue)** written in previous section is invoked.

The **Convert.ToInt32** method is used to convert the textbox string to numeric value.

Step 4 : Right click on web client project and click on 'view in Browser'. The output will be as follows -





Example 5.25.1 *Write a web service and web client for implementing addition of two numbers using .NET.*

Solution : Step 1 : Web service application

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Services;
namespace CalculatorWebService
{
   [WebService(Namespace = "http://tempuri.org/")]
   [WebServiceBinding(ConformsTo = WsiProfiles.BasicProfile1_1)]
   [System.ComponentModel.ToolboxItem(false)]

   public class WebService1 : System.Web.Services.WebService
   {
      [WebMethod]
      public int Add(int a,int b)
      {
            return a+b;
      }
   }
}
```

Step 2 : The web service client.

HTML Document for creating GUI

```
<%@ Page Language="C#" AutoEventWireup="true" CodeFile="Default.aspx.cs"
Inherits="_Default" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
   <title></title>
</head>
<body>
   <form id="form1" runat="server">
      <table>
         <tr><td><b>Enter first Number:</b></td>
            <td> <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox></td>
         </tr>
         <tr><td><b>Enter second Number:</b></td>
            <td> <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox></td>
         </tr>
         <tr><td><b>Addition is:</b></td>
            <td><asp:Label ID="LblResult" runat="server" Text=""></asp:Label> </td>
```

```
        </tr>
        <tr><td>
          <asp:Button ID="btnResult" runat="server" Text="Click Me" OnClick="btn_Click"/>
        </tr>

      </table>
   </form>
</body>
</html>
```

C# Program for handling button click event

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

public partial class _Default : System.Web.UI.Page
{
   protected void btn_Click(object sender, EventArgs e)
   {
     ServiceReference1.WebService1SoapClient client =
     new ServiceReference1.WebService1SoapClient();
     int result = client.Add(
     Convert.ToInt32(TextBox1.Text), Convert.ToInt32(TextBox2.Text));
         LblResult.Text = result.ToString();
   }
}
```

Part - III : Node.js

5.26 Overview of Node JS

SPPU : May-18, Marks 4

- NodeJS is an open source technology for server.
- Using Node.js we can run JavaScript on server.
- It runs on various platform such as Windows, Linux, Unix, and MacOS.

Uses of Node.js

- It can perform various tasks such as -
  1. It can create, open, read, delete, write and close files on the server.
  2. It can collect form data.
  3. It can also add, delete, modify data in databases.
  4. It generate dynamic web pages.

How to write and execute Node.js document ?

- For executing the Node.js scripts we need to install it. We can get it downloaded from https://nodejs.org/en.
- The installation can be done on Linux or Windows OS. It has very simple installation procedure.
- After successful installation we can now execute the Node.js document.
- The Node.js file has extension .js.

Step 1 : Open Notepad and type the following code. Save the file with some filename having extension .js

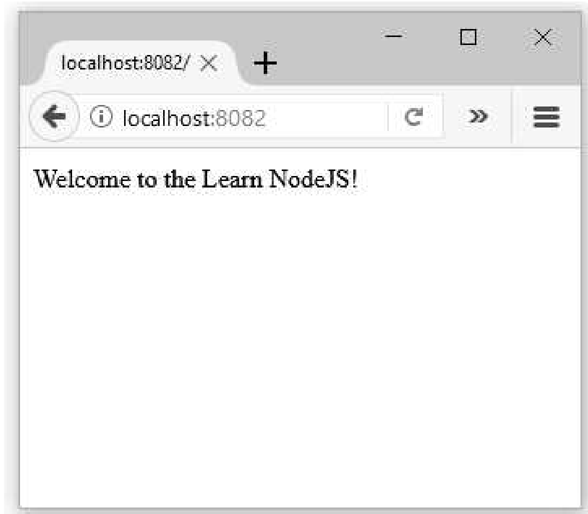
Test.js

```
var http = require('http');
http.createServer(function (req, res) {
   res.writeHead(200, {'Content-Type': 'text/html'});
   res.end("Welcome to the Learn NodeJS!");
}).listen(8082);
console.log('Server is running at http://localhost:8082/');
```

Step 2 : Open command prompt window and type the command for executing above script. The screen-shot for this execution is as follows -

Now open the web browser such as Mozilla Firefox or Chrome or Internet Explorer and in the address bar type the URL http://localhost:8082/

The output will be as follows –



Components of Node.js Application

- Module is similar to JavaScript libraries. It is basically the set of functions that need to be included in the application. For including the module we need a function **require().** For example to include the HTTP module in the application we write -

```
var http = require('http');
```

- Node.js has a built-in module called HTTP, which allows Node.js to transfer data over the Hyper Text Transfer Protocol.
- The HTTP module can create an HTTP server that listens to server ports and gives a response back to the client. Use the **createServer()** method to create an HTTP server. For example -

```
http.createServer(function (req, res) {
  res.writeHead(200, {'Content-Type': 'text/html'});
  res.end("Welcome to the Learn NodeJS!");
}).listen(8082);
```

- If the response from the HTTP server is supposed to be displayed as HTML, you should include an HTTP header with the correct content type. For example

```
res.writeHead(200, {'Content-Type': 'text/html'});
```

The status code, 200 means that all is OK, the second argument is an object containing the response headers.

- The above code can be executed and the web page can be displayed on the Web browser on particular port using **.listen(port number)**

5.26.1 Features of Node JS

- Following are some remarkable features of node.js –
  1) **Non blocking thread execution :** Non blocking means while we are waiting for a response for something during our execution of tasks, we can continue executing the next tasks in the stack.
  2) **Efficient :** The node.js is built on V8 JavaScript engine and it is very fast in code execution.
  3) **Open source packages :** The Node community is enormous and the number of permissive open source projects available to help in developing your application in much faster manner.
  4) **No buffering :** The Node.js applications never buffer the data.
  5) **Single threaded and highly scalable :** Node.js uses a single thread model with event looping. Similarly the non-blocking response of Node.js makes it highly scalable to sever large number of requests.

Review Question

1. *Write short note on NodeJS.* **SPPU : May-18, Marks 4**

5.27 Multiple Choice Questions

Q.1 In PHP, a variable starts with the sign _______.

a $	b &
c #	d @

Q.2 PHP supports the following data types :

a String	b Integer
c Array	d All the above

Q.3 In PHP, to represent exponentiation, the following operator is used :

a *	b **
c ^	d $

Q.4 A cookie is created by using the following function :

a getcookie	b setcookie
c getcookie()	d setcookie()

Q.5 The following mode opens the file for write only :

a w
b r
c w+
d a+

Q.6 How do you get information from a form that is submitted using the "get" method ?

a $_GET[];
b Request.Form;
c Request.QueryString;
d Request.GetQuery;

Q.7 Which superglobal variable holds information about header, paths and script, locations ?

a $_GLOBALS
b $-SERVER
c $-SESSION
d $_GET

Q.8 The filesize() function returns the file size in _______.

a bits
b bytes
c kilobytes
d gigabyte

Q.9 Which function sets the file filename's last-modified and last-accessed times ?

a sets()
b Set()
c touch()
d touched()

Q.10 The file extension to ASP.NET web form is _______.

a .net
b .aspx
c .docx
d .html

Q.11 ASP.NET was developed by ______.

a Google
b IBM
c Oracle
d Microsoft

Q.12 The JIT is a part of _______ in .NET framework.

a Framework Class Library (FCL)
b Languages
c Common Language Runtime (CLR)
d All of the above

Q.13 In ASP.NET application, the DLL files are stored in _______ folder.

a bin
b App_data
c App_code
d App_LocalResources

Q.14 Which of the following is the first event of ASP.NET page when the user requests a web page ?

a Init b PreInit
c Render d Control Events

Q.15 Which of the following is the page life cycle event of ASP.NET page when the user makes use of button control to invoke click events a web page ?

a Init b PreInit
c Render d Control Events

Q.16 Web services can be discovered using _______.

a UDDII b UDDI
c UDDDI d UDII

Q.17 Which of the following is a directory service where enterprises register and search for web services ?

a WSDL b UDDI
c Supporting protocols of SOAP d None of the above

Q.18 Node.js is written in _______.

a C b C++
c JavaScript d All of the above

Q.19 The extension used to store the Node.js script is _______.

a .html b .java
c .js d .node

Answer Keys for Multiple Choice Questions :

Q.1	a	**Q.2**	d	**Q.3**	b	**Q.4**	b	**Q.5**	a
Q.6	a	**Q.7**	a	**Q.8**	b	**Q.9**	c	**Q.10**	b
Q.11	d	**Q.12**	c	**Q.13**	a	**Q.14**	b	**Q.15**	d
Q.16	b	**Q.17**	b	**Q.18**	d	**Q.19**	c		

❑❑❑